

Supplementary Material for

Discovering the Network Granger Causality in Large Vector Autoregressive Models

A Proofs of Theorems

A.1 Proof of Theorem 1

Proof. Fix any $(i, j) \in \mathcal{H}$ such that $\bar{v}_i = o(1)$. Denote $m_{ij} = \sqrt{\sigma_i^2 \omega_{jj}} = \sqrt{\sigma_i^2 \boldsymbol{\omega}'_j \boldsymbol{\Sigma}_x \boldsymbol{\omega}_j}$ and $\hat{m}_{ij} = \hat{\sigma}_i \sqrt{\hat{\boldsymbol{\omega}}'_j \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j}$ or $\hat{\sigma}_i \hat{\omega}_j$. By the construction of the debiased lasso estimator with Proposition 2 and Lemma 7, the t -statistic is written as

$$\begin{aligned} \mathbf{T}_{ij} - \frac{\sqrt{T}\phi_{ij}}{\hat{m}_{ij}} &= \frac{\sqrt{T}(\hat{\phi}_{ij} - \phi_{ij})}{\hat{m}_{ij}} = \frac{z_{ij} + r_{ij}}{m_{ij}} \left\{ 1 + \left(\frac{m_{ij}}{\hat{m}_{ij}} - 1 \right) \right\} \\ &= \left\{ \frac{z_{ij}}{m_{ij}} + O(\bar{r}_i) \right\} \{1 + O(M_\omega^2 \lambda)\}, \end{aligned}$$

which holds with probability at least $1 - O((N \vee T)^{-\nu})$. Here, $\bar{r}_i$ and $M_\omega^2 \lambda$ are asymptotically negligible (with high probability) by the assumed condition.

Next, prove the asymptotic normality of z_{ij}/m_{ij} for each $(i, j) \in \mathcal{H}$. Recall that

$$\frac{z_{ij}}{m_{ij}} = \sum_{t=1}^T \xi_{Tt}^{(i,j)}, \quad \xi_{Tt}^{(i,j)} = \frac{u_{it} \mathbf{x}'_t \boldsymbol{\omega}_j}{\sqrt{T} m_{ij}}.$$

By a simple calculation, we have $\text{Var}(\xi_{Tt}^{(i,j)}) = 1/T$ and $\sum_{t=1}^T (\xi_{Tt}^{(i,j)})^2 \rightarrow_p 1$. Further-

more, by an application of Lemma 3, we obtain

$$\max_t |\xi_{Tt}^{(i,j)}| \leq T^{-1/2} \max_t \|u_{it} \mathbf{x}_t\|_\infty \|\boldsymbol{\omega}_j\|_1 / m_{ij} \lesssim M_\omega T^{-1/2} \sqrt{\log^3(N \vee T)}$$

with high probability, which implies $\max_t |\xi_{Tt}^{(i,j)}| \rightarrow_p 0$ when $M_\omega^2 \lambda = o(1)$. Thus McLeish's central limit theorem is applicable to achieve $\mathbf{T}_{ij} - \sqrt{T} \phi_{ij} / \hat{m}_{ij} \rightarrow_d N(0, 1)$ for each $(i, j) \in \mathcal{H}$ such that $\bar{r}_i + M_\omega^2 \lambda = o(1)$. This completes the proof. $\square$

A.2 Proof of Theorem 2

Proof. Throughout this proof, let $p = |\mathcal{H}| = KN^2$ denote the total number of parameters in the (augmented) coefficient matrix, Φ . Define

$$\begin{aligned} \mathcal{H}_{\leq 0} &= \mathcal{H} \cap \{(i, j) : \phi_{ij} \leq 0\}, & \mathcal{H}_{\geq 0} &= \mathcal{H} \cap \{(i, j) : \phi_{ij} \geq 0\}, \\ \mathcal{S}_{< 0} &= \mathcal{H} \cap \{(i, j) : \phi_{ij} < 0\}, & \mathcal{S}_{> 0} &= \mathcal{H} \cap \{(i, j) : \phi_{ij} > 0\}. \end{aligned}$$

By the definition of $s = |\mathcal{S}| = |\mathcal{H} \cap \{(i, j) : \phi_{ij} \neq 0\}|$, there is some sequence $\pi = \pi_{np} \in [0, 1]$ such that $|\mathcal{S}_{< 0}| = \pi s$ and $|\mathcal{S}_{> 0}| = (1 - \pi)s$. We also have $|\mathcal{H}_{\leq 0}| = p - (1 - \pi)s$ and $|\mathcal{H}_{\geq 0}| = p - \pi s$.

Case 1. Consider when (6) does not exist and $\mathbf{t}_0 = \sqrt{2 \log p}$. First, we observe that

$$\begin{aligned} \text{dFDR}(\mathbf{t}_0) &\leq \text{dFWER} = \mathbb{P} \left(\sum_{(i,j) \in \hat{\mathcal{S}}(\mathbf{t}_0)} 1\{\text{sgn}(\hat{\phi}_{ij}) \neq \text{sgn}(\phi_{ij})\} \geq 1 \right) \\ &\leq \mathbb{P} \left(\sum_{(i,j) \in \mathcal{H}_{\leq 0}} 1\{\mathbf{T}_{ij} \geq \mathbf{t}_0\} \geq 1 \right) + \mathbb{P} \left(\sum_{(i,j) \in \mathcal{H}_{\geq 0}} 1\{\mathbf{T}_{ij} \leq -\mathbf{t}_0\} \geq 1 \right). \end{aligned} \tag{A.1}$$

The first probability of (A.1) is further bounded as

$$\begin{aligned}
& \mathbb{P} \left(\sum_{(i,j) \in \mathcal{H}_{\leq 0}} 1\{\mathbf{T}_{ij} \geq \mathbf{t}_0\} \geq 1 \right) \leq \sum_{(i,j) \in \mathcal{H}_{\leq 0}} \mathbb{P}(\mathbf{T}_{ij} \geq \mathbf{t}_0) \\
& \leq \sum_{(i,j) \in \mathcal{S}^c} \mathbb{P}(\mathcal{Z}_{ij} \geq \mathbf{t}_0 - \delta_1) + \sum_{(i,j) \in \mathcal{S}_{<0}} \mathbb{P}(\mathcal{Z}_{ij} + \sqrt{T}\phi_{ij}/\hat{m}_{ij} \geq \mathbf{t}_0 - \delta_1) \\
& \quad + pO((N \vee T)^{-\nu+2}) \\
& \leq \sum_{(i,j) \in \mathcal{S}^c} \mathbb{P}(\mathcal{Z}_{ij} \geq \mathbf{t}_0 - \delta_1) + \sum_{(i,j) \in \mathcal{S}_{<0}} \mathbb{P}(\mathcal{Z}_{ij} \geq \mathbf{t}_0 - \delta_1) + pO((N \vee T)^{-\nu+2}) \\
& = (p - s + \pi s)Q(\mathbf{t}_0 - \delta_1) + pO((N \vee T)^{-\nu+2}) \\
& \leq pQ(\mathbf{t}_0 - \delta_1) + pO((N \vee T)^{-\nu+2}),
\end{aligned}$$

where the second inequality holds for some positive sequence $\delta_1 = O(T^{-\kappa})$ for some $\kappa \in (0, \kappa_1]$ by Lemma 8(a) and $Q(\mathbf{t}) = \mathbb{P}(\mathcal{Z} > \mathbf{t})$ is the upper tail probability of a standard normal random variable $\mathcal{Z}$. Because $Q(\mathbf{t}) \leq (\mathbf{t}\sqrt{2\pi})^{-1} \exp(-\mathbf{t}^2/2)$ and $\mathbf{t}_0 - \delta_1 = \sqrt{2\log p} + o(1)$, we have

$$pQ(\mathbf{t}_0 - \delta_1) \lesssim p(\sqrt{\log p})^{-1} \exp(-\log p + o(1)) = O(1/\sqrt{\log p}).$$

We also obtain $pO((N \vee T)^{-\nu+2}) = o(1)$ for $\nu > 4$. The second probability of (A.1) is bounded in the same way. We thus conclude $\text{dFDR}(\mathbf{t}_0) \leq \text{dFWER} = o(1)$.

Case 2. Consider when $\mathbf{t}_0$ is given by (6). Define

$$V = \sup_{\mathbf{t} \in [0, \bar{\mathbf{t}}]} \left| \frac{\sum_{(i,j) \in \mathcal{H}_{\leq 0}} [1\{\mathbf{T}_{ij} \geq \mathbf{t}\} - Q(\mathbf{t})] + \sum_{(i,j) \in \mathcal{H}_{\geq 0}} [1\{\mathbf{T}_{ij} \leq -\mathbf{t}\} - Q(\mathbf{t})]}{2pQ(\mathbf{t})} \right|.$$

Then we have

$$\begin{aligned}
\text{dFDP}(\mathbf{t}_0) &= \frac{|\{(i, j) \in \hat{\mathcal{S}}(\mathbf{t}_0) : \text{sgn}(\hat{\phi}_{ij}) \neq \text{sgn}(\phi_{ij})\}|}{|\hat{\mathcal{S}}(\mathbf{t}_0)| \vee 1} \\
&= \frac{2pQ(\mathbf{t}_0)}{|\hat{\mathcal{S}}(\mathbf{t}_0)| \vee 1} \frac{\sum_{(i,j) \in \mathcal{H}_{\leq 0}} 1\{\mathbf{T}_{ij} \geq \mathbf{t}_0\} + \sum_{(i,j) \in \mathcal{H}_{\geq 0}} 1\{\mathbf{T}_{ij} \leq -\mathbf{t}_0\}}{2pQ(\mathbf{t}_0)} \\
&\leq q \left[\frac{\sum_{(i,j) \in \mathcal{H}_{\leq 0}} 1\{\mathbf{T}_{ij} \geq \mathbf{t}_0\} + \sum_{(i,j) \in \mathcal{H}_{\geq 0}} 1\{\mathbf{T}_{ij} \leq -\mathbf{t}_0\}}{2pQ(\mathbf{t}_0)} \right] \\
&\leq q \left[V + \frac{(|\mathcal{H}_{\leq 0}| + |\mathcal{H}_{\geq 0}|)Q(\mathbf{t}_0)}{2pQ(\mathbf{t}_0)} \right] \leq q(V + 1).
\end{aligned}$$

Thus the dFDR and dFDP are controlled if we prove $V = o_p(1)$ in view of Fatou's lemma and Markov's inequality, respectively. Note that

$$\begin{aligned}
V &\leq \sup_{\mathbf{t} \in [0, \bar{\mathbf{t}}]} \left| \frac{\sum_{(i,j) \in \mathcal{H}_{\leq 0}} [1\{\mathbf{T}_{ij} \geq \mathbf{t}\} - Q(\mathbf{t})]}{2pQ(\mathbf{t})} \right| \\
&\quad + \sup_{\mathbf{t} \in [0, \bar{\mathbf{t}}]} \left| \frac{\sum_{(i,j) \in \mathcal{H}_{\geq 0}} [1\{\mathbf{T}_{ij} \leq -\mathbf{t}\} - Q(\mathbf{t})]}{2pQ(\mathbf{t})} \right|.
\end{aligned}$$

We only prove that the first term is $o_p(1)$ by symmetry.

To this end, consider discretization. That is, we partition $[0, \bar{\mathbf{t}}]$ into small intervals, $0 = \mathbf{t}_0 < \mathbf{t}_1 < \dots < \mathbf{t}_h = \bar{\mathbf{t}} = (2 \log p - a \log \log p)^{1/2}$, such that $\mathbf{t}_m - \mathbf{t}_{m-1} = v_p$ for $m \in \{1, \dots, h-1\}$ and $\mathbf{t}_h - \mathbf{t}_{h-1} \leq v_p$, where $v_p = (\log p \log \log p)^{-1/2}$. Then a simple calculation gives $1/h \leq v_p/\bar{\mathbf{t}} = O(1/(\log p \sqrt{\log \log p}))$. Fix arbitrary $m \in \{1, \dots, h\}$. For any $\mathbf{t} \in [\mathbf{t}_{m-1}, \mathbf{t}_m]$, we have

$$\frac{\sum_{(i,j) \in \mathcal{H}_{\leq 0}} 1\{\mathbf{T}_{ij} \geq \mathbf{t}\}}{2pQ(\mathbf{t})} \leq \frac{\sum_{(i,j) \in \mathcal{H}_{\leq 0}} 1\{\mathbf{T}_{ij} \geq \mathbf{t}_{m-1}\}}{2pQ(\mathbf{t}_{m-1})} \frac{Q(\mathbf{t}_{m-1})}{Q(\mathbf{t}_m)}$$

and

$$\frac{\sum_{(i,j) \in \mathcal{H}_{\leq 0}} 1\{\mathbf{T}_{ij} \geq \mathbf{t}\}}{2pQ(\mathbf{t})} \geq \frac{\sum_{(i,j) \in \mathcal{H}_{\leq 0}} 1\{\mathbf{T}_{ij} \geq \mathbf{t}_m\}}{2pQ(\mathbf{t}_m)} \frac{Q(\mathbf{t}_m)}{Q(\mathbf{t}_{m-1})}.$$

Lemma 7.2 of [Javanmard and Javadi \(2019\)](#) gives

$$\frac{Q(\mathbf{t}_{m-1})}{Q(\mathbf{t}_m)} \leq \frac{Q(\mathbf{t}_m - v_p)}{Q(\mathbf{t}_m)} = 1 + O(v_p + v_p \bar{\mathbf{t}}) = 1 + o(1)$$

uniformly in $m \in \{1, \dots, h\}$. Thus the proof completes if the following is true:

$$\tilde{V} := \max_{m \in \{1, \dots, h\}} \left| \frac{\sum_{(i,j) \in \mathcal{H}_{\leq 0}} [1\{\mathbf{T}_{ij} \geq \mathbf{t}_m\} - Q(\mathbf{t}_m)]}{2pQ(\mathbf{t}_m)} \right| = o_p(1).$$

Prove $\tilde{V} = o_p(1)$. Fix arbitrary $\varepsilon > 0$. Then we have

$$\begin{aligned} \mathbb{P}(\tilde{V} > \varepsilon) &\leq h \max_{m \in \{1, \dots, h\}} \mathbb{P}\left(\left| \frac{\sum_{(i,j) \in \mathcal{H}_{\leq 0}} [1\{\mathbf{T}_{ij} \geq \mathbf{t}_m\} - Q(\mathbf{t}_m)]}{2pQ(\mathbf{t}_m)} \right| > \varepsilon\right) \\ &\leq h \max_{m \in \{1, \dots, h\}} \mathbb{E} \left| \frac{\sum_{(i,j) \in \mathcal{H}_{\leq 0}} [1\{\mathbf{T}_{ij} \geq \mathbf{t}_m\} - Q(\mathbf{t}_m)]}{2pQ(\mathbf{t}_m)} \right|^2 / \varepsilon^2, \end{aligned}$$

where $h \lesssim \log p \sqrt{\log \log p}$. Consider bounding the expectation uniformly in $m \in \{1, \dots, h\}$. Denote by $\sum$ and $\sum \sum$ the summations over $(i, j) \in \mathcal{H}_{\leq 0}$ and $(i, j), (k, \ell) \in \mathcal{H}_{\leq 0}$, respectively. By a simple calculus and Lemma 8(a)(b), we obtain

$$\begin{aligned} &\mathbb{E} \left[\frac{\sum \sum [1\{\mathbf{T}_{ij} \geq \mathbf{t}_m\} - Q(\mathbf{t}_m)] [1\{\mathbf{T}_{k\ell} \geq \mathbf{t}_m\} - Q(\mathbf{t}_m)]}{4p^2Q(\mathbf{t}_m)^2} \right] \\ &= \frac{\sum \sum \mathbb{P}(\mathbf{T}_{ij} \geq \mathbf{t}_m, \mathbf{T}_{k\ell} \geq \mathbf{t}_m)}{4p^2Q(\mathbf{t}_m)^2} - \frac{|\mathcal{H}_{\leq 0}| \sum \mathbb{P}(\mathbf{T}_{ij} \geq \mathbf{t}_m)}{2p^2Q(\mathbf{t}_m)} + \frac{|\mathcal{H}_{\leq 0}|^2}{4p^2} \\ &\leq \frac{\sum \sum \mathbb{P}(\mathcal{Z}_{ij} \geq \mathbf{t}_m - \delta_1, \mathcal{Z}_{k\ell} \geq \mathbf{t}_m - \delta_1)}{4p^2Q(\mathbf{t}_m)^2} - \frac{(1 + O(s/p))Q(\mathbf{t}_m + \delta_1)}{2Q(\mathbf{t}_m)} + \frac{1}{4} \\ &=: (i) + (ii) + 1/4, \end{aligned}$$

where $(\mathcal{Z}_{ij}, \mathcal{Z}_{k\ell})$ is a bivariate standard normal random vector and $\delta_1 = O(T^{-\kappa})$ for some constant $\kappa > 0$. Thus we will conclude $\tilde{V} = o_p(1)$ if we show that $(i) \leq 1/4 + o(1/h)$ and $(ii) \leq -1/2 + o(1/h)$, where $1/h = O(1/(\log p \sqrt{\log \log p}))$.

First consider (ii) . Expand $Q(\mathbf{t}_m + \delta_1)$ around $\delta_1 = 0$ by the mean value theorem.

Then there exists δ_1^* between 0 and δ_1 such that

$$\begin{aligned}
(ii) &= -\frac{Q(\mathfrak{t}_m + \delta_1)}{2Q(\mathfrak{t}_m)}(1 + o(1)) = -\frac{Q(\mathfrak{t}_m) + Q'(\mathfrak{t}_m + \delta_1^*)\delta_1}{2Q(\mathfrak{t}_m)}(1 + o(1)) \\
&\leq -1/2 - \frac{(2\pi)^{-1/2} \exp\{-(\mathfrak{t}_m + \delta_1^*)^2/2\} \delta_1}{2(2\pi)^{-1/2} \mathfrak{t}_m^{-1} \exp\{-\mathfrak{t}_m^2/2\}}(1 + o(1)) \\
&= -1/2 - \mathfrak{t}_m \exp\{-\delta_1^*(\mathfrak{t}_m + \delta_1^*/2)\} \delta_1(1 + o(1))/2 \\
&= -1/2 + o(1/h),
\end{aligned} \tag{A.2}$$

where the last equality holds uniformly in $m \in \{1, \dots, h\}$ because δ_1 is polynomially decaying while $\mathfrak{t}_m$ is the logarithmic function for all $m \in \{1, \dots, h\}$.

Next consider (i) by decomposing the summation into two parts, $(i, j) = (k, \ell)$ and $(i, j) \neq (k, \ell)$. First we see the summation over $(i, j) = (k, \ell)$, which has p entries. We have

$$\begin{aligned}
&\frac{\sum \sum \mathbb{P}(\mathcal{Z}_{ij} \geq \mathfrak{t}_m - \delta_1, \mathcal{Z}_{k\ell} \geq \mathfrak{t}_m - \delta_1) 1\{(i, j) = (k, \ell)\}}{4p^2 Q(\mathfrak{t}_m)^2} \\
&= \frac{|\mathcal{H}_{\geq 0}|}{4p} \frac{1}{pQ(\mathfrak{t}_m)} \frac{Q(\mathfrak{t}_m - \delta_1)}{Q(\mathfrak{t}_m)} = o(1/h).
\end{aligned} \tag{A.3}$$

The last estimate is true because we have $|\mathcal{H}_{\geq 0}|/p = O(1)$ and $Q(\mathfrak{t}_m - \delta_1)/Q(\mathfrak{t}_m) = 1 + o(1/h)$ by the same reason as above, and by [Szarek and Werner \(1999\)](#),

$$\begin{aligned}
\frac{1}{pQ(\mathfrak{t}_m)} &\leq \frac{\bar{\mathfrak{t}} + (\bar{\mathfrak{t}}^2 + 4)^{1/2}}{p2(2\pi)^{-1/2} \exp\{-\bar{\mathfrak{t}}^2/2\}} \lesssim \frac{\bar{\mathfrak{t}}}{p \exp\{-\bar{\mathfrak{t}}^2/2\}} \\
&\lesssim \log^{1/2} p \cdot \exp\{-\log \log^{a/2} p\} = O(\log^{1/2-a/2} p) = o(1/h)
\end{aligned} \tag{A.4}$$

for any $a > 3$.

Finally we bound (i) with summation over $(i, j) \neq (k, \ell)$, which contains $p^2 - p$

entries. We have

$$\begin{aligned} & \frac{\sum \sum \mathbb{P}(\mathcal{Z}_{ij} \geq \mathfrak{t}_m - \delta_1, \mathcal{Z}_{k\ell} \geq \mathfrak{t}_m - \delta_1)}{4p^2 Q(\mathfrak{t}_m)^2} \\ & \leq \frac{\sum \sum \mathbb{P}(\mathcal{Z}_{ij} \geq \mathfrak{t}_m - \delta_1, \mathcal{Z}_{k\ell} \geq \mathfrak{t}_m - \delta_1)}{4p^2 Q(\mathfrak{t}_m - \delta_1)^2} \frac{Q(\mathfrak{t}_m - \delta_1)^2}{Q(\mathfrak{t}_m)^2}, \end{aligned} \quad (\text{A.5})$$

where $Q(\mathfrak{t}_m - \delta_1)^2 / Q(\mathfrak{t}_m)^2 = 1 + o(1/h)$ as above. By Lemma 9 and the inequality $1/\sqrt{1-x^2} \leq 1 + |x|/\sqrt{1-x^2}$ for any $x \in (-1, 1)$, we obtain

$$\begin{aligned} & \frac{\sum \sum \mathbb{P}(\mathcal{Z}_{ij} \geq \mathfrak{t}_m - \delta_1, \mathcal{Z}_{k\ell} \geq \mathfrak{t}_m - \delta_1)}{4p^2 Q(\mathfrak{t}_m - \delta_1)^2} \leq \frac{1}{4p^2} \sum \sum \frac{1}{\sqrt{1 - \rho_{(i,j),(k,\ell)}^2}} \\ & = 1/4 + o(1/h) + \frac{1}{4p^2} \sum \sum \frac{|\rho_{(i,j),(k,\ell)}|}{\sqrt{1 - \rho_{(i,j),(k,\ell)}^2}}. \end{aligned} \quad (\text{A.6})$$

We evaluate the sum using Condition 6 with $\log N \asymp \log p$. Denote $\mathcal{H}_w^2 = \mathcal{H}_{w1} \times \mathcal{H}_{w2}$ and $\mathcal{H}_s^2 = \mathcal{H}_{s1} \times \mathcal{H}_{s2}$. Then it is decomposed as

$$\sum_{(i,j) \in \mathcal{H}_{\leq 0}} \sum_{(k,\ell) \in \mathcal{H}_{\leq 0}} = \sum_{(i,j) \in \mathcal{H}_{\leq 0} \cap \mathcal{H}_{w1}} \sum_{(k,\ell) \in \mathcal{H}_{\leq 0} \cap \mathcal{H}_{w2}} + \sum_{(i,j) \in \mathcal{H}_{\leq 0} \cap \mathcal{H}_{s1}} \sum_{(k,\ell) \in \mathcal{H}_{\leq 0} \cap \mathcal{H}_{s2}}.$$

They are bounded by

$$\begin{aligned} & \frac{1}{4p^2} \sum_{(i,j) \in \mathcal{H}_{\leq 0} \cap \mathcal{H}_{w1}} \sum_{(k,\ell) \in \mathcal{H}_{\leq 0} \cap \mathcal{H}_{w2}} \frac{|\rho_{(i,j),(k,\ell)}|}{\sqrt{1 - \rho_{(i,j),(k,\ell)}^2}} \\ & \leq \frac{1}{4p^2} \sum_{(i,j) \in \mathcal{H}_{w1}} \sum_{(k,\ell) \in \mathcal{H}_{w2}} \frac{|\rho_{(i,j),(k,\ell)}|}{\sqrt{1 - \rho_{(i,j),(k,\ell)}^2}} \\ & \leq \frac{p^2 - p - |\mathcal{H}_{s1} \times \mathcal{H}_{s2}|}{4p^2} \frac{c/\log^2 p}{\sqrt{1 - c^2/\log^4 p}} = O(1)O(1/\log^2 p) = o(1/h) \end{aligned} \quad (\text{A.7})$$

and

$$\begin{aligned}
& \frac{1}{4p^2} \sum_{(i,j) \in \mathcal{H}_{\leq 0} \cap \mathcal{H}_{s1}} \sum_{(k,\ell) \in \mathcal{H}_{\leq 0} \cap \mathcal{H}_{s2}} \frac{|\rho_{(i,j),(k,\ell)}|}{\sqrt{1 - \rho_{(i,j),(k,\ell)}^2}} \\
& \leq \frac{1}{4p^2} \sum_{(i,j) \in \mathcal{H}_{s1}} \sum_{(k,\ell) \in \mathcal{H}_{s2}} \frac{|\rho_{(i,j),(k,\ell)}|}{\sqrt{1 - \rho_{(i,j),(k,\ell)}^2}} \\
& \leq \frac{|\mathcal{H}_{s1} \times \mathcal{H}_{s2}|}{4p^2} \frac{|\bar{\rho}|}{\sqrt{1 - \bar{\rho}^2}} = O(1/\log^2 p)O(1) = o(1/h). \tag{A.8}
\end{aligned}$$

From (A.3) and (A.5)–(A.8), we have $(i) = 1/4 + o(1/h)$. This completes the proof. $\square$

A.3 Proof of Theorem 3

Before proceeding the proof, we recall the notation. In the proof, we mainly use the t -statistics with the “sandwich” s.e., defined as

$$T_{ij}^* = \frac{\sqrt{T}\hat{\phi}_{ij}^*}{\hat{\sigma}_i^* \sqrt{\hat{\omega}_j' \hat{\Sigma}_x \hat{\omega}_j}}$$

for $(i, j) \in \hat{\mathcal{S}}_{\mathbf{L}}^c$, where

$$\begin{aligned}
\hat{\sigma}_i^{*2} &= (T - \hat{s}_i)^{-1} \sum_{t=1}^T \hat{u}_{it}^{*2}, \quad \hat{u}_{it}^{*2} = u_{it}^{*2} - 2u_{it}^* \mathbf{x}_t' \boldsymbol{\delta}_i^* + \boldsymbol{\delta}_i^{*'} \mathbf{x}_t \mathbf{x}_t' \boldsymbol{\delta}_i^*, \\
\boldsymbol{\delta}_i^* &= (\hat{\boldsymbol{\phi}}_{i\cdot}^{\mathbf{L}*} - \hat{\boldsymbol{\phi}}_{i\cdot}^{\mathbf{L}})', \quad u_{it}^{*2} = \hat{u}_{it}^2 \zeta_t^2.
\end{aligned}$$

The bootstrap debiased lasso estimator has been defined as

$$\begin{aligned}
\hat{\boldsymbol{\Phi}}^* &= \hat{\boldsymbol{\Phi}}^{\mathbf{L}*} + \left(\mathbf{Y}^* - \hat{\boldsymbol{\Phi}}^{\mathbf{L}*} \mathbf{X} \right) \mathbf{X}' \hat{\boldsymbol{\Omega}} / T \\
&= \hat{\boldsymbol{\Phi}}^{\mathbf{L}} + \mathbf{U}^* \mathbf{X}' \boldsymbol{\Omega} / T + \mathbf{U}^* \mathbf{X}' (\hat{\boldsymbol{\Omega}} - \boldsymbol{\Omega}) / T + \left(\hat{\boldsymbol{\Phi}}^{\mathbf{L}} - \hat{\boldsymbol{\Phi}}^{\mathbf{L}*} \right) (\hat{\Sigma}_x \hat{\boldsymbol{\Omega}} - \mathbf{I}).
\end{aligned}$$

By the definition, we have $\hat{\phi}_{ij}^L = 0$ for all $(i, j) \in \hat{\mathcal{S}}_L^c$, and hence

$$\hat{\phi}_{ij}^* = \mathbf{u}_i^* \mathbf{X}' \boldsymbol{\omega}_j / T + \mathbf{u}_i^* \mathbf{X}' (\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j) / T + \left(\hat{\phi}_i^L - \hat{\phi}_i^{L*} \right) (\hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j - \mathbf{e}_j)$$

for all $(i, j) \in \hat{\mathcal{S}}_L^c$. We have also defined

$$\mathbb{Q}^*(\mathbf{t}) = \frac{1}{|\hat{\mathcal{S}}_L^c|} \sum_{(i,j) \in \hat{\mathcal{S}}_L^c} \mathbb{P}^*(\mathbf{T}_{ij}^* > \mathbf{t}) \quad (\text{conditional on the observations}),$$

where $|\hat{\mathcal{S}}_L^c| = p - \hat{s}$ with $p = KN^2$, and $\mathbb{P}^*$ indicates the probability measure induced by the bootstrap.

Proof. It is sufficient to consider the case when $\mathbf{t}_0$ is computed by (9). From the beginning of *Case 2* in the proof of Theorem 2 with the same argument, the FDR is decomposed as

$$\begin{aligned} & \text{dFDP}(\mathbf{t}_0) \\ &= \frac{p \{ \mathbb{Q}^*(\mathbf{t}_0) + 1 - \mathbb{Q}^*(-\mathbf{t}_0) \}}{|\hat{\mathcal{S}}(\mathbf{t}_0)| \vee 1} \cdot \frac{2Q(\mathbf{t}_0)}{\mathbb{Q}^*(\mathbf{t}_0) + 1 - \mathbb{Q}^*(-\mathbf{t}_0)} \\ & \quad \cdot \frac{\sum_{(i,j) \in \mathcal{I}_{\leq 0}} 1\{\mathbf{T}_{ij} \geq \mathbf{t}_0\} + \sum_{(i,j) \in \mathcal{I}_{\geq 0}} 1\{\mathbf{T}_{ij} \leq -\mathbf{t}_0\}}{2pQ(\mathbf{t}_0)} \\ &\leq q \cdot \frac{2}{\mathbb{Q}^*(\mathbf{t}_0)/Q(\mathbf{t}_0) + \{1 - \mathbb{Q}^*(-\mathbf{t}_0)\}/Q(\mathbf{t}_0)} \{V + 1 + o(1)\}, \end{aligned}$$

where V is defined in the proof of Theorem 2 and has been shown $V = o_p(1)$. Hence, it suffices to prove that the event,

$$\sup_{\mathbf{t} \in [0, \bar{\mathbf{t}}]} \left| \frac{\mathbb{Q}^*(\mathbf{t})}{Q(\mathbf{t})} - 1 \right| + \sup_{\mathbf{t} \in [0, \bar{\mathbf{t}}]} \left| \frac{1 - \mathbb{Q}^*(-\mathbf{t})}{Q(\mathbf{t})} - 1 \right| = o(1),$$

occurs with high probability.

Define

$$\begin{aligned} S_{ij}^* &= T^{-1/2} \sum_{t=1}^T u_{it}^* \mathbf{x}_t' \hat{\boldsymbol{\omega}}_j, \\ R_{ij}^* &= T^{-1/2} \mathbf{u}_i^* \mathbf{X}' (\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j) + \sqrt{T} (\hat{\boldsymbol{\phi}}_i^L - \hat{\boldsymbol{\phi}}_i^{L*}) (\hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j - \mathbf{e}_j), \\ \hat{m}_{ij}^{*2} &= \hat{\sigma}_i^{*2} \hat{\boldsymbol{\omega}}_j' \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j, \quad \tilde{m}_{ij}^{*2} = T^{-1} \sum_{t=1}^T u_{it}^{*2} \hat{\boldsymbol{\omega}}_j' \mathbf{x}_t \mathbf{x}_t' \hat{\boldsymbol{\omega}}_j, \end{aligned}$$

Then by the construction, we have $\mathbf{T}_{ij}^* = (S_{ij}^* + R_{ij}^*)/\hat{m}_{ij}^*$. We first check if $\mathbf{T}_{ij}^*$ is asymptotically equivalent to the self-normalized sum, $S_{ij}^*/\tilde{m}_{ij}^*$, and then verify that it can be uniformly normally approximated in the relative error.

Define the events,

$$A_1^* = \left\{ \max_{i \in [N]} \max_{j \in [KN]} \left| \frac{\tilde{m}_{ij}^*}{\hat{m}_{ij}^*} - 1 \right| \lesssim \bar{\mu}_1 \right\}, \quad A_2^* = \left\{ \max_{i \in [N]} \max_{j \in [KN]} \left| \frac{R_{ij}^*}{\hat{m}_{ij}^*} \right| \lesssim \bar{\mu}_2 \right\},$$

where

$$\begin{aligned} \bar{\mu}_1 &= \max \{ \bar{\tau}_1, \bar{\tau}_2, \bar{\tau}_3 \}, \\ \bar{\mu}_2 &= \max \left\{ \lambda^{2-r} M_\omega^{2-2r} s_\omega, \bar{s} \lambda M_\omega \log^{3/2}(N \vee T) \right\} \end{aligned}$$

with $\bar{\tau}_1$, $\bar{\tau}_2$, and $\bar{\tau}_3$ defined in Lemmas 13, 14, and 15, respectively. Then the event,

$$A := \left\{ \hat{\mathcal{S}}_{\mathbf{L}} \supset \mathcal{S} \right\} \cap \left\{ \mathbb{P}^* (A_1^{*c}) = O((NT)^{-\nu}) \right\} \cap \left\{ \mathbb{P}^* (A_2^{*c}) = O((NT)^{-\nu}) \right\},$$

occurs with high probability by Lemmas 16, 17, and 19. By a simple computation, we have $\bar{\mu}_1 \vee \bar{\mu}_2 \asymp \bar{\mu}$ with

$$\bar{\mu} = \max \left\{ M_\omega^{3-2r} \lambda^{1-r} s_\omega \log^2(N \vee T), M_\omega \bar{s} \lambda \log^{3/2}(N \vee T), M_\omega^2 \lambda \log(N \vee T) \right\}.$$

Because $\bar{\mu}$ tends to zero polynomially, we observe that conditional on A ,

$$\begin{aligned}
\mathbb{Q}^*(\mathfrak{t}) &= \frac{1}{|\hat{\mathcal{S}}_{\mathfrak{L}}^c|} \sum_{(i,j) \in \hat{\mathcal{S}}_{\mathfrak{L}}^c} \mathbb{P}^* \left(\frac{S_{ij}^* + R_{ij}^*}{\hat{m}_{ij}^*} > \mathfrak{t} \right) \\
&= \frac{1}{|\hat{\mathcal{S}}_{\mathfrak{L}}^c|} \sum_{(i,j) \in \hat{\mathcal{S}}_{\mathfrak{L}}^c} \mathbb{P}^* \left(\frac{S_{ij}^*}{\tilde{m}_{ij}^*} \left\{ 1 + \left(\frac{\tilde{m}_{ij}^*}{\hat{m}_{ij}^*} - 1 \right) \right\} > \mathfrak{t} - \frac{R_{ij}^*}{\hat{m}_{ij}^*}, A_1^* \cap A_2^* \right) \\
&\quad + O((N \vee T)^{-\nu}) \\
&= \frac{1}{|\hat{\mathcal{S}}_{\mathfrak{L}}^c|} \sum_{(i,j) \in \hat{\mathcal{S}}_{\mathfrak{L}}^c} \mathbb{P}^* \left(\frac{S_{ij}^*}{\tilde{m}_{ij}^*} > \frac{\mathfrak{t} - O(\bar{\mu}_2)}{1 + O(\bar{\mu}_1)} \right) + O((N \vee T)^{-\nu}) \tag{A.9}
\end{aligned}$$

with high probability, where the terms, $O(\bar{\mu}_1)$, $O(\bar{\mu}_2)$, and $O((N \vee T)^{-\nu})$, depend neither on $\mathfrak{t}$ nor (i, j) . Since $(i, j) \in \hat{\mathcal{S}}_{\mathfrak{L}}^c$ implies $(i, j) \in \mathcal{S}^c$ on event A , Lemma 18 entails the normal approximation of the self-normalized sum,

$$\begin{aligned}
&\mathbb{P}^* \left(\frac{S_{ij}^*}{\tilde{m}_{ij}^*} > \frac{\mathfrak{t} - O(\bar{\mu}_2)}{1 + O(\bar{\mu}_1)} \right) \\
&= Q \left(\frac{\mathfrak{t} - O(\bar{\mu}_2)}{1 + O(\bar{\mu}_1)} \right) \left\{ 1 + O \left(\frac{M_\omega \log NT}{T^{1/2}} \right) \left(1 + \frac{\mathfrak{t} - O(\bar{\mu}_2)}{1 + O(\bar{\mu}_1)} \right)^3 \right\}, \tag{A.10}
\end{aligned}$$

with high probability. Combining (A.9) and (A.10) and dividing the both sides by $Q(\mathfrak{t})$ yield, with high probability,

$$\begin{aligned}
\frac{\mathbb{Q}^*(\mathfrak{t})}{Q(\mathfrak{t})} &= \frac{Q \left(\frac{\mathfrak{t} - O(\bar{\mu}_2)}{1 + O(\bar{\mu}_1)} \right)}{Q(\mathfrak{t})} \left\{ 1 + O \left(\frac{M_\omega \log NT}{T^{1/2}} \right) \left(1 + \frac{\mathfrak{t} - O(\bar{\mu}_2)}{1 + O(\bar{\mu}_1)} \right)^3 \right\} \\
&\quad + \frac{O((N \vee T)^{-\nu})}{Q(\mathfrak{t})}.
\end{aligned}$$

By the assumed condition, $\bar{\mu}_1$, $\bar{\mu}_2$ and $M_\omega/T^{1/2}$ decay polynomially while $\mathfrak{t} \in [0, \bar{\mathfrak{t}}]$ at most diverges logarithmically. Thus we obtain uniformly in $\mathfrak{t} \in [0, \bar{\mathfrak{t}}]$,

$$1 + O \left(\frac{M_\omega \log NT}{T^{1/2}} \right) \left(1 + \frac{\mathfrak{t} - O(\bar{\mu}_2)}{1 + O(\bar{\mu}_1)} \right)^3 = 1 + o(1)$$

and, by an application of Lemma 7.2 of [Javanmard and Javadi \(2019\)](#),

$$\frac{Q\left(\frac{\mathfrak{t} - O(\bar{\mu}_2)}{1 + O(\bar{\mu}_1)}\right)}{Q(\mathfrak{t})} = 1 + O(\bar{\mu}_2)(1 + \mathfrak{t}) + O(\bar{\mu}_1)(1 + \mathfrak{t}^2) = 1 + o(1)$$

uniformly in $\mathfrak{t} \in [0, \bar{\mathfrak{t}}]$. By (A.4), we have

$$\sup_{\mathfrak{t} \in [0, \bar{\mathfrak{t}}]} \frac{O((N \vee T)^{-\nu})}{Q(\mathfrak{t})} = \frac{O((N \vee T)^{-\nu})}{Q(\bar{\mathfrak{t}})} = O((N \vee T)^{-\nu+2}).$$

Consequently, it holds that

$$\sup_{\mathfrak{t} \in [0, \bar{\mathfrak{t}}]} \left| \frac{Q^*(\mathfrak{t})}{Q(\mathfrak{t})} - 1 \right| = o(1)$$

with high probability for any $\nu > 2$. The same argument is applied to showing $|\{1 - Q^*(-\mathfrak{t})\}/Q(\mathfrak{t}) - 1| = o(1)$ uniformly in $\mathfrak{t} \in [0, \bar{\mathfrak{t}}]$. This completes the proof. $\square$

A.4 Proof of Theorem 4

Proof. We use the same notation as in the proof of Theorem 2. Let $\bar{\mathfrak{t}}_0 = \sqrt{2 \log p}$ denote the upper bound of the critical value, $\mathfrak{t}_0$. By the definition of power and monotonicity of probability, we have

$$\begin{aligned} \text{Power}(\mathfrak{t}_0) &= \mathbb{E} \left[\frac{|\{(i, j) \in \hat{\mathcal{S}}(\mathfrak{t}_0) : \text{sgn}(\hat{\phi}_{ij}) = \text{sgn}(\phi_{ij})\}|}{s \vee 1} \right] \\ &= \frac{1}{s} \sum_{(i, j) \in \mathcal{S}_{<0}} \mathbb{P}(\mathbf{T}_{ij} \leq -\mathfrak{t}_0) + \frac{1}{s} \sum_{(i, j) \in \mathcal{S}_{>0}} \mathbb{P}(\mathbf{T}_{ij} \geq \mathfrak{t}_0) \\ &\geq \frac{1}{s} \sum_{(i, j) \in \mathcal{S}_{<0}} \mathbb{P}(\mathbf{T}_{ij} \leq -\bar{\mathfrak{t}}_0) + \frac{1}{s} \sum_{(i, j) \in \mathcal{S}_{>0}} \mathbb{P}(\mathbf{T}_{ij} \geq \bar{\mathfrak{t}}_0). \end{aligned}$$

Consider the second probability in the lower bound. Proposition 2 gives

$$\mathbf{T}_{ij} = \sqrt{T} \hat{\phi}_{ij} / \hat{m}_{ij} = (\sqrt{T} \phi_{ij} + z_{ij} + r_{ij}) / \hat{m}_{ij},$$

where $\phi_{ij} = 0$ if and only if $(i, j) \in \mathcal{S}^c$. It holds that

$$\begin{aligned}
& \max_{(i,j) \in \mathcal{S}_{>0}} \mathbb{P}(\mathbf{T}_{ij} \leq \bar{\mathbf{t}}_0) \\
& \leq \max_{(i,j) \in \mathcal{S}_{>0}} \mathbb{P}\left(\frac{z_{ij} + r_{ij}}{\hat{m}_{ij}} + \frac{m_{ij}}{\hat{m}_{ij}} \frac{\sqrt{T}\phi_{ij}^0}{m_{ij}} \leq \bar{\mathbf{t}}_0, \frac{m_{ij}}{\hat{m}_{ij}} > \frac{1}{2}\right) + \max_{(i,j) \in \mathcal{S}_{>0}} \mathbb{P}\left(\frac{m_{ij}}{\hat{m}_{ij}} \leq \frac{1}{2}\right) \\
& \leq \max_{(i,j) \in \mathcal{S}_{>0}} \mathbb{P}\left(\mathcal{Z}_{ij} \leq \bar{\mathbf{t}}_0 - \min_{(i,j) \in \mathcal{S}_{>0}} \frac{\sqrt{T}\phi_{ij}^0}{2m_{ij}} + \delta_1\right) + O((N \vee T)^{-\nu}) \\
& \leq \Phi\left(-\sqrt{2\log p} + \delta_1\right) + O((N \vee T)^{-\nu}),
\end{aligned}$$

where the second inequality follows from Lemmas 7 and 8, and the third inequality holds by $\bar{\mathbf{t}}_0 = \sqrt{2\log p}$ and Condition 8. Since $\delta_1\sqrt{\log p} = o(1)$ by the assumption, the Gaussian probability in the upper bound tends to zero. The same result is obtained for the first probability. Thus the power goes to unity because $|\mathcal{S}_{<0}| + |\mathcal{S}_{>0}| = s$. This completes the proof. $\square$

A.5 Proof of Theorem 5

Proof of (a). For any $(i, j) \in \hat{\mathcal{S}}_R(h^*)$, we have $qE_{h(i,j)}/|\mathcal{H}| \geq 1/|\hat{\mathcal{S}}_R(h^*)|$ by the property of e-BH. Thus the FDR of $\hat{\mathcal{S}} := \hat{\mathcal{S}}_R(h^*)$ is bounded by

$$\begin{aligned}
\text{FDR} &= \mathbb{E}\left[\sum_{(i,j) \in \mathcal{S}^c} \frac{1\{(i,j) \in \hat{\mathcal{S}}\}}{|\hat{\mathcal{S}}|}\right] \leq \mathbb{E}\left[\sum_{(i,j) \in \mathcal{S}^c} \frac{qE_{h(i,j)}1\{(i,j) \in \hat{\mathcal{S}}\}}{|\mathcal{H}|}\right] \\
&\leq \frac{1}{|\mathcal{H}|} \sum_{(i,j) \in \mathcal{S}^c} q \mathbb{E}[E_{h(i,j)}] \leq q \max_{(i,j) \in \mathcal{S}^c} \mathbb{E}\left[\frac{f(\mathbf{T}_{ij})}{\mathbb{E}f(\mathcal{Z})}\right],
\end{aligned}$$

where $\mathcal{Z} \sim N(0, 1)$. Theorem 1 with Conditions 1–5 and $\bar{v} = o(1)$ achieves $f(\mathbf{T}_{ij}) \rightarrow_d f(\mathcal{Z})$ for all $(i, j) \in \mathcal{S}^c$ for any continuous function f by the continuous mapping theorem. Therefore, for every $(i, j) \in \mathcal{S}^c$, the assumed uniform integrability of $f(\mathbf{T}_{ij})$ implies $\mathbb{E}[f(\mathbf{T}_{ij})] \rightarrow \mathbb{E}[f(\mathcal{Z})]$. This makes the upper bound of FDR be $q + o(1)$, which completes the proof. $\square$

Proof of (b). For any $(i, j) \in \hat{\mathcal{S}}_R(h^*)$, we have $qE_{h(i,j)} \geq |\mathcal{H}|/|\hat{\mathcal{S}}_R(h^*)|$. This implies $qE_{(h^*)}/|\mathcal{H}| \geq 1/|\hat{\mathcal{S}}|$. Therefore, the power is lower bounded by

$$\begin{aligned} \text{Power} &= \mathbb{E} \left[\sum_{(i,j) \in \mathcal{S}} \frac{1\{(i,j) \in \hat{\mathcal{S}}\}}{s} \right] = \frac{1}{s} \sum_{(i,j) \in \mathcal{S}} \mathbb{P}(E_{h(i,j)} \geq E_{(h^*)}) \\ &\geq \frac{1}{s} \sum_{(i,j) \in \mathcal{S}} \mathbb{P}\left(E_{h(i,j)} \geq \frac{|\mathcal{H}|}{q}\right) \geq \min_{(i,j) \in \mathcal{S}} \mathbb{P}\left(f(\mathbf{T}_{ij}) \geq \frac{\mathbb{E} f(\mathcal{Z})|\mathcal{H}|}{q}\right) \\ &\geq \min_{(i,j) \in \mathcal{S}} \mathbb{P}(|\mathbf{T}_{ij}| \gtrsim f^{-1}(|\mathcal{H}|)), \end{aligned}$$

where f^{-1} exists by the monotonicity. From Theorem 1 and its proof, $|\mathbf{T}_{ij}|/\sqrt{T}$ is uniformly tight for every $(i, j) \in \mathcal{S}$ when $\bar{v} = o(1)$. Therefore, the lower bound of Power tends to unity by the condition of f . This completes the proof. $\square$

B Proofs of Propositions

B.1 Proof of Proposition 1

Proof. Derive the non-asymptotic error bound for the Lasso estimator. First define two events:

$$\mathcal{E}_1 = \left\{ \|T^{-1}\mathbf{X}\mathbf{U}'\|_{\max} \leq \lambda/2 \right\}, \quad \mathcal{E}_2 = \left\{ \left\| \hat{\Sigma}_x - \Sigma_x \right\|_{\max} \leq b\lambda/2 \right\},$$

where $\lambda = 8bc_{uu}\sqrt{2(\nu+7)^3 \log^3(N \vee T)/T}$ with any positive constant ν . We work on event $\mathcal{E}_1 \cap \mathcal{E}_2$ since Lemmas 3 and 4 guarantee that $\mathcal{E}_1 \cap \mathcal{E}_2$ occurs with probability at least $1 - O((N \vee T)^{-\nu})$.

Define $\boldsymbol{\delta}_{i\cdot} = \hat{\boldsymbol{\phi}}_{i\cdot}^{\mathbf{L}} - \boldsymbol{\phi}_{i\cdot}$. Because $\hat{\boldsymbol{\phi}}_{i\cdot}^{\mathbf{L}}$ minimizes the objective function, we have

$$(2T)^{-1} \|\mathbf{y}_{i\cdot} - \hat{\boldsymbol{\phi}}_{i\cdot}^{\mathbf{L}} \mathbf{X}\|_2^2 + \lambda \|\hat{\boldsymbol{\phi}}_{i\cdot}^{\mathbf{L}}\|_1 \leq (2T)^{-1} \|\mathbf{y}_{i\cdot} - \boldsymbol{\phi}_{i\cdot} \mathbf{X}\|_2^2 + \lambda \|\boldsymbol{\phi}_{i\cdot}\|_1,$$

which is equivalently written as

$$(2T)^{-1} \|\boldsymbol{\delta}_i \mathbf{X}\|_2^2 \leq T^{-1} \mathbf{u}_i \mathbf{X}' \boldsymbol{\delta}'_i + \lambda \|\boldsymbol{\phi}_i\|_1 - \lambda \|\hat{\boldsymbol{\phi}}_i^L\|_1.$$

By Hölder's inequality and the triangle inequality, we have

$$\begin{aligned} (2T)^{-1} \|\boldsymbol{\delta}_i \mathbf{X}\|_2^2 &\leq |T^{-1} \mathbf{u}_i \mathbf{X}' \boldsymbol{\delta}'_i| + \lambda \|\boldsymbol{\phi}_i\|_1 - \lambda \|\hat{\boldsymbol{\phi}}_i^L\|_1 \\ &\leq \|T^{-1} \mathbf{u}_i \mathbf{X}'\|_\infty \|\boldsymbol{\delta}_i\|_1 + \lambda \|\boldsymbol{\delta}_i\|_1. \end{aligned}$$

On event $\mathcal{E}_1$, we thus obtain the upper bound of $\|\boldsymbol{\delta}_i \mathbf{X}\|_2^2$ as

$$T^{-1} \|\boldsymbol{\delta}_i \mathbf{X}\|_2^2 \leq 3\lambda \|\boldsymbol{\delta}_i\|_1. \quad (\text{A.11})$$

Next we bound $\|\boldsymbol{\delta}_i \mathbf{X}\|_2^2$ from below. Lemma 5 states that $\boldsymbol{\delta}_i$ lies in $\mathcal{D} = \{\boldsymbol{\delta}_i \in \mathbb{R}^{1 \times KN} : \|\boldsymbol{\delta}_{i\mathcal{S}_i^c}\|_1 \leq 3\|\boldsymbol{\delta}_{i\mathcal{S}_i}\|_1\}$ on $\mathcal{E}_1$. Hence under $\mathcal{E}_1$ and $\mathcal{E}_2$, Lemma 6 entails

$$T^{-1} \|\boldsymbol{\delta}_i \mathbf{X}\|_2^2 / \|\boldsymbol{\delta}_i\|_2^2 \geq \gamma - 16s_i \|\hat{\boldsymbol{\Sigma}}_x - \boldsymbol{\Sigma}_x\|_{\max} \geq \gamma - 8bs_i\lambda. \quad (\text{A.12})$$

By (A.11), (A.12), Lemma 5, and the Cauchy–Schwarz inequality, we have

$$\begin{aligned} (\gamma - 8bs_i\lambda) \|\boldsymbol{\delta}_i\|_2^2 &\leq 3\lambda \|\boldsymbol{\delta}_i\|_1 = 3\lambda \|\boldsymbol{\delta}_{i\mathcal{S}_i}\|_1 + 3\lambda \|\boldsymbol{\delta}_{i\mathcal{S}_i^c}\|_1 \\ &\leq 12\lambda \|\boldsymbol{\delta}_{i\mathcal{S}_i}\|_1 \leq 12s_i^{1/2} \lambda \|\boldsymbol{\delta}_{i\mathcal{S}_i}\|_2 \leq 12s_i^{1/2} \lambda \|\boldsymbol{\delta}_i\|_2. \end{aligned}$$

This concludes that

$$\|\boldsymbol{\delta}_i\|_2 \leq \frac{12s_i^{1/2} \lambda}{\gamma - 8bs_i\lambda}.$$

Next we derive the prediction error bound. By Lemma 5 and the Cauchy–Schwarz

inequality again, the error bound in the element-wise ℓ_1 -norm is given by

$$\begin{aligned}\|\boldsymbol{\delta}_{i\cdot}\|_1 &= \|\boldsymbol{\delta}_{i\mathcal{S}_i}\|_1 + \|\boldsymbol{\delta}_{i\mathcal{S}_i^c}\|_1 \\ &\leq 4\|\boldsymbol{\delta}_{i\mathcal{S}_i}\|_1 \leq 4s_i^{1/2}\|\boldsymbol{\delta}_{i\mathcal{S}_i}\|_2 \leq 4s_i^{1/2}\|\boldsymbol{\delta}_{i\cdot}\|_2 \leq \frac{48s_i\lambda}{\gamma - 8bs_i\lambda}.\end{aligned}\tag{A.13}$$

From (A.11) and (A.13), the prediction error bound is obtained as

$$T^{-1}\|\boldsymbol{\delta}_{i\cdot}\mathbf{X}\|_2^2 \leq 3\lambda\|\boldsymbol{\delta}_{i\cdot}\|_1 \leq \frac{144s_i\lambda^2}{\gamma - 8bs_i\lambda}.$$

This completes the proof. $\square$

B.2 Proof of Proposition 2

Proof. The first assertion is trivial by the construction of the debiased lasso estimator.

We derive the upper bound of $|r_{ij}|$. Observe that for each $i \in [N]$,

$$\begin{aligned}|r_{ij}| &\leq \|T^{-1/2}\mathbf{u}_i\mathbf{X}'\|_\infty \|\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j\|_1 + \sqrt{T} \left\| \hat{\boldsymbol{\phi}}_{i\cdot}^L - \boldsymbol{\phi}_{i\cdot} \right\|_1 \left\| \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j - \mathbf{e}_j \right\|_\infty \\ &\leq \|T^{-1/2}\mathbf{U}\mathbf{X}'\|_{\max} \max_j \|\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j\|_1 + \sqrt{T} \left\| \hat{\boldsymbol{\phi}}_{i\cdot}^L - \boldsymbol{\phi}_{i\cdot} \right\|_1 \left\| \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\Omega}} - \mathbf{I}_{KN} \right\|_{\max}\end{aligned}$$

uniformly in $j \in [KN]$. Let $\lambda_1 = b\|\boldsymbol{\Omega}\|_{\ell_1}\lambda/2$, and consider the event

$$\{\|\mathbf{X}\mathbf{U}'/T\|_{\max} \leq \lambda/2\} \cap \{\|\hat{\boldsymbol{\Sigma}}_x - \boldsymbol{\Sigma}_x\|_{\max} \leq \lambda_1/\|\boldsymbol{\Omega}\|_{\ell_1}\}.$$

Then this occurs with probability at least $1 - O((N \vee T)^{-\nu})$ by Lemmas 3 and 4 with the proof of Proposition 1. On this event, the proof of Theorem 6 in Cai et al. (2011) establishes the bounds, $\max_j \|\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j\|_1 \lesssim (\|\boldsymbol{\Omega}\|_{\ell_1}\lambda_1)^{1-r} s_\omega$ and $\|\hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\Omega}} - \mathbf{I}_{KN}\|_{\max} \leq \lambda_1$, under Condition 4. Therefore, together with the Lasso bound derived in Proposition

1 and $\lambda_1 \lesssim M_\omega \lambda$, we obtain

$$\begin{aligned}
\max_{j \in [KN]} |r_{ij}| &\lesssim \sqrt{T} \lambda (\|\mathbf{\Omega}\|_{\ell_1} \lambda_1)^{1-r} s_\omega / 2 + \frac{48 s_i \sqrt{T} \lambda \lambda_1}{\gamma - 8 b s_i \lambda} \\
&\lesssim \sqrt{T} \lambda M_\omega^{1-r} (M_\omega \lambda)^{1-r} s_\omega + s_i \sqrt{T} \lambda^2 M_\omega \\
&\lesssim (s_\omega M_\omega^{2-2r} \lambda^{1-r} + s_i M_\omega \lambda) \sqrt{\log^3(N \vee T)} (=:\bar{r}_i)
\end{aligned}$$

for each $i \in [N]$ such that $s_i \lambda = o(1)$. This completes the proof. $\square$

C Lemmas and their Proofs

Lemma 1. *If Condition 1 is true, then for any $i, j \in [N]$ and $s, t \in [T]$, there exist constants $c_{uu}, c_m > 0$ such that*

$$\begin{aligned}
(a) \quad &\mathbb{P} \left(\left| \frac{1}{T} \sum_{t=1}^T (u_{it} u_{jt} - \mathbb{E}[u_{it} u_{jt}]) \right| > x \right) \leq 2 \exp \left\{ -\frac{1}{2} \left(\frac{T x^2}{c_{uu}^2} \wedge \frac{T x}{c_{uu}} \right) \right\}, \\
(b) \quad &\mathbb{E} \left[\max_{i \in [N]} \max_{t \in [T]} |u_{it}|^m \right] \leq c_m \log^{m/2}(N \vee T),
\end{aligned}$$

where $m \in \mathbb{N}$ is an arbitrary fixed constant.

Proof of Lemma 1. Prove (a). Since u_{it} is sub-Gaussian by Condition 1, Lemma 2.7.7 of [Vershynin \(2018\)](#) entails that for any $i, j \in [N]$, $\{u_{it} u_{jt} - \mathbb{E}[u_{it} u_{jt}]\}_t$ is a sequence of i.i.d. (centered) sub-exponential random variables. Thus the result directly follows by Bernstein's inequality ([Vershynin, 2018](#), Theorem 2.8.1).

Prove (b). For any fixed $m \in \mathbb{N}$, the sub-Gaussian tail property for u_{it} in Condition 1 implies for all $x > 0$,

$$\mathbb{P}(|u_{it}|^m > x) = \mathbb{P}(|u_{it}| > x^{1/m}) \leq 2 \exp(-x^{2/m}/c_u).$$

Using this tail probability with the union bound, we have

$$\begin{aligned}
\mathbb{E} \left[\max_{i,t \in \mathbb{N}} \frac{|u_{it}|^m}{(1 + \log it)^{m/2}} \right] &\leq \int_0^\infty \mathbb{P} \left(\max_{i,t \in \mathbb{N}} \frac{|u_{it}|^m}{(1 + \log it)^{m/2}} > x \right) dx \\
&\leq \int_0^{(2c_u)^{m/2}} dx + 2 \sum_{i=1}^\infty \sum_{t=1}^\infty \int_{(2c_u)^{m/2}}^\infty \exp \left(-\frac{x^{2/m}(1 + \log i + \log t)}{c_u} \right) dx \\
&\leq (2c_u)^{m/2} + 2 \sum_{i=1}^\infty i^{-2} \sum_{t=1}^\infty t^{-2} \int_{(2c_u)^{m/2}}^\infty \exp \left(-\frac{x^{2/m}}{c_u} \right) dx,
\end{aligned}$$

where the upper bound is further bounded by some positive constants. Thus there exists some constant $M > 0$ such that

$$\begin{aligned}
M &\geq \mathbb{E} \left[\max_{i,t \in \mathbb{N}} \frac{|u_{it}|^m}{(1 + \log it)^{m/2}} \right] \\
&\geq \mathbb{E} \left[\max_{i \in [N], t \in [T]} \frac{|u_{it}|^m}{(1 + \log it)^{m/2}} \right] \geq \mathbb{E} \left[\max_{i \in [N], t \in [T]} |u_{it}|^m \right] \frac{1}{(1 + \log NT)^{m/2}}.
\end{aligned}$$

Replacing the constant factor appropriately gives the result. This completes the proof. $\square$

Lemma 2. *If Condition 2 is true, then there exists a constant $\delta \in (0, 1)$ such that for any monotonically divergent sequence $r_T > 0$ with sufficiently large T ,*

$$\sum_{\ell=r_T}^\infty \|\mathbf{B}_\ell\|_\infty \leq \frac{\delta^{r_T}}{1 - \delta}. \tag{A.14}$$

In particular, the summability condition in (11) holds.

Proof of Lemma 2. The VAR(K) model is written as the VAR(1) companion form

with coefficient

$$\mathbf{A} = \begin{pmatrix} \Phi_1 & \Phi_2 & \dots & \Phi_{K-1} & \Phi_K \\ \mathbf{I}_N & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_N & \dots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{I}_N & \mathbf{0} \end{pmatrix}.$$

Condition 2 entails that the spectral radius of $\mathbf{A}$, defined as $\rho(\mathbf{A}) = \max_{j \in [KN]} |\lambda_j(\mathbf{A})|$, is strictly less than one uniformly in N . This implies that the VAR(1) model is invertible to the VMA(∞). Therefore, we have the representation $\mathbf{y}_t = \sum_{\ell=0}^{\infty} \mathbf{B}_\ell \mathbf{u}_{t-\ell}$, where $\mathbf{B}_0 = \mathbf{I}_N$ and $\mathbf{B}_\ell = \mathbf{J}' \mathbf{A}^\ell \mathbf{J}$ with $\mathbf{J}' = (\mathbf{I}_N, \mathbf{0}_{N \times (KN-N)})$ for $\ell = 1, 2, \dots$. Note that

$$\sum_{\ell=0}^{\infty} \|\mathbf{B}_\ell\|_\infty \leq \sum_{\ell=0}^{\infty} \|\mathbf{J}\|_\infty^2 \|\mathbf{A}^\ell\|_\infty = \sum_{\ell=0}^{\infty} \|\mathbf{A}^\ell\|_\infty.$$

Again, since we have $\rho(\mathbf{A}) < 1$ uniformly in N by Condition 2, we can always pick up a small constant $\varepsilon > 0$ such that $\delta := \rho(\mathbf{A}) + \varepsilon$ lies in $(0, 1)$. We apply the Gelfand formula (Horn and Johnson, 2012); for this choice of ε , there exists T^* such that $\|\mathbf{A}^{r_T}\|_\infty \leq \delta^{r_T}$ for all $T \geq T^*$, where $r_T > 0$ is any monotonically diverging sequence as $T \rightarrow \infty$. Therefore, for any $T \geq T^*$ we have

$$\sum_{\ell=r_T}^{\infty} \|\mathbf{A}^\ell\|_\infty \leq \sum_{\ell=r_T}^{\infty} \delta^\ell = \frac{\delta^{r_T}}{1 - \delta},$$

which proves (A.14). Finally, because $\limsup_{\ell \rightarrow \infty} \|\mathbf{A}^\ell\|_\infty^{1/\ell} \leq \delta < 1$, the summability condition in (11) holds by the root test. This completes the proof. $\square$

Lemma 3. *Let $b = \sum_{\ell=0}^{\infty} \|\mathbf{B}_\ell\|_\infty$. If Conditions 1 and 2 are true, then for any $\bar{c}_1 > 0$*

and arbitrary divergent sequence $r_T > 0$ with sufficiently large T , we have for all $x > 0$,

$$\begin{aligned} & \mathbb{P}(\|T^{-1}\mathbf{X}\mathbf{U}'\|_{\max} > x) \\ & \leq 2KN^2 \exp\left(-\frac{x^2 T}{2\bar{c}_1^2}\right) + 2r_T K N^2 T \exp\left(-\frac{\bar{c}_1}{4bc_{uu}}\right) + 2c_1^2 K T \frac{\delta^{r_T} \log N}{\bar{c}_1(1-\delta)}, \end{aligned}$$

where constants $\delta \in (0, 1)$ and $c_{uu}, c_1 > 0$ are given by Lemmas 1 and 2, respectively.

In particular, if we set $x = \sqrt{2\bar{c}_1^2(\nu+3)\log(N \vee T)/T}$, $\bar{c}_1 = 4bc_{uu}(\nu+5)\log(N \vee T)$, and $r_T = T$ with any constant $\nu > 0$, then the upper bound becomes $O((N \vee T)^{-\nu})$.

Proof of Lemma 3. Following the definition, we have for any $x, \bar{c}_1 > 0$,

$$\begin{aligned} & \mathbb{P}(\|T^{-1}\mathbf{X}\mathbf{U}'\|_{\max} > x) = \mathbb{P}\left(\max_{k \in [K]} \left\|T^{-1} \sum_{t=1}^T \mathbf{y}_{t-k} \mathbf{u}'_t\right\|_{\max} > x\right) \\ & \leq \mathbb{P}\left(\max_{k \in [K]} \left\|T^{-1} \sum_{t=1}^T \mathbf{y}_{t-k} \mathbf{u}'_t\right\|_{\max} > x \mid \max_{k \in [K]} \max_{t \in [T]} \|\mathbf{y}_{t-k} \mathbf{u}'_t\|_{\max} \leq \bar{c}_1\right) \\ & \quad + \mathbb{P}\left(\max_{k \in [K]} \max_{t \in [T]} \|\mathbf{y}_{t-k} \mathbf{u}'_t\|_{\max} > \bar{c}_1\right). \end{aligned}$$

Consider the first term. Note for any $k \in [K]$ that $\{\mathbf{y}_{t-k} \mathbf{u}'_t\}$ forms a martingale difference sequence with respect to the filtration, $\mathcal{F}_t = \sigma\{\mathbf{u}_{t-s} : s = 0, 1, \dots\}$. Thus the union bound and the Azuma-Hoeffding inequality (Bercu et al., 2015) give

$$\begin{aligned} & \mathbb{P}\left(\max_{k \in [K]} \left\|T^{-1} \sum_{t=1}^T \mathbf{y}_{t-k} \mathbf{u}'_t\right\|_{\max} > x \mid \max_{k \in [K]} \max_{t \in [T]} \|\mathbf{y}_{t-k} \mathbf{u}'_t\|_{\max} \leq \bar{c}_1\right) \\ & \leq KN^2 \max_{k \in [K]} \max_{i,j \in [N]} \mathbb{P}\left(\left|T^{-1} \sum_{t=1}^T y_{i,t-k} u_{jt}\right| > x \mid \max_{k \in [K]} \max_{t \in [T]} \|\mathbf{y}_{t-k} \mathbf{u}'_t\|_{\max} \leq \bar{c}_1\right) \\ & \leq 2KN^2 \exp\left(-\frac{x^2 T}{2\bar{c}_1^2}\right). \end{aligned}$$

For the second term, applying the triangle and Hölder's inequalities yields

$$\|\mathbf{y}_{t-k} \mathbf{u}'_t\|_{\max} = \left\| \sum_{\ell=0}^{\infty} \mathbf{B}_{\ell} \mathbf{u}_{t-k-\ell} \mathbf{u}'_t \right\|_{\max} \leq \left(\sum_{\ell=0}^{r-1} + \sum_{\ell=r}^{\infty} \right) \|\mathbf{B}_{\ell}\|_{\infty} \|\mathbf{u}_{t-k-\ell} \mathbf{u}'_t\|_{\max},$$

and hence,

$$\begin{aligned}
& \mathbb{P} \left(\max_{k \in [K]} \max_{t \in [T]} \|\mathbf{y}_{t-k} \mathbf{u}'_t\|_{\max} > \bar{c}_1 \right) \\
& \leq \mathbb{P} \left(\max_{k \in [K]} \max_{t \in [T]} \sum_{\ell=0}^{r-1} \|\mathbf{B}_\ell\|_{\infty} \|\mathbf{u}_{t-k-\ell} \mathbf{u}'_t\|_{\max} > \bar{c}_1/2 \right) \\
& \quad + \mathbb{P} \left(\max_{k \in [K]} \max_{t \in [T]} \sum_{\ell=r}^{\infty} \|\mathbf{B}_\ell\|_{\infty} \|\mathbf{u}_{t-k-\ell} \mathbf{u}'_t\|_{\max} > \bar{c}_1/2 \right).
\end{aligned}$$

Consider the probability with $\sum_{\ell=0}^{r-1}$. The union bound and the summability condition in Lemma 2 yield

$$\begin{aligned}
& \mathbb{P} \left(\max_{k \in [K]} \max_{t \in [T]} \sum_{\ell=0}^{r-1} \|\mathbf{B}_\ell\|_{\infty} \|\mathbf{u}_{t-k-\ell} \mathbf{u}'_t\|_{\max} > \bar{c}_1/2 \right) \\
& \leq rKT \max_{k \in [K]} \max_{t \in [T]} \max_{\ell=0, \dots, r-1} \mathbb{P} \left(\|\mathbf{u}_{t-k-\ell} \mathbf{u}'_t\|_{\max} > \frac{\bar{c}_1}{2 \sum_{\ell=0}^{r-1} \|\mathbf{B}_\ell\|_{\infty}} \right) \\
& \leq rKN^2T \max_{k \in [K+r-1]} \max_{t \in [T]} \max_{i,j \in [N]} \mathbb{P} \left(|u_{i,t-k} u_{jt}| > \frac{\bar{c}_1}{2b} \right) \\
& \leq 2rKN^2T \exp \left(-\frac{\bar{c}_1}{4bc_{uu}} \right),
\end{aligned}$$

where the last inequality holds by Lemma 1(a) since $u_{i,t-k} u_{jt}$ is sub-exponential (Ver-shynin, 2018, Proposition 2.7.1 and Lemma 2.7.7). Consider the probability with $\sum_{\ell=r}^{\infty}$. By the union bound and the Markov inequality, we have

$$\begin{aligned}
& \mathbb{P} \left(\max_{k \in [K]} \max_{t \in [T]} \sum_{\ell=r}^{\infty} \|\mathbf{B}_\ell\|_{\infty} \|\mathbf{u}_{t-k-\ell} \mathbf{u}'_t\|_{\max} > \bar{c}_1/2 \right) \\
& \leq 2KT \sum_{\ell=r}^{\infty} \|\mathbf{B}_\ell\|_{\infty} \mathbb{E} [\|\mathbf{u}_{t-k-\ell} \mathbf{u}'_t\|_{\max}] / \bar{c}_1 \\
& \leq 2KT \frac{\delta^r}{\bar{c}_1(1-\delta)} \mathbb{E} [\|\mathbf{u}_t\|_{\max}]^2 \leq 2c_1^2KT \frac{\delta^r \log N}{\bar{c}_1(1-\delta)},
\end{aligned}$$

where the last inequality holds by Lemma 1(b). Combining these upper bounds completes the proof. $\square$

Lemma 4. Let $b = \sum_{\ell=0}^{\infty} \|\mathbf{B}_\ell\|_\infty$. If Conditions 1 and 2 are true, then for any $\bar{c}_2 \geq c_{uu}$ and arbitrary divergent sequence $r_T > 0$ with sufficiently large T , we have for all $x > 0$,

$$\begin{aligned} & \mathbb{P} \left(\|T^{-1} \mathbf{X} \mathbf{X}' - \mathbb{E}[T^{-1} \mathbf{X} \mathbf{X}']\|_{\max} > x \right) \\ & \leq 2K^2 r_T^2 N^2 \exp \left(-\frac{x^2 T}{8b^4 \bar{c}_2^2} \right) + 2K^2 r_T^2 N^2 T \exp \left(-\frac{\bar{c}_2}{2c_{uu}} \right) + 6K^2 b c_2 \frac{\delta^{r_T} \log N}{x(1-\delta)}, \end{aligned}$$

where constants $\delta \in (0, 1)$ and $c_{uu}, c_2 > 0$ are given by Lemmas 1 and 2, respectively. In particular, if we set $x = \sqrt{8(\nu+6)b^4 \bar{c}_2^2 \log(N \vee T)/T}$, $\bar{c}_2 = 2(\nu+7)c_{uu} \log(N \vee T)$, and $r_T = T$ with any constant $\nu > 0$, the upper bound becomes $O((N \vee T)^{-\nu})$.

Proof of Lemma 4. For $g, h \in [K]$, define

$$\mathbf{W}_{g,h} = T^{-1} \sum_{t=1}^T (\mathbf{u}_{t-g} \mathbf{u}'_{t-h} - \mathbb{E}[\mathbf{u}_{t-g} \mathbf{u}'_{t-h}]).$$

By the VMA(∞) representation in Lemma 2, we have

$$\mathbf{y}_{t-g} \mathbf{y}'_{t-h} = \sum_{\ell=0}^{\infty} \sum_{m=0}^{\infty} \mathbf{B}_\ell \mathbf{u}_{t-g-\ell} \mathbf{u}'_{t-h-m} \mathbf{B}'_m,$$

so that

$$\begin{aligned} & \|T^{-1} \mathbf{X} \mathbf{X}' - \mathbb{E}[T^{-1} \mathbf{X} \mathbf{X}']\|_{\max} \leq \max_{g,h \in [K]} \left\| T^{-1} \sum_{t=1}^T (\mathbf{y}_{t-g} \mathbf{y}'_{t-h} - \mathbb{E}[\mathbf{y}_{t-g} \mathbf{y}'_{t-h}]) \right\|_{\max} \\ & \leq \max_{g,h \in [K]} \sum_{\ell=0}^{\infty} \sum_{m=0}^{\infty} \|\mathbf{B}_\ell \mathbf{W}_{g+\ell, h+m} \mathbf{B}'_m\|_{\max} \\ & \leq \max_{g,h \in [K]} \left(\sum_{\ell=0}^{r-1} + \sum_{\ell=r}^{\infty} \right) \left(\sum_{m=0}^{r-1} + \sum_{m=r}^{\infty} \right) \|\mathbf{B}_\ell\|_\infty \|\mathbf{W}_{g+\ell, h+m} \mathbf{B}'_m\|_{\max} \\ & \leq \max_{g,h \in [K]} \left(\sum_{\ell=0}^{r-1} \sum_{m=0}^{r-1} + 3 \sum_{\ell=0}^{\infty} \sum_{m=r}^{\infty} \right) \|\mathbf{B}_\ell\|_\infty \|\mathbf{B}_m\|_\infty \|\mathbf{W}_{g+\ell, h+m}\|_{\max}. \end{aligned}$$

Therefore, we obtain

$$\begin{aligned}
& \mathbb{P} \left(\|T^{-1} \mathbf{X} \mathbf{X}' - \mathbb{E}[T^{-1} \mathbf{X} \mathbf{X}']\|_{\max} > x \right) \\
& \leq \mathbb{P} \left(\max_{g,h \in [K]} \sum_{\ell=0}^{r-1} \sum_{m=0}^{r-1} \|\mathbf{B}_\ell\|_\infty \|\mathbf{B}_m\|_\infty \|\mathbf{W}_{g+\ell,h+m}\|_{\max} > x/2 \right) \\
& \quad + \mathbb{P} \left(\max_{g,h \in [K]} 3 \sum_{\ell=0}^{\infty} \sum_{m=r}^{\infty} \|\mathbf{B}_\ell\|_\infty \|\mathbf{B}_m\|_\infty \|\mathbf{W}_{g+\ell,h+m}\|_{\max} > x/2 \right).
\end{aligned}$$

We consider the first probability. The union bound gives

$$\begin{aligned}
& \mathbb{P} \left(\max_{g,h \in [K]} \sum_{\ell=0}^{r-1} \sum_{m=0}^{r-1} \|\mathbf{B}_\ell\|_\infty \|\mathbf{B}_m\|_\infty \|\mathbf{W}_{g+\ell,h+m}\|_{\max} > x/2 \right) \\
& \leq K^2 \max_{g,h \in [K]} \mathbb{P} \left(\max_{\ell,m=0,\dots,r-1} \|\mathbf{W}_{g+\ell,h+m}\|_{\max} \sum_{\ell=0}^{r-1} \sum_{m=0}^{r-1} \|\mathbf{B}_\ell\|_\infty \|\mathbf{B}_m\|_\infty > x/2 \right) \\
& \leq K^2 r^2 \max_{g,h \in [K]} \max_{\ell,m=0,\dots,r-1} \mathbb{P} \left(\|\mathbf{W}_{g+\ell,h+m}\|_{\max} > x/(2b^2) \right) \\
& = K^2 r^2 \max_{k \in [K+r-1]} \mathbb{P} \left(\|\mathbf{W}_{k,1}\|_{\max} > x/(2b^2) \right).
\end{aligned}$$

If $k = 1$, each component of $\mathbf{W}_{1,1}$ is a sample average of the i.i.d. random variables, $\{u_{it}u_{jt} - \mathbb{E}[u_{it}u_{jt}]\}_t$. Clearly this is a martingale difference sequence with respect to filtration $\mathcal{F}_t = \sigma\{u_{i,t-s}, u_{j,t-s} : s = 0, 1, \dots\}$. If $k \geq 2$, we have $\mathbb{E}[u_{i,t-k}u_{j,t-1}] = 0$ and the sequence, $\{u_{i,t-k+1}u_{jt}\}_t$, is also martingale difference with respect to $\mathcal{F}_t$. Therefore, the Azuma-Hoeffding inequality ([Bercu et al., 2015](#)) with the conditioning argument and the union bound as in the proof of Lemma 3, we have

$$\begin{aligned}
& \max_{k \in [K+r-1]} \mathbb{P} \left(\|\mathbf{W}_{k,1}\|_{\max} > x/(2b^2) \right) \\
& \leq \max_{k \in [K+r-1]} \mathbb{P} \left(\|\mathbf{W}_{k,1}\|_{\max} > x/(2b^2) \mid \max_{t \in [T]} \|\mathbf{u}_{t-k} \mathbf{u}'_{t-1} - \mathbb{E}[\mathbf{u}_{t-k} \mathbf{u}'_{t-1}]\|_{\max} \leq \bar{c}_2 \right) \\
& \quad + \max_{k \in [K+r-1]} \mathbb{P} \left(\max_{t \in [T]} \|\mathbf{u}_{t-k} \mathbf{u}'_{t-1} - \mathbb{E}[\mathbf{u}_{t-k} \mathbf{u}'_{t-1}]\|_{\max} > \bar{c}_2 \right) \\
& \leq 2N^2 \exp \left(-\frac{x^2 T}{8b^4 \bar{c}_2^2} \right) + 2N^2 T \exp \left(-\frac{\bar{c}_2}{2c_{uu}} \right).
\end{aligned}$$

We next consider the second probability. The union bound and the Markov inequality with Lemma 2 give

$$\begin{aligned}
& \mathbb{P} \left(3 \max_{g,h \in [K]} \sum_{\ell=0}^{\infty} \sum_{m=r}^{\infty} \|\mathbf{B}_{\ell}\|_{\infty} \|\mathbf{B}_m\|_{\infty} \|\mathbf{W}_{g+\ell,h+m}\|_{\max} > x/2 \right) \\
& \leq 6K^2 \max_{g,h \in [K]} \sum_{\ell=0}^{\infty} \sum_{m=r}^{\infty} \|\mathbf{B}_{\ell}\|_{\infty} \|\mathbf{B}_m\|_{\infty} \mathbb{E} \|\mathbf{W}_{g+\ell,h+m}\|_{\max} / x \\
& \leq 6K^2 \frac{b\delta^r}{x(1-\delta)} \max_{k \in \{1,2\}} \mathbb{E} [\|\mathbf{u}_{t-k}\mathbf{u}'_{t-1} - \mathbb{E}[\mathbf{u}_{t-k}\mathbf{u}'_{t-1}]\|_{\max}].
\end{aligned}$$

By Lemma 1, the last expectation is evaluated as

$$\max_{k \in \{1,2\}} \mathbb{E} [\|\mathbf{u}_{t-k}\mathbf{u}'_{t-1} - \mathbb{E} \mathbf{u}_{t-k}\mathbf{u}'_{t-1}\|_{\max}] \leq c_2 \log N.$$

Combining the obtained results so far yields the desired inequality. This completes the proof. $\square$

Lemma 5. Let $\delta'_i \in \mathbb{R}^{KN}$ denote the i -th column vector of $\Delta' \in \mathbb{R}^{KN \times N}$ with $\Delta = \hat{\Phi} - \Phi^0$. If inequality (A.11) is true, then it holds that

$$\|\delta_{i\mathcal{S}_i^c}\|_1 \leq 3\|\delta_{i\mathcal{S}_i}\|_1.$$

Proof of Lemma 5. Note that $\phi = \phi_{\mathcal{S}}$ and $\mathbf{v} = \mathbf{v}_{\mathcal{S}} + \mathbf{v}_{\mathcal{S}^c}$ for any KN -dimensional vector $\mathbf{v}$. Hence the lower bound of $\|\hat{\phi}_i\|_1$ is computed as

$$\begin{aligned}
\|\hat{\phi}_i\|_1 &= \|\phi_{i\cdot} + \delta_{i\cdot}\|_1 = \|\phi_{i\mathcal{S}_i} + \delta_{i\mathcal{S}_i} + \delta_{i\mathcal{S}_i^c}\|_1 \geq \|\phi_{i\mathcal{S}_i} + \delta_{i\mathcal{S}_i^c}\|_1 - \|\delta_{i\mathcal{S}_i}\|_1 \\
&= \|\phi_{i\mathcal{S}_i}\|_1 + \|\delta_{i\mathcal{S}_i^c}\|_1 - \|\delta_{i\mathcal{S}_i}\|_1 = \|\phi_{i\cdot}\|_1 + \|\delta_{i\mathcal{S}_i^c}\|_1 - \|\delta_{i\mathcal{S}_i}\|_1,
\end{aligned}$$

where the last equality holds by decomposability of the ℓ_1 -norm. Thus we obtain

$$\|\phi_{i\cdot}\|_1 - \|\hat{\phi}_i\|_1 \leq \|\delta_{i\mathcal{S}_i}\|_1 - \|\delta_{i\mathcal{S}_i^c}\|_1.$$

From (A.11), we have

$$\begin{aligned}
0 &\leq (2T)^{-1} \|\boldsymbol{\delta}_i \mathbf{X}\|_2^2 \leq 2^{-1} \lambda \|\boldsymbol{\delta}_i\|_1 + \lambda \|\boldsymbol{\phi}_i\|_1 - \lambda \|\hat{\boldsymbol{\phi}}_i\|_1 \\
&\leq 2^{-1} \lambda \|\boldsymbol{\delta}_{i\mathcal{S}_i}\|_1 + 2^{-1} \lambda \|\boldsymbol{\delta}_{i\mathcal{S}_i^c}\|_1 + \lambda \|\boldsymbol{\delta}_{i\mathcal{S}_i}\|_1 - \lambda \|\boldsymbol{\delta}_{i\mathcal{S}_i^c}\|_1 \\
&= (3/2) \lambda \|\boldsymbol{\delta}_{i\mathcal{S}_i}\|_1 - 2^{-1} \lambda \|\boldsymbol{\delta}_{i\mathcal{S}_i^c}\|_1,
\end{aligned}$$

giving the desired inequality, $\|\boldsymbol{\delta}_{i\mathcal{S}_i^c}\|_1 \leq 3\|\boldsymbol{\delta}_{i\mathcal{S}_i}\|_1$. This completes the proof. $\square$

Lemma 6. *Suppose the same conditions as Proposition 1. We have*

$$\min_{\boldsymbol{\delta}_i \in \mathcal{D}_i} \frac{T^{-1} \|\boldsymbol{\delta}_i \mathbf{X}\|_F^2}{\|\boldsymbol{\delta}_i\|_2^2} \geq \gamma - 16s_i \left\| T^{-1} \mathbf{X} \mathbf{X}' - \boldsymbol{\Sigma}_x \right\|_{\max},$$

where $\mathcal{D}_i = \{\boldsymbol{\delta} \in \mathbb{R}^{KN} : \|\boldsymbol{\delta}_{\mathcal{S}_i^c}\|_1 \leq 3\|\boldsymbol{\delta}_{\mathcal{S}_i}\|_1\}$, and $\gamma > 0$ is some constant.

Proof of Lemma 6. Let $\boldsymbol{\delta}'_i \in \mathbb{R}^{KN}$ denote the i th column vector of $\boldsymbol{\Delta}' \in \mathbb{R}^{KN \times N}$. We see that

$$\frac{T^{-1} \|\boldsymbol{\delta}_i \mathbf{X}\|_2^2}{\|\boldsymbol{\delta}_i\|_2^2} = \frac{\boldsymbol{\delta}_i \cdot \mathbb{E}[T^{-1} \mathbf{X} \mathbf{X}'] \boldsymbol{\delta}'_i}{\boldsymbol{\delta}_i \cdot \boldsymbol{\delta}'_i} - \frac{\boldsymbol{\delta}_i \cdot (\mathbb{E}[T^{-1} \mathbf{X} \mathbf{X}'] - T^{-1} \mathbf{X} \mathbf{X}') \boldsymbol{\delta}'_i}{\boldsymbol{\delta}_i \cdot \boldsymbol{\delta}'_i}. \quad (\text{A.15})$$

Then it is sufficient to show that uniformly in $\boldsymbol{\delta}_i \in \mathcal{D}_i$ the first term is bounded from below by some constant and the second term converges to zero with high probability.

We consider the first term of (A.15). In view of the Rayleigh quotient with Condition 3, we obtain

$$\min_{\boldsymbol{\delta}_i \in \mathcal{D}_i} \frac{\boldsymbol{\delta}_i \cdot \mathbb{E}[T^{-1} \mathbf{X} \mathbf{X}'] \boldsymbol{\delta}'_i}{\boldsymbol{\delta}_i \cdot \boldsymbol{\delta}'_i} \geq \gamma. \quad (\text{A.16})$$

We next consider the second term of (A.15). By applying Hölder's inequality twice, the numerator is bounded as

$$\begin{aligned}
|\boldsymbol{\delta}'_i \cdot (\mathbb{E}[T^{-1} \mathbf{X} \mathbf{X}'] - T^{-1} \mathbf{X} \mathbf{X}') \boldsymbol{\delta}'_i| &\leq \|\boldsymbol{\delta}_i \cdot (\mathbb{E}[T^{-1} \mathbf{X} \mathbf{X}'] - T^{-1} \mathbf{X} \mathbf{X}')\|_{\max} \|\boldsymbol{\delta}_i\|_1 \\
&\leq \|\mathbb{E}[T^{-1} \mathbf{X} \mathbf{X}'] - T^{-1} \mathbf{X} \mathbf{X}'\|_{\max} \|\boldsymbol{\delta}_i\|_1^2.
\end{aligned} \quad (\text{A.17})$$

We further compute the upper bound of $\|\boldsymbol{\delta}_{i\cdot}\|_1^2$. Lemma 5 yields

$$\|\boldsymbol{\delta}_{i\cdot}\|_1^2 = (\|\boldsymbol{\delta}_{\mathcal{S}_i}\|_1 + \|\boldsymbol{\delta}_{\mathcal{S}_i^c}\|_1)^2 \leq 16\|\boldsymbol{\delta}_{\mathcal{S}_i}\|_1^2 \leq 16s_i\|\boldsymbol{\delta}_{\mathcal{S}_i}\|_2^2 \leq 16s_i\|\boldsymbol{\delta}_{i\cdot}\|_2^2. \quad (\text{A.18})$$

Combining (A.17) and (A.18) and rearranging the terms yield

$$\max_{\boldsymbol{\delta}_{i\cdot} \in \mathcal{D}} \frac{\boldsymbol{\delta}_{i\cdot} (\mathbb{E}[T^{-1}\mathbf{X}\mathbf{X}'] - T^{-1}\mathbf{X}\mathbf{X}') \boldsymbol{\delta}_{i\cdot}'}{\|\boldsymbol{\delta}_{i\cdot}\|_2^2} \leq 16s_i \|\mathbb{E}[T^{-1}\mathbf{X}\mathbf{X}'] - T^{-1}\mathbf{X}\mathbf{X}'\|_{\max}. \quad (\text{A.19})$$

From (A.15), (A.16), and (A.19), we obtain

$$\min_{\boldsymbol{\delta}_{i\cdot} \in \mathcal{D}} \frac{T^{-1}\|\boldsymbol{\delta}_{i\cdot}\mathbf{X}\|_2^2}{\|\boldsymbol{\delta}_{i\cdot}\|_2^2} \geq \gamma - 16s_i \|\mathbb{E}[T^{-1}\mathbf{X}\mathbf{X}'] - T^{-1}\mathbf{X}\mathbf{X}'\|_{\max}.$$

This completes the proof. $\square$

Lemma 7. *If Conditions 1–5 are true, then the following inequality holds for all $i \in [N]$ such that $\bar{r}_i = o(1)$ with probability at least $1 - O((N \vee T)^{-\nu})$:*

$$\begin{aligned} (a) \quad & \max_{j \in [KN]} \left| \hat{\sigma}_i^2 \hat{\boldsymbol{\omega}}_j' \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j - \sigma_i^2 \omega_j^2 \right| \lesssim M_\omega^2 \lambda, \\ (b) \quad & \max_{j \in [KN]} \left| \hat{\sigma}_i^2 \hat{\omega}_j^2 - \sigma_i^2 \omega_j^2 \right| \lesssim M_\omega^2 \lambda, \\ (c) \quad & \max_{j \in [KN]} \left| \frac{\sigma_i \omega_j}{\hat{\sigma}_i \sqrt{\hat{\boldsymbol{\omega}}_j' \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j}} - 1 \right| \lesssim \frac{M_\omega^2 \lambda}{\left| \gamma^2 - O(\sqrt{M_\omega^2 \lambda}) \right|}, \\ (d) \quad & \max_{j \in [KN]} \left| \frac{\sigma_i \omega_j}{\hat{\sigma}_i \hat{\omega}_j} - 1 \right| \lesssim \frac{M_\omega^2 \lambda}{\left| \gamma^2 - O(\sqrt{M_\omega^2 \lambda}) \right|}. \end{aligned}$$

In consequence, if $\bar{v}_i = \bar{r}_i + M_\omega^2 \lambda$ is $o(1)$, all the upper bounds are $o(1)$.

Proof of Lemma 7. For any $(i, j) \in \mathcal{H}$, we consider two bounds:

$$\begin{aligned} (a) \quad & \left| \hat{\sigma}_i^2 \hat{\boldsymbol{\omega}}_j' \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j - \sigma_i^2 \omega_j^2 \right| \leq (|\hat{\sigma}_i^2 - \sigma_i^2| + \sigma_i^2) \left| \hat{\boldsymbol{\omega}}_j' \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j - \omega_j^2 \right| + |\hat{\sigma}_i^2 - \sigma_i^2| \omega_j^2, \\ (b) \quad & \left| \hat{\sigma}_i^2 \hat{\omega}_j^2 - \sigma_i^2 \omega_j^2 \right| \leq (|\hat{\sigma}_i^2 - \sigma_i^2| + \sigma_i^2) \left| \hat{\omega}_j^2 - \omega_j^2 \right| + |\hat{\sigma}_i^2 - \sigma_i^2| \omega_j^2. \end{aligned}$$

To complete the proof, we derive upper bounds of

$$(i) := \max_j |\hat{\boldsymbol{\omega}}_j' \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j - \omega_j^2|, \quad (i)' := \max_j |\hat{\omega}_j^2 - \omega_j^2|, \quad \text{and} \quad (ii) := |\hat{\sigma}_i^2 - \sigma_i^2|.$$

First, $(i)'$ is bounded as

$$(i)' \lesssim \max_j \|\boldsymbol{\omega}_j\|_{\lambda_1} \lesssim M_\omega^2 \lambda$$

by Theorem 6 of [Cai et al. \(2011\)](#). Next, bound (i) . Since $w_{jj} = \boldsymbol{\omega}_j' \mathbf{e}_j$, we have

$$\begin{aligned} (i) &\leq \max_j |(\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j)' \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j| + |\boldsymbol{\omega}_j' (\mathbf{e}_j - \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j)| \\ &\leq \max_j |(\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j)' (\hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j - \mathbf{e}_j)| + \max_j |(\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j)' \mathbf{e}_j| + \max_j |\boldsymbol{\omega}_j' (\mathbf{e}_j - \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j)| \\ &\leq \max_j \|\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j\|_1 \|\hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j - \mathbf{e}_j\|_\infty + \max_j \|\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j\|_\infty + \max_j \|\boldsymbol{\omega}_j\|_1 \|\mathbf{e}_j - \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j\|_\infty \\ &\leq \max_j \|\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j\|_1 \lambda_1 + 2M_\omega \lambda_1. \end{aligned}$$

By the proof of Proposition 2, it holds with probability at least $1 - O((N \vee T)^{-\nu})$ that

$$\max_j \|\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j\|_1 \lesssim (\max_j \|\boldsymbol{\omega}_j\|_{\ell_1} \lambda_1)^{1-r} s_\omega \lesssim M_\omega^{2-2r} \lambda^{1-r} s_\omega,$$

where we have used $\lambda_1 = b \max_j \|\boldsymbol{\omega}_j\|_1 \lambda / 2$. Then Conditions 4 and 5 yield

$$(i) \lesssim M_\omega^{3-2r} \lambda^{2-r} s_\omega + M_\omega^2 \lambda.$$

Finally bound (ii) . We have

$$(ii) \leq T^{-1} \left| \sum_{t=1}^T (\hat{u}_{it}^2 - u_{it}^2) \right| + \left| T^{-1} \sum_{t=1}^T (u_{it}^2 - \mathbb{E} u_{it}^2) \right|.$$

Consider the first term. Because $\hat{\mathbf{u}}_{i\cdot} - \mathbf{u}_{i\cdot} = -(\hat{\boldsymbol{\phi}}_{i\cdot}^L - \boldsymbol{\phi}_{i\cdot})\mathbf{X}$, we see that

$$\begin{aligned}
T^{-1} \left| \sum_{t=1}^T (\hat{u}_{it}^2 - u_{it}^2) \right| &\leq T^{-1} \left| \sum_{t=1}^T (\hat{u}_{it} - u_{it})^2 \right| + 2T^{-1} \left| \sum_{t=1}^T (\hat{u}_{it} - u_{it}) u_{it} \right| \\
&\leq T^{-1} \left\| (\hat{\boldsymbol{\phi}}_{i\cdot}^L - \boldsymbol{\phi}_{i\cdot})\mathbf{X} \right\|_2^2 + 2T^{-1} |(\hat{\boldsymbol{\phi}}_{i\cdot}^L - \boldsymbol{\phi}_{i\cdot})\mathbf{X}\mathbf{u}'_{i\cdot}| \\
&\leq T^{-1} \left\| (\hat{\boldsymbol{\phi}}_{i\cdot}^L - \boldsymbol{\phi}_{i\cdot})\mathbf{X} \right\|_2^2 + 2T^{-1} \|\hat{\boldsymbol{\phi}}_{i\cdot}^L - \boldsymbol{\phi}_{i\cdot}\|_1 \|\mathbf{X}\mathbf{U}'\|_{\max} \\
&\lesssim s_i \lambda^2 + s_i \lambda \lambda = 2s_i \lambda^2,
\end{aligned}$$

where the last inequality holds by Proposition 1 (b)(c) and Lemma 3. To evaluate the second term, putting $x = \sqrt{2\nu c_{uu}^2 T^{-1} \log N}$ in Lemma 1 (a) yields

$$\mathbb{P} \left(\max_i \left| \frac{1}{T} \sum_{t=1}^T (u_{it}^2 - \mathbb{E} u_{it}^2) \right| \gtrsim \sqrt{\frac{\log N}{T}} = o(\lambda) \right) = O(N^{-\nu}).$$

Combining these inequalities gives (ii) $\lesssim s_i \lambda^2 + o(\lambda) = o(\lambda)$ since $s_i \lambda = o(1)$ by the assumed condition $\bar{r}_i = o(1)$.

Consequently we obtain

$$\begin{aligned}
(a) \quad \{ (ii) + \sigma_i^2 \} (i) + (ii) \omega_j^2 &\lesssim (M_\omega^2 \lambda + M_\omega^{3-2r} \lambda^{2-r} s_\omega) + o(\lambda) \\
&\lesssim M_\omega^2 \lambda + \min_{i \in [N]} \bar{r}_i^2
\end{aligned}$$

and

$$(b) \quad \{ (ii) + \sigma_i^2 \} (i)' + (ii) \omega_j^2 \lesssim M_\omega^2 \lambda + o(\lambda) \asymp M_\omega^2 \lambda.$$

Denote $m_{ij} = \sigma_i \sqrt{\boldsymbol{\omega}'_j \boldsymbol{\Sigma}_x \boldsymbol{\omega}_j} = \sigma_i \omega_j$ and $\hat{m}_{ij}$ is either $\hat{\sigma}_i \sqrt{\hat{\boldsymbol{\omega}}'_j \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j}$ or $\hat{\sigma}_i \hat{\omega}_j$. For any r such that $|\hat{m}_{ij}^2 - m_{ij}^2| \leq r$, we have $\hat{m}_{ij}^2 \geq m_{ij}^2 - r$, and hence, $\hat{m}_{ij} \geq m_{ij} - \sqrt{r}$.

Therefore,

$$\begin{aligned}
r &\geq |m_{ij}^2 - \hat{m}_{ij}^2| = |m_{ij}\hat{m}_{ij} + \hat{m}_{ij}^2| \left| \frac{m_{ij}}{\hat{m}_{ij}} - 1 \right| \\
&\geq |m_{ij}(m_{ij} - \sqrt{r}) + m_{ij}^2 - r| \left| \frac{m_{ij}}{\hat{m}_{ij}} - 1 \right| \\
&= |2m_{ij}^2 - O(\sqrt{r})| \left| \frac{m_{ij}}{\hat{m}_{ij}} - 1 \right|,
\end{aligned}$$

which gives the inequality,

$$\left| \frac{m_{ij}}{\hat{m}_{ij}} - 1 \right| \lesssim \frac{r}{|m_{ij}^2 - O(\sqrt{r})|}.$$

Since $\min_{i,j} m_{ij}^2 \geq \gamma^2$ by Condition 5, this completes the proof. $\square$

Lemma 8. Suppose $\nu > 2$ and $\max_{i \in [N]} \bar{v}_i = O(T^{-\kappa_1})$ for some constants $\kappa_1 \in (0, 1/2)$.

If Conditions 1–5 are assumed, then for each $(i, j), (k, \ell) \in \mathcal{H}$ and any $x > 0$, the following inequalities hold:

$$\begin{aligned}
(a) \quad &\mathbb{P}(\mathcal{Z}_{ij} \geq x + \delta_1) - O((N \vee T)^{-\nu+2}) \\
&\leq \mathbb{P}(\mathcal{T}_{ij} - \sqrt{T}\phi_{ij}/\hat{m}_{ij} \geq x) \leq \mathbb{P}(\mathcal{Z}_{ij} \geq x - \delta_1) + O((N \vee T)^{-\nu+2}), \\
(b) \quad &\mathbb{P}(\mathcal{T}_{ij} - \sqrt{T}\phi_{ij}/\hat{m}_{ij} \geq x, \mathcal{T}_{k\ell} - \sqrt{T}\phi_{k\ell}/\hat{m}_{k\ell} \geq x) \\
&\leq \mathbb{P}(\mathcal{Z}_{ij} \geq x - \delta_1, \mathcal{Z}_{k\ell} \geq x - \delta_1) + O((N \vee T)^{-\nu+2}),
\end{aligned}$$

where $(\mathcal{Z}_{ij}, \mathcal{Z}_{k\ell})$ is a bivariate standard normal random vector with the covariance (correlation) $\rho_{(i,j),(k,\ell)} = \sigma_{ik}\omega_{j\ell}/(\sigma_i\sigma_k\omega_j\omega_\ell)$ and $\delta_1 = O(T^{-\kappa})$ for some $\kappa \in (0, \kappa_1]$ is some positive sequence.

Proof of Lemma 8. Prove (b). Fix (i, j) and (k, ℓ) arbitrary and let $\tilde{z}_{ij} = T^{-1/2} \sum_{t=1}^T \xi_t^{(i,j)}$

with $\xi_t^{(i,j)} = u_{it}\mathbf{x}'_t\boldsymbol{\omega}_j/m_{ij}$ and $m_{ij} = \sigma_i\omega_j$. From Theorem 1 we have

$$\begin{aligned} \mathbb{P}(\mathbf{T}_{ij} - \sqrt{T}\phi_{ij}/\hat{m}_{ij} \geq x, \mathbf{T}_{k\ell} - \sqrt{T}\phi_{k\ell}/\hat{m}_{k\ell} \geq x) \\ \leq \mathbb{P}(\tilde{z}_{ij} \geq x - \bar{v}_i, \tilde{z}_{k\ell} \geq x - \bar{v}_k) + O((N \vee T)^{-\nu+2}). \end{aligned}$$

Note that $(\tilde{z}_{ij}, \tilde{z}_{k\ell}) = T^{-1/2} \sum_{t=1}^T (\xi_t^{(i,j)}, \xi_t^{(k,\ell)})$ is the sum of the square integrable martingale difference sequence with respect to the filtration, $\mathcal{F}_t = \sigma\{\mathbf{u}_{t-s}, \mathbf{x}_{t-s+1} : s = 0, 1, \dots\}$. (Kifer, 2013, Theorem 3.2) establishes the strong approximation that without changing its distribution the vector sequence $(\xi_t^{(i,j)}, \xi_t^{(k,\ell)})$ can be redefined on a richer probability space where there exists a bivariate standard normal random vector $(\mathcal{Z}_{ij}, \mathcal{Z}_{k\ell})$ with the covariance (correlation) $\rho_{ijk\ell} = \sigma_{ik}\omega_{j\ell}/(m_{ij}m_{k\ell})$ such that $(\tilde{z}_{ij}, \tilde{z}_{k\ell}) = (\mathcal{Z}_{ij}, \mathcal{Z}_{k\ell}) + O(T^{-\kappa_2})$ holds a.s. for some $\kappa_2 > 0$, provided that the sufficient conditions (3.4) and (3.5) in his paper are true. Therefore, the proof of (b) completes if these two conditions are verified.

Check Kifer's condition (3.4). For any (i, j) and (k, ℓ) , we obtain

$$\begin{aligned} & \left| T^{-1} \sum_{t=1}^T \left(\mathbb{E}[\xi_t^{(i,j)} \xi_t^{(k,\ell)} \mid \mathcal{F}_{t-1}] - \frac{\sigma_{ik}\boldsymbol{\omega}'_j \boldsymbol{\Sigma}_x \boldsymbol{\omega}_\ell}{m_{ij}m_{k\ell}} \right) \right| \\ &= \frac{|\sigma_{ik}|}{m_{ij}m_{k\ell}} \left| \boldsymbol{\omega}'_j T^{-1} \sum_{t=1}^T (\mathbf{x}_t \mathbf{x}'_t - \boldsymbol{\Sigma}_x) \boldsymbol{\omega}_\ell \right| \\ &\leq \frac{\|\boldsymbol{\omega}_j\|_1 \|\boldsymbol{\omega}_\ell\|_1}{\gamma^2} \|T^{-1} \mathbf{X} \mathbf{X}' - T^{-1} \mathbb{E} \mathbf{X} \mathbf{X}'\|_{\max}. \end{aligned}$$

Thus Lemma 4 with Conditions 4 and 5 entails that the upper bound of this inequality becomes $O(M_\omega^2 T^{-1/2} \log^{3/2}(N \vee T))$ with probability at least $1 - O((N \vee T)^{-\nu})$. By the assumed condition, we have

$$M_\omega^2 T^{-1/2} \log^{3/2}(N \vee T) \lesssim \max_{i \in [N]} \bar{v}_i = O(T^{-\kappa_1})$$

for some constant $\kappa_1 \in (0, 1/2)$. This verifies condition (3.4).

Check Kifer's condition (3.5). By Condition 5, we observe that

$$\mathbb{E} \left[\|(\xi_t^{(i,j)}, \xi_t^{(k,\ell)})\|_2^4 \right] \lesssim \max_{i,j} \mathbb{E} \left[|\xi_t^{(i,j)}|^4 \right] \leq \max_{i,j} \frac{\mathbb{E}[u_{it}^4]}{\sigma_i^4 \omega_j^4} \max_j \mathbb{E} [(\mathbf{x}'_t \boldsymbol{\omega}_j)^4],$$

where $\max_i \max_t \mathbb{E}[u_{it}^4] < \infty$ by Condition 1 and $\min_{i,j} \sigma_i^4 \omega_j^4 \geq \gamma^4$. Without loss of generality, suppose $K = 1$. By the VMA(∞) representation in (11), we obtain

$$\mathbb{E} [(\mathbf{x}'_t \boldsymbol{\omega}_j)^4] \lesssim \sum_{\ell=0}^{\infty} \mathbb{E} [(\boldsymbol{\omega}'_j \mathbf{B}_\ell \mathbf{u}_{t-\ell})^4] + \left(\sum_{\ell=0}^{\infty} \mathbb{E} [(\boldsymbol{\omega}'_j \mathbf{B}_\ell \mathbf{u}_{t-\ell})^2] \right)^2 =: (I) + (II).$$

Recall that $\boldsymbol{\Sigma}_x = \mathbb{E}[\mathbf{x}_t \mathbf{x}'_t] = \sum_{\ell=0}^{\infty} \mathbf{B}_\ell \boldsymbol{\Sigma}_u \mathbf{B}'_\ell$. Thus we have

$$(II) = \left(\sum_{\ell=0}^{\infty} \boldsymbol{\omega}'_j \mathbf{B}_\ell \boldsymbol{\Sigma}_u \mathbf{B}'_\ell \boldsymbol{\omega}_j \right)^2 = (\boldsymbol{\omega}'_j \boldsymbol{\Sigma}_x \boldsymbol{\omega}_j)^2 = \omega_j^4 \leq 1/\gamma^2.$$

Khintchine's inequality for a weighted sum of i.i.d. subG random variables ([Vershynin, 2018](#), Exercise 2.6.5) yields

$$\begin{aligned} (I) &= \sum_{\ell=0}^{\infty} \mathbb{E} [(\boldsymbol{\omega}'_j \mathbf{B}_\ell \mathbf{u}_{t-\ell})^4] \lesssim \sum_{\ell=0}^{\infty} (\boldsymbol{\omega}'_j \mathbf{B}_\ell \boldsymbol{\Sigma}_u \mathbf{B}'_\ell \boldsymbol{\omega}_j)^2 \\ &\leq \left(\sum_{\ell=0}^{\infty} \boldsymbol{\omega}'_j \mathbf{B}_\ell \boldsymbol{\Sigma}_u \mathbf{B}'_\ell \boldsymbol{\omega}_j \right)^2 = \omega_j^4 \leq 1/\gamma^2. \end{aligned}$$

Therefore for given constant $\kappa_1 \in (0, 1/2)$, we have

$$\begin{aligned} &\sum_{t=1}^{\infty} t^{\kappa_1-1} \mathbb{E} \left[\|(\xi_t^{(i,j)}, \xi_t^{(k,\ell)})\|^2 \mathbf{1} \left\{ \|(\xi_t^{(i,j)}, \xi_t^{(k,\ell)})\|^2 \geq 1/t^{\kappa_1-1} \right\} \right] \\ &\leq \sum_{t=1}^{\infty} t^{2\kappa_1-2} \mathbb{E} \left[\|(\xi_t^{(i,j)}, \xi_t^{(k,\ell)})\|^4 \right] \lesssim \frac{\mathbb{E}[u_{it}^4]}{\gamma^8} \sum_{t=1}^{\infty} t^{2\kappa_1-2} < \infty, \end{aligned}$$

which verifies condition (3.5). This completes the proof of (b).

Prove (a). Note that

$$\begin{aligned}\mathbb{P}(\mathbf{T}_{ij} \geq x) &\geq \mathbb{P}(\tilde{z}_{ij} \geq x + \bar{v}_i) - \mathbb{P}\left(\max_{i,j} |v_{ij}| > \max_{i \in [N]} \bar{v}_i\right) \\ &= \mathbb{P}\left(\tilde{z}_{ij} \geq x + \max_{i \in [N]} \bar{v}_i\right) - O((N \vee T)^{-\nu+2}),\end{aligned}$$

where the last equality follows from Theorem 1. The remainder of the proof of (a) is the same as that of (b). This completes all the proofs. $\square$

Lemma 9. *For any bivariate standard normal random vector $(\mathcal{Z}_1, \mathcal{Z}_2)$ with correlation $\rho \in (-1, 1)$, it holds that*

$$\sup_{z \in [0, \infty)} \frac{\mathbb{P}(\mathcal{Z}_1 \geq z, \mathcal{Z}_2 \geq z)}{Q(z)^2} \leq \frac{1}{\sqrt{1 - \rho^2}}.$$

Proof of Lemma 9. Let $(x, y) \mapsto \phi(x, y; \rho)$ denote the density function of the bivariate standard normal random vector with correlation $\rho \in (-1, 1)$. A simple calculation yields for any $z \geq 0$,

$$\begin{aligned}\mathbb{P}(\mathcal{Z}_{ij} \geq z, \mathcal{Z}_{k\ell} \geq z) &= \int_z^\infty \int_z^\infty \phi(x, y; \rho) \, dx \, dy \\ &= \frac{1}{\sqrt{1 - \rho^2}} \int_z^\infty \int_z^\infty \phi(x, y; 0) \exp\left\{-\frac{2 - \rho^2}{2(1 - \rho^2)}(x^2 + y^2) + \frac{2\rho}{2(1 - \rho^2)}xy\right\} \, dx \, dy \\ &\leq \frac{1}{\sqrt{1 - \rho^2}} \int_z^\infty \int_z^\infty \phi(x, y; 0) \, dx \, dy \max_{0 \leq x, y < \infty} \exp\left\{-\frac{2 - \rho^2}{2(1 - \rho^2)}(x^2 + y^2) + \frac{2\rho}{2(1 - \rho^2)}xy\right\}.\end{aligned}$$

Note that $\int_z^\infty \int_z^\infty \phi(x, y; 0) \, dx \, dy = Q(z)^2$ and

$$\max_{0 \leq x, y < \infty} \left\{-\frac{2 - \rho^2}{2(1 - \rho^2)}(x^2 + y^2) + \frac{2\rho}{2(1 - \rho^2)}xy\right\} \leq \max_{0 \leq x, y < \infty} \frac{-2 + \rho + \rho^2}{2(1 - \rho^2)}(x^2 + y^2) = 0$$

uniformly in $\rho \in (-1, 1)$. Combining the results gives the desired uniform bound. $\square$

Lemma 10. *If Conditions 1 and 2 are true, then we have*

$$\mathbb{P}\left(\max_{i \in [N]} \max_{t \in [T]} \max_{k \in [K]} |y_{i,t-k}|^m \gtrsim \sqrt{\log^m NT}\right) = O((N \vee T)^{-\nu}).$$

Proof of Lemma 10. For $\mathbf{y}_t = \sum_{\ell=0}^{\infty} \mathbf{B}_\ell \mathbf{u}_{t-\ell}$, let $\mathbf{b}_{\ell,i}$ denote the i th column vector of $\mathbf{B}'_\ell$ and $b = \sum_{\ell=T}^{\infty} \|\mathbf{B}_\ell\|_\infty$ as before. By the truncation argument as in the proof of Lemma 3, we have for any $x > 0$,

$$\begin{aligned}
& \mathbb{P} \left(\max_{i \in [N]} \max_{t \in [T]} \max_{k \in [K]} |y_{i,t-k}| > x \right) \\
& \leq \mathbb{P} \left(\max_{t \in [T]} \max_{k \in [K]} \sum_{\ell=0}^{\infty} \|\mathbf{B}_\ell\|_\infty \|\mathbf{u}_{t-k-\ell}\|_\infty > x \right) \\
& \leq \mathbb{P} \left(\max_{t \in [T]} \max_{k \in [K]} \sum_{\ell=0}^{T-1} \|\mathbf{B}_\ell\|_\infty \|\mathbf{u}_{t-k-\ell}\|_\infty > x/2 \right) \\
& \quad + \mathbb{P} \left(\max_{t \in [T]} \max_{k \in [K]} \sum_{\ell=T}^{\infty} \|\mathbf{B}_\ell\|_\infty \|\mathbf{u}_{t-k-\ell}\|_\infty > x/2 \right) \\
& \leq \mathbb{P} \left(\max_{t \in [T]} \max_{\ell \in [T]} \|\mathbf{u}_{t-2\ell}\|_\infty > \frac{x}{2b} \right) + \frac{2}{x} \sum_{\ell=T}^{\infty} \|\mathbf{B}_\ell\|_\infty \mathbb{E} \left[\max_{t \in [T]} \|\mathbf{u}_{t-\ell}\|_\infty \right] \\
& \leq 3NT \exp \left(-\frac{x^2}{4c_u b^2} \right) + \frac{2c_1 \delta^T}{x(1-\delta)} \log^{1/2} NT,
\end{aligned}$$

where $c_1 > 0$, $\delta \in (0, 1)$, and $c_u > 0$ have been given in Lemmas 1(b), 2, and Condition 1, respectively. Thus for any finite $m \in \mathbb{N}$, we have

$$\begin{aligned}
& \mathbb{P} \left(\max_{i \in [N]} \max_{t \in [T]} \max_{k \in [K]} |y_{i,t-k}|^m > x \right) = \mathbb{P} \left(\max_{i \in [N]} \max_{t \in [T]} \max_{k \in [K]} |y_{i,t-k}| > x^{1/m} \right) \\
& \lesssim NT \exp \left(-\frac{x^{2/m}}{4c_u b^2} \right) + \frac{\delta^T}{x^{1/m}(1-\delta)} \log^{1/2} NT.
\end{aligned}$$

Setting $x = \{4c_u b^2(\nu + 2) \log(N \vee T)\}^{m/2}$ for any $\nu > 0$ leads to the result. $\square$

Lemma 11. Define $\lambda^* \asymp \sqrt{T^{-1} \log^3(N \vee T)}$. If Conditions 1–3 and 7 are true and $\bar{s}\lambda = o(1)$ holds, then the event,

$$\mathbb{P}^* \left(\|\mathbf{T}^{-1} \mathbf{U}^* \mathbf{X}'\|_{\max} \gtrsim \lambda^* \right) = O((N \vee T)^{-\nu}),$$

occurs with probability at least $1 - O((N \vee T)^{-\nu})$.

Proof of Lemma 11. Recall the notation that $\mathbf{U}^* = (\hat{\mathbf{u}}_1 \zeta_1, \dots, \hat{\mathbf{u}}_T \zeta_T)$ and $\hat{\mathbf{u}}_t = \mathbf{y}_t -$

$\hat{\Phi}_L \mathbf{x}_t = \mathbf{u}_t + (\Phi - \hat{\Phi}_L) \mathbf{x}_t$. We have

$$\begin{aligned} & \mathbb{P}^* \left(\left\| T^{-1} \sum_{t=1}^T \mathbf{x}_t \mathbf{u}_t^{*'} \right\|_{\max} > x \right) \\ & \leq \mathbb{P}^* \left(\left\| T^{-1} \sum_{t=1}^T \mathbf{x}_t \mathbf{u}_t' \zeta_t \right\|_{\max} > x/2 \right) + \mathbb{P}^* \left(\left\| T^{-1} \sum_{t=1}^T \mathbf{x}_t (\hat{\mathbf{u}}_t - \mathbf{u}_t)' \zeta_t \right\|_{\max} > x/2 \right). \end{aligned} \quad (\text{A.20})$$

Bound the first term of (A.20). Since ζ_t is a sequence of i.i.d. subG random variables by Condition 7, applying the union bound and Hoeffding's inequality under $\mathbb{P}^*$ yields for any $x > 0$,

$$\begin{aligned} & \mathbb{P}^* \left(\left\| T^{-1} \sum_{t=1}^T \mathbf{x}_t \mathbf{u}_t' \zeta_t \right\|_{\max} > x/2 \right) \\ & \leq 2KN^2 \max_{i,j \in [N]} \max_{k \in [K]} \exp \left(- \frac{x^2 T}{4c_\zeta T^{-1} \sum_{t=1}^T y_{i,t-k}^2 u_{jt}^2} \right) \\ & \leq 2KN^2 \exp \left(- \frac{x^2 T}{4c_\zeta \max_{i,j \in [N]} \max_{t \in [T]} \max_{k \in [K]} y_{i,t-k}^2 u_{jt}^2} \right) \\ & \lesssim 2KN^2 \exp \left(- \frac{x^2 T}{4c_\zeta \log^2 NT} \right), \end{aligned}$$

where the last inequality holds with probability at least $1 - O((N \vee T)^{-\nu})$ by Lemma 10.

By Proposition 1(b) and Lemma 10, the second term of (A.20) is bounded as

$$\begin{aligned}
& \mathbb{P}^* \left(\left\| T^{-1} \sum_{t=1}^T \mathbf{x}_t (\hat{\mathbf{u}}_t - \mathbf{u}_t)' \zeta_t \right\|_{\max} > x/2 \right) \\
&= \mathbb{P}^* \left(\left\| T^{-1} \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \zeta_t (\Phi - \hat{\Phi}_L)' \right\|_{\max} > x/2 \right) \\
&\leq \mathbb{P}^* \left(\left\| T^{-1} \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \zeta_t \right\|_{\max} \max_{i \in [N]} \left\| \hat{\phi}_{i\cdot}^L - \phi_{i\cdot} \right\|_1 > x/2 \right) \\
&\leq \mathbb{P}^* \left(\left\| T^{-1} \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \zeta_t \right\|_{\max} \bar{s} \lambda \gtrsim x/2 \right) \\
&\leq 2K^2 N^2 \max_{k, \ell \in [K]} \max_{i, j \in [N]} \exp \left(- \frac{x^2 T}{4c_\zeta \bar{s}^2 \lambda^2 T^{-1} \sum_{t=1}^T y_{i, t-k}^2 y_{j, t-\ell}^2} \right) \\
&\lesssim 2K^2 N^2 \exp \left(- \frac{x^2 T}{4c_\zeta \bar{s}^2 \lambda^2 \log^2 NT} \right),
\end{aligned}$$

with probability at least $1 - O((N \vee T)^{-\nu})$. Combining the obtained results with taking

$$x = 2\sqrt{c_\zeta(\nu + 2)T^{-1} \log^3 NT} \ (\asymp \lambda^*)$$

gives the desired statement since $\bar{s} \lambda = o(1)$. This completes the proof. $\square$

Lemma 12. Recall $\lambda^* \asymp \sqrt{T^{-1} \log^3(N \vee T)}$, defined in Lemma 11. If Conditions 1–3 and 7 are true, then the event,

$$\mathbb{P}^* \left(\max_{i \in [N]} \left\| \hat{\phi}_{i\cdot}^{L*} - \hat{\phi}_{i\cdot}^L \right\|_1 \gtrsim \bar{s} \lambda^* \right) = O((N \vee T)^{-\nu+1})$$

occurs with high probability.

Proof of Lemma 12. Define $\boldsymbol{\delta}_{i\cdot}^* = \hat{\phi}_{i\cdot}^{L*} - \hat{\phi}_{i\cdot}^L$. Derive the non-asymptotic error bound for the Lasso estimator. For λ^* such that $b\lambda/2 \leq \lambda^* \asymp \sqrt{T^{-1} \log^3(N \vee T)}$, define two events,

$$\mathcal{E}_1^* = \left\{ \left\| T^{-1} \mathbf{X} \mathbf{U}^{*'} \right\|_{\max} \leq \lambda^*/2 \right\}, \quad \mathcal{E}_2^* = \left\{ \left\| T^{-1} \mathbf{X} \mathbf{X}' - \mathbb{E}[T^{-1} \mathbf{X} \mathbf{X}'] \right\|_{\max} \leq \lambda^* \right\}.$$

Lemmas 11 and 4 guarantee that the event,

$$\{\mathbb{P}^*(\mathcal{E}_1^{*c}) = O((N \vee T)^{-\nu})\} \cap \mathcal{E}_2^*,$$

occurs with probability at least $1 - O((N \vee T)^{-\nu})$. Then, following the proof of Proposition 1, we obtain the desired error bound, $\|\boldsymbol{\delta}_i^*\|_1 \lesssim \hat{s}_i \lambda^*$, on event $\mathcal{E}_1^* \cap \mathcal{E}_2^*$. Finally an application of Theorem 1(b) of Zhu and Liu (2020) gives $\hat{s}_i \lesssim s_i \leq \bar{s}$ with high probability. The assertion is obtained by the union bound. This completes the proof. $\square$

Lemma 13. Recall $\lambda^* \asymp \sqrt{T^{-1} \log^3(N \vee T)}$, defined in Lemma 11. If Conditions 1–3 and 7 are true and $\bar{s}\lambda = o(1)$ holds, then for $\bar{\tau}_1 = \lambda^* + \bar{s}^2 \lambda^{*2}$ the event,

$$\mathbb{P}^* \left(\max_{i \in [N]} |\hat{\sigma}_i^{*2} - \hat{\sigma}_i^2| \gtrsim \bar{\tau}_1 \right) = O((N \vee T)^{-\nu+1}),$$

occurs with probability at least $1 - O((N \vee T)^{-\nu+1})$.

Proof of Lemma 13. Recall that $u_{it}^* = \hat{u}_{it} \zeta_t$, $\sigma_i^{*2} = T^{-1} \sum_{t=1}^T u_{it}^{*2}$, and $\hat{\sigma}_i^2 = T^{-1} \sum_{t=1}^T \hat{u}_{it}^2$. By the construction, we have $\sigma_i^{*2} - \hat{\sigma}_i^2 = T^{-1} \sum_{t=1}^T \hat{u}_{it}^2 (\zeta_t^2 - 1)$, where $\{\hat{u}_{it}^2 (\zeta_t^2 - 1)\}_t$ (for each $i \in [N]$) is a sequence of independent centered sub-exponential random variables under $\mathbb{P}^*$. Thus for some constant $c_\zeta > 0$, Bernstein's inequality with the union bound gives for any $x \in (0, 1]$,

$$\begin{aligned} \mathbb{P}^* \left(\max_{i \in [N]} |\sigma_i^{*2} - \hat{\sigma}_i^2| > x \right) &= \mathbb{P}^* \left(\max_{i \in [N]} \left| T^{-1} \sum_{t=1}^T \hat{u}_{it}^2 (\zeta_t^2 - 1) \right| > x \right) \\ &\leq 2N \max_i \exp \left(-\frac{x^2 T}{c_\zeta T^{-1} \sum_{t=1}^T \hat{u}_{it}^4} \right) \leq 2N \exp \left(-\frac{x^2 T}{c_\zeta \max_i \max_t \hat{u}_{it}^4} \right). \end{aligned}$$

Denote by $\boldsymbol{\delta}_i \in \mathbb{R}^{KN}$ the i th column vector of $\boldsymbol{\Delta}' = (\hat{\boldsymbol{\Phi}}^L - \boldsymbol{\Phi}^0)'$. Then we have

$$\hat{u}_{it}^4 = (u_{it} - \mathbf{x}_t' \boldsymbol{\delta}_i)^4 \lesssim u_{it}^4 + (\mathbf{x}_t' \boldsymbol{\delta}_i)^4 \lesssim u_{it}^4 + \|\mathbf{x}_t\|_\infty^4 \max_{i \in [N]} \|\boldsymbol{\delta}_i\|_1^4.$$

Therefore, by Proposition 1(b) and Lemma 10, it holds that $\max_i \max_t u_{it}^4 \lesssim \log^2(NT)$ and $\max_t \|\mathbf{x}_t\|_\infty^4 \max_{i \in [N]} \|\boldsymbol{\delta}_i\|_1^4 \lesssim \bar{s} \lambda \log^2(NT)$ occur with probability at least $1 - O((N \vee T)^{-\nu+1})$. Therefore, taking

$$x \asymp \sqrt{(\nu + 1)T^{-1} \log^3(NT)} \asymp \lambda^*$$

leads to the upper bound,

$$\mathbb{P}^* \left(\max_i |\sigma_i^{*2} - \hat{\sigma}_i^2| \gtrsim \lambda^* \right) \lesssim (N \vee T)^{-\nu},$$

which holds with probability at least $1 - O((N \vee T)^{-\nu+1})$.

Recall $\boldsymbol{\delta}_i^* = \hat{\boldsymbol{\phi}}_{i\cdot}^{\text{L}*} - \hat{\boldsymbol{\phi}}_{i\cdot}^{\text{L}}$. Then we have

$$\begin{aligned} & \mathbb{P}^* \left(\max_i |\hat{\sigma}_i^{*2} - \hat{\sigma}_i^2| > x \right) \\ &= \mathbb{P}^* \left(\max_i \left| \sigma_i^{*2} - \hat{\sigma}_i^2 - 2T^{-1} \sum_{t=1}^T u_{it}^* \mathbf{x}_t' \boldsymbol{\delta}_i^* + \boldsymbol{\delta}_i^{*'} \hat{\boldsymbol{\Sigma}}_x \boldsymbol{\delta}_i^* \right| > x \right) \\ &\leq \mathbb{P}^* \left(\max_i |\sigma_i^{*2} - \hat{\sigma}_i^2| + 2 \|T^{-1} \mathbf{U}^{*'} \mathbf{X}\|_{\max} \max_i \|\boldsymbol{\delta}_i^*\|_1 + \max_i \|\boldsymbol{\delta}_i^*\|_1^2 \left\| \hat{\boldsymbol{\Sigma}}_x \right\|_{\max} > x \right), \end{aligned}$$

where $\|\hat{\boldsymbol{\Sigma}}_x\|_{\max} = O(1)$ with probability at least $1 - O((N \vee T)^{-\nu})$ by Lemma 4 and Condition 5. By Lemmas 11 and 12, setting

$$x \asymp \lambda^* + \bar{s} \lambda^{*2} + \bar{s}^2 \lambda^{*2} \asymp \lambda^* + \bar{s}^2 \lambda^{*2} = \bar{\tau}_1$$

with the union bound yields the desired result. This completes the proof. $\square$

Lemma 14. *If Conditions 1–3 and 7 are true, then for $\bar{\tau}_2 = M_\omega^2 \lambda$ the event,*

$$\mathbb{P}^* \left(\left| T^{-1} \sum_{t=1}^T (u_{it}^{*2} - \hat{u}_{it}^2) \hat{\boldsymbol{\omega}}_j' \mathbf{x}_t \mathbf{x}_t' \hat{\boldsymbol{\omega}}_j \right| \gtrsim \bar{\tau}_2 \right) = O((NT)^{-\nu}),$$

occurs with probability at least $1 - O((N \vee T)^{-\nu})$.

Proof of Lemma 14. Because $\mathbb{E}^* u_{it}^{*2} = \hat{u}_{it}^2$, $(u_{it}^{*2} - \hat{u}_{it}^2) \hat{\omega}'_j \mathbf{x}_t \mathbf{x}'_t \hat{\omega}_j$ is a sequence of i.i.d. sub-exponential random variables under $\mathbb{P}^*$, Bernstein's inequality is applied. For any $x \in (0, 1]$, we obtain

$$\begin{aligned} \mathbb{P}^* \left(\left| T^{-1} \sum_{t=1}^T (u_{it}^{*2} - \hat{u}_{it}^2) \hat{\omega}'_j \mathbf{x}_t \mathbf{x}'_t \hat{\omega}_j \right| > x \right) &\lesssim \exp \left(- \frac{x^2 T}{T^{-1} \sum_{t=1}^T (\hat{\omega}'_j \mathbf{x}_t \mathbf{x}'_t \hat{\omega}_j)^2} \right) \\ &\lesssim \exp \left(- \frac{x^2 T}{\max_j \max_t \|\hat{\omega}_j\|_1^4 \|\mathbf{x}_t\|_\infty^4} \right) \\ &\lesssim \exp \left(- \frac{x^2 T}{M_\omega^4 \log^2(NT)} \right), \end{aligned}$$

where the last inequality holds with probability at least $1 - O((N \vee T)^{-\nu})$. Thus taking

$$x = \sqrt{\nu M_\omega^4 T^{-1} \log^3(NT)} \asymp M_\omega^2 \lambda = \bar{\tau}_2$$

leads to the upper bound to be $O((NT)^{-\nu})$. This completes the proof. $\square$

Lemma 15. *If Conditions 1–3 are true and $\bar{s}\lambda + M_\omega^2 \lambda = o(1)$ holds, then for*

$$\bar{\tau}_3 := \max \left\{ M_\omega^{3-2r} \lambda^{1-r} s_\omega \log^2(N \vee T), \bar{s}\lambda \log(N \vee T), M_\omega^2 \lambda \log(N \vee T) \right\},$$

we have

$$\mathbb{P} \left(\left| T^{-1} \sum_{t=1}^T (\hat{u}_{it}^2 - \hat{\sigma}_i^2) \hat{\omega}'_j \mathbf{x}_t \mathbf{x}'_t \hat{\omega}_j \right| \gtrsim \bar{\tau}_3 \right) = O((N \vee T)^{-\nu}).$$

Proof of Lemma 15. We have

$$\begin{aligned} &\mathbb{P} \left(\left| T^{-1} \sum_{t=1}^T (\hat{u}_{it}^2 - \hat{\sigma}_i^2) \hat{\omega}'_j \mathbf{x}_t \mathbf{x}'_t \hat{\omega}_j \right| > x \right) \\ &\leq \mathbb{P} \left(\left| T^{-1} \sum_{t=1}^T (u_{it}^2 - \sigma_i^2) \omega'_j \mathbf{x}_t \mathbf{x}'_t \omega_j \right| > x/4 \right) \\ &\quad + \mathbb{P} \left(\max_i \max_t |u_{it}^2 - \sigma_i^2| |\mathbf{x}'_t (\hat{\omega}_j \hat{\omega}'_j - \omega_j \omega'_j) \mathbf{x}_t| > x/4 \right) \\ &\quad + \mathbb{P} \left(\max_i \max_t |\hat{u}_{it}^2 - u_{it}^2| \hat{\omega}'_j \hat{\Sigma}_x \hat{\omega}_j > x/4 \right) + \mathbb{P} \left(\max_i |\sigma_i^2 - \hat{\sigma}_i^2| \hat{\omega}'_j \hat{\Sigma}_x \hat{\omega}_j > x/4 \right) \end{aligned}$$

We first evaluate the second to fourth probabilities. By (the proof of) Lemma 7 with Condition 5, the inequalities

$$\begin{aligned}
& \max_i \max_j \max_t |u_{it}^2 - \sigma_i^2| \mathbf{x}_t' (\hat{\boldsymbol{\omega}}_j \hat{\boldsymbol{\omega}}_j' - \boldsymbol{\omega}_j \boldsymbol{\omega}_j') \mathbf{x}_t \\
& \leq \max_i \max_j \max_t |u_{it}^2 + \sigma_i^2| \|\mathbf{x}_t\|_\infty^2 \|\hat{\boldsymbol{\omega}}_j (\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j)' + (\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j) \boldsymbol{\omega}_j'\|_1 \\
& \lesssim \|\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j\|_1 \|\boldsymbol{\omega}_j\|_1 \log^2(N \vee T) \lesssim M_\omega^{3-2r} \lambda^{1-r} s_\omega \log^2(N \vee T), \\
& \max_i \max_j \max_t |\hat{u}_{it}^2 - u_{it}^2| \hat{\boldsymbol{\omega}}_j' \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j \\
& \lesssim 2 \max_i \max_t |u_{it}| \|\mathbf{x}_t\|_\infty \|\boldsymbol{\delta}_i\|_1 + \max_t \|\mathbf{x}_t \mathbf{x}_t'\|_\infty \max_i \|\boldsymbol{\delta}_i\|_1^2 \\
& \lesssim \bar{s} \lambda \log(N \vee T), \\
& \max_i \max_j |\hat{\sigma}_i^2 - \sigma_i^2| \hat{\boldsymbol{\omega}}_j' \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j \lesssim \bar{s} \lambda^2
\end{aligned}$$

simultaneously hold with probability at least $1 - O((N \vee T)^{-\nu})$.

Next we evaluate the first probability. For $\bar{c}_3 = M_\omega^2 \log^2(N \vee T)$, the Azuma-Hoeffding inequality with the union bound and Lemma 10 yield

$$\begin{aligned}
& \mathbb{P} \left(\max_{i,j} \left| T^{-1} \sum_{t=1}^T (u_{it}^2 - \sigma_i^2) \boldsymbol{\omega}_j' \mathbf{x}_t \mathbf{x}_t' \boldsymbol{\omega}_j \right| > x \right) \\
& \leq \mathbb{P} \left(\max_{i,j} \left| T^{-1} \sum_{t=1}^T (u_{it}^2 - \sigma_i^2) \boldsymbol{\omega}_j' \mathbf{x}_t \mathbf{x}_t' \boldsymbol{\omega}_j \right| > x \mid \max_{i,j,t} |u_{it}^2 - \sigma_i^2| \boldsymbol{\omega}_j' \mathbf{x}_t \mathbf{x}_t' \boldsymbol{\omega}_j \lesssim \bar{c}_3 \right) \\
& \quad + \mathbb{P} \left(\max_{i,j,t} |u_{it}^2 - \sigma_i^2| \boldsymbol{\omega}_j' \mathbf{x}_t \mathbf{x}_t' \boldsymbol{\omega}_j \gtrsim \bar{c}_3 \right) \\
& \leq 2KN^2 \exp \left(-\frac{x^2 T}{2\bar{c}_3^2} \right) + O((N \vee T)^{-\nu}),
\end{aligned}$$

where the upper bound becomes $O((N \vee T)^{-\nu})$ by setting

$$x = 2(\nu + 2)\bar{c}_3 T^{-1/2} \log^{1/2}(N \vee T) \asymp M_\omega^2 \lambda \log(N \vee T).$$

If we set

$$\begin{aligned} x &\asymp \max \left\{ M_\omega^{3-2r} \lambda^{1-r} s_\omega \log^2(N \vee T), \bar{s} \lambda \log(N \vee T), \bar{s} \lambda^2, M_\omega^2 \lambda \log(N \vee T) \right\} \\ &= \max \left\{ M_\omega^{3-2r} \lambda^{1-r} s_\omega \log^2(N \vee T), \bar{s} \lambda \log(N \vee T), M_\omega^2 \lambda \log(N \vee T) \right\} = \bar{\tau}_3, \end{aligned}$$

the result follows. This completes the proof. $\square$

Lemma 16. *If Conditions 1–5 and 7 are true and $\bar{s} \lambda + M_\omega^2 \lambda = o(1)$ holds, then for*

$$\bar{\mu}_1 := \max \{ \bar{\tau}_1, \bar{\tau}_2, \bar{\tau}_3 \},$$

the event,

$$\mathbb{P}^* \left(\max_{i \in [N]} \max_{j \in [KN]} \left| \frac{\tilde{m}_{ij}^*}{\hat{m}_{ij}^*} - 1 \right| \gtrsim \bar{\mu}_1 \right) = O((N \vee T)^{-\nu}),$$

occurs with probability at least $1 - O((N \vee T)^{-\nu})$.

Proof of Lemma 16. For any $x > 0$, a simple calculus yields

$$\begin{aligned} &\mathbb{P}^* \left(\max_{i \in [N]} \max_{j \in [KN]} \left| \frac{\tilde{m}_{ij}^*}{\hat{m}_{ij}^*} - 1 \right| > x \right) \\ &\leq \mathbb{P}^* \left(\max_i \max_j |\tilde{m}_{ij}^{*2} - \hat{m}_{ij}^{*2}| > x \hat{m}_{ij}^* (\tilde{m}_{ij}^* + \hat{m}_{ij}^*) \right) \\ &\leq \mathbb{P}^* \left(\max_i \max_j |\tilde{m}_{ij}^{*2} - \hat{m}_{ij}^{*2}| > x \min_i \min_j \hat{m}_{ij}^{*2} \right) \\ &\leq \mathbb{P}^* \left(\max_i \max_j |\tilde{m}_{ij}^{*2} - \hat{m}_{ij}^{*2}| \gtrsim \gamma^3 x \right) + \mathbb{P}^* \left(\min_i \min_j \hat{m}_{ij}^{*2} \lesssim \gamma^3 \right) \end{aligned}$$

We see that $\tilde{m}_{ij}^{*2} \geq 0$ a.s. and, by Lemmas 7 and 13 with Condition 5, we obtain

$$\begin{aligned} \hat{m}_{ij}^{*2} &= \hat{\sigma}_i^{*2} \hat{\omega}'_j \hat{\Sigma}_x \hat{\omega}_j \geq \hat{\sigma}_i^2 \hat{\omega}'_j \hat{\Sigma}_x \hat{\omega}_j - |\hat{\sigma}_i^{*2} - \hat{\sigma}_i^2| \hat{\omega}'_j \hat{\Sigma}_x \hat{\omega}_j \\ &\gtrsim \sigma_i^2 \omega_j^2 - M_\omega^2 \lambda - \bar{\tau}_1 (\omega_j^2 - M_\omega^2 \lambda) \geq \gamma^2 - O(M_\omega^2 \lambda). \end{aligned}$$

uniformly in $(i, j) \in [N] \times [KN]$ with probability at least $1 - O((N \vee T)^{-\nu+1})$. Thus

it suffices to evaluate the upper bound of

$$\begin{aligned} & \mathbb{P}^* \left(\max_i \max_j |\tilde{m}_{ij}^{*2} - \hat{m}_{ij}^{*2}| > x \right) \\ & \leq \mathbb{P}^* \left(\max_i \max_j |\tilde{m}_{ij}^{*2} - \hat{m}_{ij}^2| > x/2 \right) + \mathbb{P}^* \left(\max_i \max_j |\hat{m}_{ij}^{*2} - \hat{m}_{ij}^2| > x/2 \right). \end{aligned} \quad (\text{A.21})$$

The second probability of (A.21) is bounded by the same argument just above; we obtain

$$\mathbb{P}^* \left(\max_i \max_j |\hat{m}_{ij}^{*2} - \hat{m}_{ij}^2| \gtrsim \bar{\tau}_1 \right) = O((N \vee T)^{-\nu+1}), \quad (\text{A.22})$$

which occurs with probability at least $1 - O((N \vee T)^{-\nu+1})$. Next bound the first probability of (A.21). Set

$$x_2 := c\bar{\tau}_2 + \tau_{ij,3} \quad \text{with} \quad \bar{\tau}_2 = M_\omega^2 \lambda, \quad \tau_{ij,3} = \left| T^{-1} \sum_{t=1}^T (\hat{u}_{it}^2 - \hat{\sigma}_i^2) \hat{\omega}'_j \mathbf{x}_t \mathbf{x}'_t \hat{\omega}_j \right|$$

for some constant $c > 0$. Then Lemma 14 establishes that the event,

$$\begin{aligned} & \mathbb{P}^* (|\tilde{m}_{ij}^{*2} - \hat{m}_{ij}^2| > x_2) \\ & \leq \mathbb{P}^* \left(\left| T^{-1} \sum_{t=1}^T (u_{it}^{*2} - \hat{u}_{it}^2) \hat{\omega}'_j \mathbf{x}_t \mathbf{x}'_t \hat{\omega}_j \right| + \left| T^{-1} \sum_{t=1}^T (\hat{u}_{it}^2 - \hat{\sigma}_i^2) \hat{\omega}'_j \mathbf{x}_t \mathbf{x}'_t \hat{\omega}_j \right| > x_2 \right) \\ & \leq \mathbb{P}^* \left(\left| T^{-1} \sum_{t=1}^T (u_{it}^{*2} - \hat{u}_{it}^2) \hat{\omega}'_j \mathbf{x}_t \mathbf{x}'_t \hat{\omega}_j \right| \gtrsim \bar{\tau}_2 \right) = O((N \vee T)^{-\nu}), \end{aligned} \quad (\text{A.23})$$

holds with probability at least $1 - O((N \vee T)^{-\nu})$. By Lemma 15, $\tau_{ij,3}$ is bounded by

$$\tau_{ij,3} \lesssim \bar{\tau}_3 = \max \{ M_\omega^{3-2r} \lambda^{1-r} s_\omega \log^2(N \vee T), \bar{s} \lambda \log(N \vee T), M_\omega^2 \lambda \log(N \vee T) \}$$

with probability at least $1 - O((N \vee T)^{-\nu})$.

Combine the two probabilities. Since $\lambda \asymp \lambda^*$, taking

$$x = \max \{\bar{\tau}_1, x_2\} \asymp \max \{\bar{\tau}_1, \bar{\tau}_2, \bar{\tau}_3\} = \bar{\mu}_1$$

in (A.21) establishes the desired inequality up to some positive constant factor. This completes the proof. $\square$

Lemma 17. *If Conditions 1–5 and 7 are true, then for*

$$\bar{\mu}_2 := \max \left\{ \lambda^{2-r} M_\omega^{2-2r} s_\omega, \bar{s} \lambda M_\omega \log^{3/2}(N \vee T) \right\},$$

the event,

$$\mathbb{P}^* \left(\max_{i \in [N]} \max_{j \in [KN]} \left| \frac{R_{ij}^*}{\hat{m}_{ij}^*} \right| \gtrsim \bar{\mu}_2 \right) = O((N \vee T)^{-\nu}),$$

occurs with probability at least $1 - O((N \vee T)^{-\nu})$.

Proof of Lemma 17. Since we have established that $\hat{m}_{ij}^*$ is uniformly lower bounded by a positive constant by Lemmas 7 and 13 with Condition 5 and $M_\omega^2 \lambda = o(1)$ in the proof of Lemma 16. Thus it is sufficient to derive the upper bound of $\max_i \max_j |R_{ij}^*|$. Observe that

$$|R_{ij}^*| \leq \|T^{-1/2} \mathbf{u}_i^* \mathbf{X}'\|_\infty \|\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j\|_1 + \|\sqrt{T}(\hat{\boldsymbol{\phi}}_i^L - \hat{\boldsymbol{\phi}}_i^{L*})\|_1 \|\hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j - \mathbf{e}_j\|_\infty.$$

By the proof of Proposition 2, the event,

$$\left\{ \max_{j \in [N]} \|\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j\|_1 \lesssim (M_\omega^2 \lambda)^{1-r} s_\omega \right\} \cap \left\{ \max_{j \in [N]} \|\hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j - \mathbf{e}_j\|_\infty \lesssim M_\omega \lambda \right\},$$

occurs with probability at least $1 - O((N \vee T)^{-\nu})$. Conditional on this event with

setting

$$x = (\lambda^* \max_j \|\hat{\omega}_j - \omega_j\|_1) \vee (\sqrt{T} \bar{s} \lambda^* \max_j \|\hat{\Sigma}_x \hat{\omega}_j - \mathbf{e}_j\|_\infty),$$

we have by Lemmas 11 and 12,

$$\begin{aligned} & \mathbb{P}^* \left(\max_{i,j} |R_{ij}^*| > x \right) \\ & \leq \mathbb{P}^* \left(\max_i \|T^{-1/2} \mathbf{u}_i^* \mathbf{X}'\|_\infty \max_j \|\hat{\omega}_j - \omega_j\|_1 > x/2 \right) \\ & \quad + \mathbb{P}^* \left(\max_i \|\sqrt{T}(\hat{\phi}_i^L - \hat{\phi}_i^{L*})\|_1 \max_j \|\hat{\Sigma}_x \hat{\omega}_j - \mathbf{e}_j\|_\infty > x/2 \right) \\ & \leq \mathbb{P}^* \left(\max_i \|T^{-1/2} \mathbf{u}_i^* \mathbf{X}'\|_\infty \gtrsim \lambda^* \right) + \mathbb{P}^* \left(\max_i \|\hat{\phi}_i^L - \hat{\phi}_i^{L*}\|_1 \gtrsim \bar{s} \lambda^* \right) \\ & = O((N \vee T)^{-\nu+1}) \end{aligned}$$

with high probability. Finally we see that by the proof of Proposition 2

$$\begin{aligned} x &= (\lambda^* \max_j \|\hat{\omega}_j - \omega_j\|_1) \vee (\sqrt{T} \bar{s} \lambda^* \max_j \|\hat{\Sigma}_x \hat{\omega}_j - \mathbf{e}_j\|_\infty) \\ &\lesssim (\lambda^{2-r} M_\omega^{2-2r} s_\omega) \vee \left(\bar{s} \lambda M_\omega \log^{3/2}(N \vee T) \right) = \bar{\mu}_2, \end{aligned}$$

with high probability. This completes the proof. $\square$

Lemma 18. *If Conditions 1–5 and 7, and $M_\omega \leq T^{1/2}/\log^3(N \vee T)$ are true, then the events,*

$$\mathbb{P}^* \left(\frac{S_{ij}^*}{\tilde{m}_{ij}^*} > t \right) = Q(t) \left\{ 1 + O \left(\frac{M_\omega \log NT}{T^{1/2}} \right) (1+t)^3 \right\}$$

and

$$\mathbb{P}^* \left(\frac{S_{ij}^*}{\tilde{m}_{ij}^*} < -t \right) = Q(t) \left\{ 1 + O \left(\frac{M_\omega \log NT}{T^{1/2}} \right) (1+t)^3 \right\}$$

for $t \in [0, \bar{t}]$ uniformly in $(i, j) \in \mathcal{H}$, occur with probability at least $1 - O((N \vee T)^{-\nu})$.

Proof of Lemma 18. Apply (Jing et al., 2003, Theorem 2.3) to the self-normalized sum, $S_{ij}^*/\tilde{m}_{ij}^*$ under $\mathbb{P}^*$. By Lemmas 15 and 7 with Conditions 5 and 7, we have

$$\begin{aligned} B_{ij}^2 &:= T^{-1} \sum_{t=1}^T \mathbb{E}^* u_{it}^{*2} (\mathbf{x}'_t \hat{\boldsymbol{\omega}}_j)^2 = T^{-1} \sum_{t=1}^T \hat{u}_{it}^2 (\mathbf{x}'_t \hat{\boldsymbol{\omega}}_j)^2 \\ &\geq \hat{\sigma}_t^2 \hat{\boldsymbol{\omega}}_j' \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j - O(\bar{\tau}_3) \geq \gamma^3 (1 - o(1)) \end{aligned}$$

with probability at least $1 - O((N \vee T)^{-\nu})$. Furthermore, since $\mathbb{E}^* |\zeta|^3$ is bounded by a finite constant under Condition 7, we have

$$\begin{aligned} L_{ij} &:= T^{-3/2} \sum_{t=1}^T \mathbb{E}^* |u_{it}^*|^3 |\mathbf{x}'_t \hat{\boldsymbol{\omega}}_j|^3 \asymp T^{-3/2} \sum_{t=1}^T |\hat{u}_{it}|^3 |\mathbf{x}'_t \hat{\boldsymbol{\omega}}_j|^3 \\ &\leq T^{-3/2} \sum_{t=1}^T |\hat{u}_{it}|^2 |\mathbf{x}'_t \hat{\boldsymbol{\omega}}_j|^2 \max_t |\hat{u}_{it}| |\mathbf{x}'_t \hat{\boldsymbol{\omega}}_j| = T^{-1/2} B_{ij}^2 \max_t |\hat{u}_{it}| |\mathbf{x}'_t \hat{\boldsymbol{\omega}}_j|, \end{aligned}$$

where Proposition 1 and (the proofs of) Lemmas 7 and 10 give

$$\begin{aligned} \max_t |\hat{u}_{it}| |\mathbf{x}'_t \hat{\boldsymbol{\omega}}_j| &\lesssim \max_t (|u_{it}| + \|\mathbf{x}_t\|_\infty \|\boldsymbol{\delta}_i\|_1) \|\mathbf{x}_t\|_\infty \|\hat{\boldsymbol{\omega}}_j\|_1 \\ &\leq \left(\max_i \max_t |u_{it}| + \max_t \|\mathbf{x}_t\|_\infty \|\boldsymbol{\delta}_i\|_1 \right) \max_t \|\mathbf{x}_t\|_\infty \max_j \|\hat{\boldsymbol{\omega}}_j\|_1 \lesssim M_\omega \log(NT) \end{aligned}$$

with probability at least $1 - O((N \vee T)^{-\nu})$. Thus

$$d_{ij} := \frac{B_{ij}}{L_{ij}^{1/3}} \gtrsim \frac{T^{1/6} B_{ij}^{1/3}}{M_\omega^{1/3} \log^{1/3}(NT)} \geq \frac{T^{1/6} \gamma^{1/2} (1 + o(1))}{M_\omega^{1/3} \log^{1/3}(NT)} \geq \log^{2/3}(N \vee T),$$

where the lower bound holds uniformly in (i, j) since $M_\omega \leq T^{1/2} / \log^3(N \vee T)$ is implied by the condition $\bar{\mu} = o(1)$. Therefore, by (Jing et al., 2003, Theorem 2.3), we conclude that

$$\mathbb{P}^* \left(\frac{S_{ij}^*}{\tilde{m}_{ij}^*} > \mathfrak{t} \right) = Q(\mathfrak{t}) \left\{ 1 + O \left(\frac{M_\omega \log NT}{T^{1/2}} \right) (1 + \mathfrak{t})^3 \right\}$$

holds with high probability for $\mathfrak{t} \in [0, \bar{\mathfrak{t}}]$, where $\bar{\mathfrak{t}} \asymp \log^{1/2}(N \vee T) = o(d_{ij})$, uniformly

in $(i, j) \in \mathcal{H}$. This completes the proof. $\square$

Lemma 19. *Let $\nu > 4$. If Conditions 1–5 and 7 are true, then the event, $\mathcal{S} \subset \widehat{\mathcal{S}}_L$, occurs with probability at least $1 - O((N \vee T)^{-\nu+1})$.*

Proof of Lemma 19. For each $i \in [N]$, the Karush-Kuhn-Tucker (KKT) condition of the minimization problem in (2) is given by

$$\mathbf{X} \left(\mathbf{y}_i - \hat{\phi}_i \mathbf{X} \right)' / T = \boldsymbol{\psi}_i \lambda,$$

where $\boldsymbol{\psi}_i = (\psi_{i1}, \dots, \psi_{iN})'$ is defined by $\psi_{ij} = \text{sgn } \hat{\phi}_{ij}$ for $\hat{\phi}_{ij} \neq 0$ and $\psi_{ij} \in [-1, 1]$ for $\hat{\phi}_{ij} = 0$. By the definition of $\mathbf{y}_i$ and some algebra, the KKT condition is equivalently denoted as

$$\boldsymbol{\Omega}(T^{-1} \mathbf{X} \mathbf{X}' - \boldsymbol{\Sigma}_x)(\boldsymbol{\phi}_i - \hat{\boldsymbol{\phi}}_i)' + (\boldsymbol{\phi}_i - \hat{\boldsymbol{\phi}}_i)' + T^{-1} \boldsymbol{\Omega} \mathbf{X} \mathbf{u}_i' = \boldsymbol{\Omega} \boldsymbol{\psi}_i \lambda.$$

We Take the ℓ_∞ -norm. Then by the triangle and Hölder's inequality, we obtain

$$\begin{aligned} \|\boldsymbol{\phi}_i - \hat{\boldsymbol{\phi}}_i\|_\infty &\leq \|T^{-1} \boldsymbol{\Omega} \mathbf{X} \mathbf{u}_i'\|_\infty + \|\boldsymbol{\Omega}(T^{-1} \mathbf{X} \mathbf{X}' - \boldsymbol{\Sigma}_x)(\boldsymbol{\phi}_i - \hat{\boldsymbol{\phi}}_i)'\|_\infty + \|\boldsymbol{\Omega} \boldsymbol{\psi}_i\|_\infty \lambda \\ &\leq \|\boldsymbol{\Omega}\|_\infty \left(\|T^{-1} \mathbf{X} \mathbf{u}_i'\|_\infty + \|T^{-1} \mathbf{X} \mathbf{X}' - \boldsymbol{\Sigma}_x\|_{\max} \|\boldsymbol{\phi}_i - \hat{\boldsymbol{\phi}}_i\|_1 + \lambda \right). \end{aligned}$$

By Proposition 1(b), Condition 4, and Lemmas 3 and 4, we have

$$\|\boldsymbol{\phi}_i - \hat{\boldsymbol{\phi}}_i\|_\infty \lesssim M_\omega \sqrt{T^{-1} \log^3(N \vee T)} =: u_n$$

with probability at least $1 - O((N \vee T)^{-\nu})$.

The event, $\{\mathcal{S}_i \subset \widehat{\mathcal{S}}_i^L\}$, is equivalent to the event that $\hat{\phi}_{ij}^L \neq 0$ holds whenever

$j \in \mathcal{S}_i$. This together with the triangle inequality implies that

$$\begin{aligned}
\mathbb{P}(\mathcal{S}_i \subset \widehat{\mathcal{S}}_i^{\mathbf{L}}) &= \mathbb{P}\left(\min_{j \in \mathcal{S}_i} |\hat{\phi}_{ij}^{\mathbf{L}}| > 0\right) \\
&\geq \mathbb{P}\left(\min_{j \in \mathcal{S}_i} \left\{|\phi_{ij}| - |\hat{\phi}_{ij}^{\mathbf{L}} - \phi_{ij}|\right\} > 0\right) \\
&\geq \mathbb{P}\left(\max_{j \in \mathcal{S}_i} |\hat{\phi}_{ij}^{\mathbf{L}} - \phi_{ij}| < \min_{j \in \mathcal{S}_i} |\phi_{ij}|\right) \\
&\geq \mathbb{P}\left(\|\hat{\phi}_{i\cdot}^{\mathbf{L}} - \phi_{i\cdot}\|_{\infty} < \min_{j \in \mathcal{S}_i} |\phi_{ij}|\right).
\end{aligned}$$

Therefore, if $u_n \ll \min_{j \in \mathcal{S}_i} |\phi_{ij}|$, then $\mathcal{S}_i \subset \widehat{\mathcal{S}}_i^{\mathbf{L}}$ occurs with probability at least $1 - O((N \vee T)^{-\nu})$. This result holds for all $i \in [N]$. Then we can conclude

$$\mathbb{P}(\mathcal{S} \subset \widehat{\mathcal{S}}^{\mathbf{L}}) \geq 1 - \sum_{i \in [N]} \mathbb{P}(\mathcal{S}_i \not\subset \widehat{\mathcal{S}}_i^{\mathbf{L}}) \geq 1 - NO((N \vee T)^{-\nu}).$$

the lower bound converges to one since $\nu > 4$. This establishes the result. $\square$

D Precision Matrix Estimation

The construction of the debiased lasso estimator requires a precision matrix estimator. In a low dimensional setting, it is obtained as the inverse of the sample covariance matrix of $\mathbf{X}$, but it becomes less accurate or even infeasible as the dimensionality tends to large. To this end, we need some regularized estimator. In this paper we employ the CLIME method of [Cai et al. \(2011\)](#) for the estimation of $\boldsymbol{\Omega}$.

Define $\hat{\boldsymbol{\Theta}} = (\hat{\theta}_{ij}) \in \mathbb{R}^{KN \times KN}$ as the solution to the following minimization problem:

$$\min \|\boldsymbol{\Theta}\|_1 \quad \text{subject to} \quad \|\hat{\boldsymbol{\Sigma}}_{x,\varepsilon} \boldsymbol{\Theta} - \mathbf{I}_{KN}\|_{\max} \leq \lambda_1,$$

where λ_1 is a positive regularization parameter and $\hat{\boldsymbol{\Sigma}}_{x,\varepsilon} = \hat{\boldsymbol{\Sigma}}_x + \varepsilon \mathbf{I}_{KN}$ for $\varepsilon \geq 0$. Note

that $\hat{\Sigma}_{x,0} = \hat{\Sigma}_x$. The CLIME estimator of Ω is then obtained by symmetrization with

$$\hat{\omega}_{ij} = \hat{\omega}_{ji} = \hat{\theta}_{ij}1\{|\hat{\theta}_{ij}| \leq |\hat{\theta}_{ji}|\} + \hat{\theta}_{ji}1\{|\hat{\theta}_{ij}| > |\hat{\theta}_{ji}|\}.$$

The non-asymptotic error bound is derived in Lemma 7, which is of independent interest. For related theoretical results of stationary time series, see Shu and Nan (2019). In our numerical analysis, we use `fastclime` by Pang et al. (2014), which is available at <https://github.com/cran/fastclime>.

When the lasso is debiased (desparsified) in a linear regression context, the *nodewise regression* (van de Geer et al., 2014) is frequently used as an alternative. However, it might not be suitable for the VAR models since it repeats the regression of each variable in $\mathbf{X}$ on the others, which includes regressions of a “past” variable on “future” ones.

E Additional Results on Robustification

In this section, we show that $|\mathbf{T}_{ij}|^p$ and $\exp(c|\mathbf{T}_{ij}|^\alpha)$ are UI for some fixed constants $p, c, \alpha > 0$ for arbitrarily fixed $(i, j) \in \mathcal{S}^c$. In pursuit of the goal, we prove

$$\sup_{T, N, K} \mathbb{P}(|\mathbf{T}_{ij}| > x) \lesssim \exp(-\delta x^\alpha + \log x) \quad (\text{A.24})$$

for any (large) $x > 0$ and some $\delta, \alpha > 0$, which is sufficient for $\exp(c|\mathbf{T}_{ij}|^\alpha)$ to be UI for any $c \in [0, \delta)$ by Lemma 21 below. Furthermore, this result implies the UI of $|\mathbf{T}_{ij}|^p$ for any constant $p > 0$ by Lemma 22.

Before starting the proof of (A.24), we assume the following:

- A. Conditions 2–5 in Section 4.
- B. The error term, $\{\mathbf{u}_t\}$, is a sequence of i.i.d. Gaussian random vectors with mean zero and covariance matrix Σ_u .
- C. The parameter space of m_{ij} is lower bounded by some positive constant.

D. $T \geq \{(s_\omega^{4/(1-2r)} M_\omega^{8(1-r)/(1-2r)}) \vee s_i^4\} \log^{10}(N \vee T)$ and N is at most a polynomial order of T .

Condition B is used for establishing Lemma 23, where a concentration inequality for a Gaussian martingale is applied. Condition C prevents $\hat{m}_{ij}$ from being unboundedly close to zero in probability. These two conditions help streamline the proof.

Proof of (A.24). We have

$$\begin{aligned} \mathbb{P}(|\mathbf{T}_{ij}| > x) &= \mathbb{P}\left(\left|\frac{z_{ij} + r_{ij}}{m_{ij}}\right| \left|\frac{m_{ij}}{\hat{m}_{ij}}\right| > x\right) \\ &\leq \mathbb{P}(|z_{ij}/m_{ij}| > x/2) + \mathbb{P}(|r_{ij}| \gtrsim x/2). \end{aligned} \quad (\text{A.25})$$

We bound each probability as an exponential function of x . By Lemmas 23 and 25(b), the first probability of (A.25) is evaluated as

$$\begin{aligned} \mathbb{P}(|z_{ij}/m_{ij}| > x) &\lesssim \exp\left(-\frac{x^2}{2(x+1)}\right) + \mathbb{P}\left(\left\|\hat{\Sigma}_x - \Sigma_x\right\|_{\max} \gtrsim x/M_\omega^2\right) \\ &\lesssim \exp\left(-\frac{x}{2(1+1/x)}\right) + K^2 N^2 T^3 x^2 \exp\left(-\frac{x^{2/3} T^{1/3}}{M_\omega^{4/3}}\right). \end{aligned} \quad (\text{A.26})$$

The second probability of (A.25) is

$$\begin{aligned} &\mathbb{P}(|r_{ij}| \gtrsim x) \\ &\leq \mathbb{P}\left(\left\|\frac{1}{T^{3/4}} \mathbf{u}_i \mathbf{X}'\right\|_\infty \left\|T^{1/4}(\hat{\omega}_j - \omega_j)\right\|_1 + \left\|T^{1/4}(\hat{\phi}_i^L - \phi_i)\right\|_1 \left\|T^{1/4}(\hat{\Sigma}_x \hat{\omega}_j - \mathbf{e}_j)\right\|_\infty \gtrsim x\right) \\ &\leq \mathbb{P}\left(\left\|\frac{1}{T} \mathbf{u}_i \mathbf{X}'\right\|_\infty \gtrsim \frac{x^{1/2}}{T^{1/4}}\right) + \mathbb{P}\left(\left\|\hat{\omega}_j - \omega_j\right\|_1 \gtrsim \frac{x^{1/2}}{T^{1/4}}\right) \\ &\quad + \mathbb{P}\left(\left\|\hat{\phi}_i^L - \phi_i\right\|_1 \gtrsim \frac{x^{1/2}}{T^{1/4}}\right) + \mathbb{P}\left(\left\|\hat{\Sigma}_x \hat{\omega}_j - \mathbf{e}_j\right\|_\infty \gtrsim \frac{x^{1/2}}{T^{1/4}}\right). \end{aligned}$$

By Lemma 24, the last three terms are evaluated as

$$\begin{aligned}
& \mathbb{P} \left(\|\hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j\|_1 \gtrsim \frac{x^{1/2}}{T^{1/4}} \right) + \mathbb{P} \left(\left\| \hat{\boldsymbol{\phi}}_{i\cdot}^{\mathbf{L}} - \boldsymbol{\phi}_{i\cdot} \right\|_1 \gtrsim \frac{x^{1/2}}{T^{1/4}} \right) + \mathbb{P} \left(\left\| \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j - \mathbf{e}_j \right\|_\infty \gtrsim \frac{x^{1/2}}{T^{1/4}} \right) \\
& \lesssim \mathbb{P} \left(\left\| \frac{1}{T} \mathbf{u}_i \cdot \mathbf{X}' \right\|_\infty \gtrsim \frac{x^{1/(2-2r)}}{T^{1/(4-4r)} s_\omega^{1/(1-r)} M_\omega^2} \right) + \mathbb{P} \left(\left\| \hat{\boldsymbol{\Sigma}}_x - \boldsymbol{\Sigma}_x \right\|_{\max} > \frac{x^{1/(2-2r)}}{T^{1/(4-4r)} s_\omega^{1/(1-r)} M_\omega^2} \right) \\
& \quad + \mathbb{P} \left(\left\| \frac{1}{T} \mathbf{u}_i \cdot \mathbf{X}' \right\|_\infty \gtrsim \frac{x^{1/2}}{T^{1/4} (s_i \vee M_\omega)} \right) + \mathbb{P} \left(\left\| \hat{\boldsymbol{\Sigma}}_x - \boldsymbol{\Sigma}_x \right\|_{\max} \gtrsim \frac{x^{1/2}}{T^{1/4} (s_i \vee M_\omega)} \right).
\end{aligned}$$

Thus by Lemma 25, we obtain

$$\begin{aligned}
& \mathbb{P}(|r_{ij}| \gtrsim x) \\
& \lesssim K^2 N^2 T^3 x \left\{ \exp \left(-\frac{x^{1/(3-3r)} T^{(1-2r)/(6-6r)}}{s_\omega^{2/(3-3r)} M_\omega^{4/3}} \right) + \exp \left(-\frac{x^{1/3} T^{1/6}}{(s_i \vee M_\omega)^{2/3}} \right) \right\}.
\end{aligned} \tag{A.27}$$

Combining (A.25) with (A.26) and (A.27) gives

$$\begin{aligned}
& \mathbb{P}(|\mathbf{T}_{ij}| \gtrsim x) \lesssim \exp \left(-\frac{x}{2(1+1/x)} \right) + K^2 N^2 T^3 x^2 \exp \left(-\frac{x^{2/3} T^{1/3}}{M_\omega^{4/3}} \right) \\
& \quad + K^2 N^2 T^3 x \left\{ \exp \left(-\frac{x^{1/(3-3r)} T^{(1-2r)/(6-6r)}}{s_\omega^{2/(3-3r)} M_\omega^{4/3}} \right) + \exp \left(-\frac{x^{1/3} T^{1/6}}{(s_i \vee M_\omega)^{2/3}} \right) \right\} \\
& \lesssim K^2 N^2 T^3 x \left\{ \exp \left(-\frac{x^{1/(3-3r)} T^{(1-2r)/(6-6r)}}{s_\omega^{2/(3-3r)} M_\omega^{4/3}} \right) + \exp \left(-\frac{x^{1/3} T^{1/6}}{(s_i \vee M_\omega)^{2/3}} \right) \right\}.
\end{aligned}$$

Under the condition, $T \geq \{(s_\omega^{4/(1-2r)} M_\omega^{8(1-r)/(1-2r)}) \vee s_i^4\} \log^{10}(N \vee T)$, the upper bound reduces to

$$\mathbb{P}(|\mathbf{T}_{ij}| \gtrsim x) \lesssim \exp \left(-x^{1/3} \log^{1+\varepsilon}(N \vee T) + \log x + 7 \log(N \vee T) \right)$$

for some $\varepsilon > 0$. Taking the supremum over (N, T, K) with the assumed condition yields

$$\sup_{N, T, K} \mathbb{P}(|\mathbf{T}_{ij}| \gtrsim x) \lesssim \exp \left(-x^{1/3} + \log x \right).$$

This completes the proof. $\square$

E.1 Lemmas and their proofs

Lemma 20. *Any positive random sequence $\{X_n\}$ that satisfies*

$$\sup_n \mathbb{P}(X_n > x) \lesssim x^{-r} \log^s x$$

for some $r > 1$ and $s > 0$ is UI.

Proof of Lemma 20. For any $\epsilon \in (1, r)$, we have

$$\sup_n \mathbb{E}[X_n^\epsilon] = \sup_n \int_0^\infty \mathbb{P}(X_n > x^{1/\epsilon}) dx \lesssim 1 + \epsilon^{-s} \int_1^\infty x^{-r/\epsilon} \log^s x dx.$$

Since $r/\epsilon > 1$, the integral is finite. This condition implies that $\{X_n\}$ is UI; see (Billingsley, 1999, pp. 31–32). This completes the proof. $\square$

Lemma 21. *For any positive random sequence $\{X_n\}$, if*

$$\mathbb{P}(X_n > x) \lesssim \exp(-\delta x^\alpha + \log x)$$

holds for any $x > 0$ and some $\alpha, \delta > 0$, then $\exp(cX_n^\alpha)$ is UI for any $c \in [0, \delta)$.

Proof of Lemma 21. We have

$$\begin{aligned} \mathbb{P}(\exp(cX_n^\alpha) > x) &= \mathbb{P}(X_n > (\log x^{1/c})^{1/\alpha}) \\ &\lesssim \exp(-\delta \log x^{1/c} + \log(\log x^{1/c})^{1/\alpha}) \asymp x^{-\delta/c} \log^{1/\alpha} x. \end{aligned}$$

Since $c \in [0, \delta)$ is assumed, we have $\delta/c > 1$. Thus Lemma 20 establishes the UI of $\exp(cX_n^\alpha)$. This completes the proof. $\square$

Lemma 22. *For any positive random sequence $\{X_n\}$, if $\exp(cX_n^\alpha)$ is UI for given constants $c, \alpha > 0$, then X_n^p is also UI for any constant $p > 0$.*

Proof of Lemma 22. For any $x > 0$, we have $x^p \leq (2p/c)^{p/\alpha} \exp(cx^\alpha)$. Thus, for any $M > 0$, we obtain

$$\begin{aligned} & \sup_n \mathbb{E} [X_n^p 1\{X_n^p \geq M\}] \\ & \leq \sup_n \mathbb{E} [(2p/c)^{p/\alpha} \exp(cX_n^\alpha) 1\{(2p/c)^{p/\alpha} \exp(cX_n^\alpha) \geq M\}]. \end{aligned}$$

Since $(2p/c)^{p/\alpha}$ is a constant, and hence $(2p/c)^{p/\alpha} \exp(cX_n^\alpha)$ is UI by the assumption, the upper bound can be made arbitrarily small by taking large M . This establishes the UI of X_n^p . $\square$

Lemma 23. *For any $x > 0$, we have*

$$\mathbb{P}(|z_{ij}/m_{ij}| > x) \lesssim \exp\left(-\frac{x^2}{2(x+1)}\right) + \mathbb{P}\left(\left\|\hat{\Sigma}_x - \Sigma_x\right\|_{\max} > \omega_j x / M_\omega^2\right).$$

Proof of Lemma 23. Define

$$\begin{aligned} M_T &= z_{ij}/m_{ij} = \frac{1}{\sqrt{T}} \sum_{t=1}^T \frac{u_{it} \mathbf{x}'_t \boldsymbol{\omega}_j}{m_{ij}}, \\ \langle M \rangle_T &= \frac{1}{T} \sum_{t=1}^T \mathbb{E} \left[\frac{u_{it}^2 (\mathbf{x}'_t \boldsymbol{\omega}_j)^2}{m_{ij}^2} \mid \mathcal{F}_{t-1} \right] = \frac{\boldsymbol{\omega}'_j \hat{\Sigma}_x \boldsymbol{\omega}_j}{\boldsymbol{\omega}'_j \Sigma_x \boldsymbol{\omega}_j}. \end{aligned}$$

By (Bercu et al., 2015, Theorem 3.25) under the assumed Gaussianity of u_{it} , we have

$$\begin{aligned} \mathbb{P}(M_T > x) &\leq \mathbb{P}(M_T > x, \langle M \rangle_T \leq x+1) + \mathbb{P}(\langle M \rangle_T > x+1) \\ &\leq \exp\left(-\frac{x^2}{2(x+1)}\right) + \mathbb{P}\left(\frac{\boldsymbol{\omega}'_j \hat{\Sigma}_x \boldsymbol{\omega}_j}{\boldsymbol{\omega}'_j \Sigma_x \boldsymbol{\omega}_j} > x+1\right). \end{aligned}$$

By Hölder's inequality and $\|\boldsymbol{\omega}_j\|_1 \leq M_\omega$ in Condition 4, the second probability of this

upper bound is

$$\begin{aligned} \mathbb{P} \left(\frac{\boldsymbol{\omega}'_j \hat{\boldsymbol{\Sigma}}_x \boldsymbol{\omega}_j}{\boldsymbol{\omega}'_j \boldsymbol{\Sigma}_x \boldsymbol{\omega}_j} > x + 1 \right) &= \mathbb{P} \left(\boldsymbol{\omega}'_j \left(\hat{\boldsymbol{\Sigma}}_x - \boldsymbol{\Sigma}_x \right) \boldsymbol{\omega}_j > \omega_j x \right) \\ &\leq \mathbb{P} \left(\left\| \hat{\boldsymbol{\Sigma}}_x - \boldsymbol{\Sigma}_x \right\|_{\max} > \omega_j x / M_\omega^2 \right) \end{aligned}$$

This completes the proof. $\square$

Lemma 24. *For any $\lambda > 0$, we have the following:*

$$\begin{aligned} (a) \quad & \mathbb{P} \left(\left\| \hat{\boldsymbol{\omega}}_j - \boldsymbol{\omega}_j \right\|_1 > \lambda \right) \\ & \lesssim \mathbb{P} \left(\left\| \frac{1}{T} \mathbf{u}_i \mathbf{X}' \right\|_{\infty} \gtrsim \frac{\lambda^{1/(1-r)}}{s_\omega^{1/(1-r)} M_\omega^2} \right) + \mathbb{P} \left(\left\| \hat{\boldsymbol{\Sigma}}_x - \boldsymbol{\Sigma}_x \right\|_{\max} > \frac{\lambda^{1/(1-r)}}{s_\omega^{1/(1-r)} M_\omega^2} \right), \\ (b) \quad & \mathbb{P} \left(\left\| \hat{\boldsymbol{\phi}}_i^L - \boldsymbol{\phi}_i \right\|_1 > \lambda \right) \\ & \lesssim \mathbb{P} \left(\left\| \frac{1}{T} \mathbf{u}_i \mathbf{X}' \right\|_{\infty} \gtrsim \frac{\lambda}{s_i} \right) + \mathbb{P} \left(\left\| \hat{\boldsymbol{\Sigma}}_x - \boldsymbol{\Sigma}_x \right\|_{\max} \gtrsim \frac{\lambda}{s_i} \right), \\ (c) \quad & \mathbb{P} \left(\left\| \hat{\boldsymbol{\Sigma}}_x \hat{\boldsymbol{\omega}}_j - \mathbf{e}_j \right\|_{\infty} > \lambda \right) \\ & \lesssim \mathbb{P} \left(\left\| \frac{1}{T} \mathbf{u}_i \mathbf{X}' \right\|_{\infty} \gtrsim \frac{\lambda}{M_\omega} \right) + \mathbb{P} \left(\left\| \hat{\boldsymbol{\Sigma}}_x - \boldsymbol{\Sigma}_x \right\|_{\max} \gtrsim \frac{\lambda}{M_\omega} \right). \end{aligned}$$

Proof of Lemma 24. The proofs of (a) and (c) complete by the proof of Proposition 2.

Similarly, the proof of Proposition 1 gives the proof of (b). $\square$

Lemma 25. *For any large $y > 0$, we have the following:*

$$\begin{aligned} (a) \quad & \mathbb{P} \left(\left\| \frac{1}{T} \mathbf{u}_i \mathbf{X}' \right\|_{\infty} > y \right) \lesssim K N^2 T^2 y \exp \left(-y^{2/3} T^{1/3} \right), \\ (b) \quad & \mathbb{P} \left(\left\| \hat{\boldsymbol{\Sigma}}_x - \boldsymbol{\Sigma}_x \right\|_{\max} > y \right) \lesssim K^2 N^2 T^3 y^2 \exp \left(-y^{2/3} T^{1/3} \right). \end{aligned}$$

Proof of Lemma 25. In Lemma 3, setting $\bar{c}_1 = y^{2/3}T^{1/3}$ and $r_T = yT$ yields

$$\begin{aligned} & \mathbb{P} \left(\left\| \frac{1}{T} \mathbf{u}_i \mathbf{X}' \right\|_{\infty} > y \right) \\ & \lesssim K N^2 \exp(-y^{2/3}T^{1/3}) + K N^2 T^2 y \exp(-y^{2/3}T^{1/3}) + K T^{2/3} \frac{\delta^{yT} \log N}{y^{2/3}} \\ & \lesssim K N^2 T^2 y \exp(-y^{2/3}T^{1/3}), \end{aligned}$$

where the last inequality holds for any $\delta \in (0, 1)$ as long as $y > 0$ is sufficiently large.

Similarly in Lemma 4, setting $\bar{c}_2 = y^{2/3}T^{1/3}$ and $r_T = yT$ yields

$$\begin{aligned} & \mathbb{P} \left(\left\| \hat{\Sigma}_x - \Sigma_x \right\|_{\max} > y \right) \\ & \lesssim K^2 N^2 T^2 y^2 \exp(-y^{2/3}T^{1/3}) + K^2 N^2 T^3 y^2 \exp(-y^{2/3}T^{1/3}) + K^2 \frac{\delta^{yT} \log N}{y^{2/3}T^{1/3}} \\ & \lesssim K^2 N^2 T^3 y^2 \exp(-y^{2/3}T^{1/3}). \end{aligned}$$

This completes the proof. □

F Additional Experimental Results

First, we report the results of a multiple test at $q = 0.1$ in Figure F1 to the results at $q = 0.2$ in Figure 3(d). Using the larger value of q , the overall red and blue color in Figure F1 becomes weaker than in 3(d), suggesting that all the elements tend to be selected less frequently.

Figure F2 shows a heatmap of Φ used for Table 1 over three different degrees of sparsity (controlled by d), given N . Figure F3 shows the Φ in the additional design. As can be seen from the cross shape, the new design differs from the first design in the following aspects: (i) more intensive clustering with fewer nodes; (ii) higher sparsity. We believe that considering these two different networks sheds light on the robustness properties of our inference method for different network structures.

Now we report the results with Φ_1 design 2 corresponding to Tables 1 and 2. The

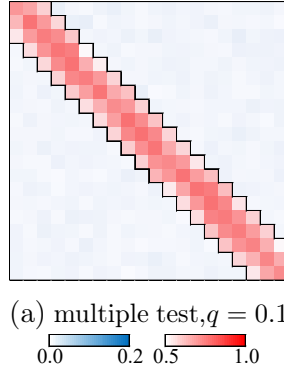


Figure F1: Heatmap of selection frequencies in Φ_1

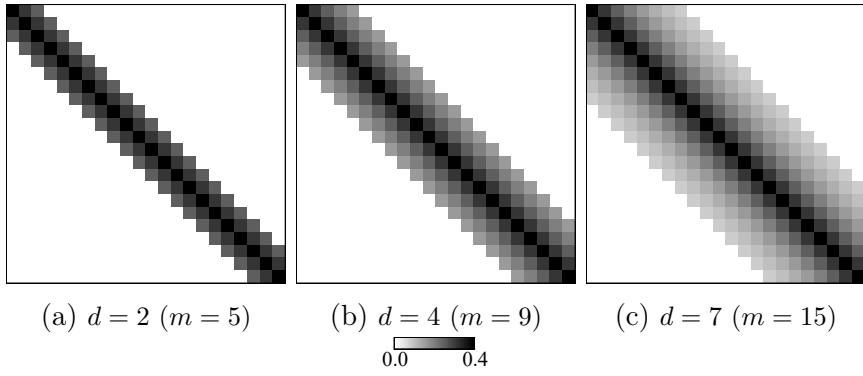


Figure F2: Heatmap of absolute value of Φ_1 , Design 1

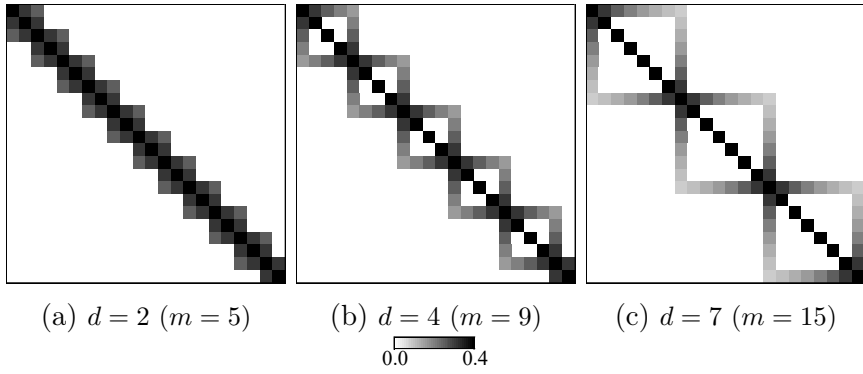


Figure F3: Heatmap of absolute value of Φ_1 , Design 2

DGPs are identical to the ones used in Tables 1 and 2, except for Φ_1 . The results are reported in Tables F1 and F2 below. As can be seen, the results are qualitatively very similar to those in Tables 1 and 2, in the sense that the asymptotic threshold often fails to control the FDR for mixture normal errors, but the bootstrap and e-BH thresholds control much better. The e-BH has a slight loss of power compared to the

bootstrap threshold due to the conservative type I error.

Table F1: Directional FDR and power using asymptotic and bootstrap thresholds for $q = 0.1$, with cross-sectionally uncorrelated errors, Φ_1 Design 2

$m = 2$ ($\max s_i = 5$)									
$T = 200$					$T = 300$				
	asymptotic		bootstrap		asymptotic		bootstrap		
	dFDR	PWR	dFDR	PWR	dFDR	PWR	dFDR	PWR	
Standard Normal Error									
$N = 50$	10.3	98.3	7.6	97.8	9.5	99.9	7.7	99.8	
$N = 100$	11.5	96.2	7.6	95.2	10.4	99.6	7.9	99.5	
$N = 200$	11.8	93.9	7.3	92.2	11.2	99.3	8.1	99.1	
$N = 300$	11.0	92.0	6.3	89.2	10.6	99.0	8.2	98.8	
Mixture Normal Error									
$N = 50$	13.5	95.5	7.4	93.5	11.7	99.4	7.7	99.1	
$N = 100$	15.5	92.1	6.8	88.1	13.5	98.4	7.7	97.7	
$N = 200$	14.3	89.4	6.4	84.8	13.7	97.7	7.5	96.7	
$N = 300$	16.2	87.0	5.3	79.2	14.1	97.1	7.7	96.0	
$m = 4$ ($\max s_i = 9$)									
$T = 200$					$T = 300$				
	asymptotic		bootstrap		asymptotic		bootstrap		
	dFDR	PWR	dFDR	PWR	dFDR	PWR	dFDR	PWR	
Standard Normal Error									
$N = 50$	10.5	88.2	7.2	86.0	9.7	96.2	7.6	95.6	
$N = 100$	11.1	83.0	7.4	80.5	10.2	93.7	7.8	92.9	
$N = 200$	10.8	77.6	7.2	74.9	10.3	91.1	8.2	90.2	
$N = 300$	11.5	73.0	5.8	67.7	10.4	88.6	8.1	87.6	
Mixture Normal Error									
$N = 50$	14.7	83.6	7.4	77.7	12.2	93.9	7.8	92.1	
$N = 100$	16.2	78.7	7.2	72.3	13.9	90.5	8.2	87.9	
$N = 200$	15.8	73.8	6.5	66.1	13.7	87.8	7.9	84.8	
$N = 300$	18.6	69.5	4.9	56.3	15.7	85.1	7.9	81.0	
$m = 7$ ($\max s_i = 15$)									
$T = 200$					$T = 300$				
	asymptotic		bootstrap		asymptotic		bootstrap		
	dFDR	PWR	dFDR	PWR	dFDR	PWR	dFDR	PWR	
Standard Normal Error									
$N = 50$	11.5	64.6	7.0	61.0	10.4	75.5	7.5	73.3	
$N = 100$	11.9	61.7	7.6	58.8	10.8	72.8	8.2	71.1	
$N = 200$	12.2	54.6	6.2	50.6	11.7	67.5	7.5	65.1	
$N = 300$	12.3	51.0	4.7	44.9	11.0	64.0	7.6	62.0	
Mixture Normal Error									
$N = 50$	19.5	62.3	2.0	44.2	15.3	72.9	8.3	68.0	
$N = 100$	20.1	60.0	6.5	49.7	16.2	70.2	9.1	65.9	
$N = 200$	18.7	51.9	4.2	39.0	16.3	64.1	7.2	58.6	
$N = 300$	22.7	49.6	1.4	29.5	18.7	61.6	7.6	55.6	

Table F2: FDR and power using e-BH thresholds for $q = 0.1$, with cross-sectionally uncorrelated errors, Φ_1 Design 2

$m = 2$ ($\max s_i = 5$)									
$f(x)$	$T = 200$				$T = 300$				
	$ x ^p$		$\exp(c x ^\alpha)$		$ x ^p$		$\exp(c x ^\alpha)$		
	FDR	PWR	FDR	PWR	FDR	PWR	FDR	PWR	
Standard Normal Error									
$N = 50$	1.9	95.2	1.3	94.1	1.4	99.4	0.9	99.1	
$N = 100$	2.2	91.4	1.5	90.0	1.7	98.7	1.1	98.3	
$N = 200$	2.0	86.6	1.4	85.0	1.6	97.8	1.1	97.4	
$N = 300$	1.3	82.1	1.1	80.4	1.2	96.9	0.9	96.4	
Mixture Normal Error									
$N = 50$	3.1	90.1	2.3	88.5	2.3	97.9	1.6	97.4	
$N = 100$	3.8	84.7	2.7	82.8	2.8	96.0	2.0	95.3	
$N = 200$	2.6	78.8	2.0	76.8	2.4	94.2	1.8	93.4	
$N = 300$	2.6	74.6	2.1	72.9	2.1	92.8	1.6	91.9	
$m = 4$ ($\max s_i = 9$)									
$f(x)$	$T = 200$				$T = 300$				
	$ x ^p$		$\exp(c x ^\alpha)$		$ x ^p$		$\exp(c x ^\alpha)$		
	FDR	PWR	FDR	PWR	FDR	PWR	FDR	PWR	
Standard Normal Error									
$N = 50$	2.1	78.3	1.4	76.0	1.6	91.2	1.1	89.9	
$N = 100$	2.0	72.2	1.4	70.1	1.6	87.5	1.1	86.0	
$N = 200$	1.4	64.4	1.0	62.4	1.3	83.3	0.9	81.6	
$N = 300$	1.3	58.3	1.0	56.7	1.1	79.0	0.8	77.5	
Mixture Normal Error									
$N = 50$	3.8	72.6	2.8	70.0	2.8	87.4	2.0	85.6	
$N = 100$	4.0	67.4	2.9	64.9	3.1	82.9	2.1	81.0	
$N = 200$	2.9	59.5	2.2	57.3	2.4	78.3	1.7	76.4	
$N = 300$	3.0	53.7	2.5	52.2	2.5	74.0	1.9	72.6	
$m = 7$ ($\max s_i = 15$)									
$f(x)$	$T = 200$				$T = 300$				
	$ x ^p$		$\exp(c x ^\alpha)$		$ x ^p$		$\exp(c x ^\alpha)$		
	FDR	PWR	FDR	PWR	FDR	PWR	FDR	PWR	
Standard Normal Error									
$N = 50$	2.1	54.0	1.5	51.9	1.7	65.7	1.1	63.4	
$N = 100$	1.9	51.5	1.3	49.6	1.7	63.1	1.1	61.2	
$N = 200$	1.5	43.4	1.1	42.2	1.5	57.6	1.1	56.2	
$N = 300$	1.1	39.1	0.9	38.5	0.9	53.2	0.7	52.4	
Mixture Normal Error									
$N = 50$	5.4	51.4	4.0	49.4	3.7	62.6	2.6	60.5	
$N = 100$	5.1	49.2	3.9	47.4	3.7	60.2	2.7	58.2	
$N = 200$	3.3	39.5	2.7	38.3	2.8	53.2	2.1	51.7	
$N = 300$	3.6	36.6	3.2	36.0	2.8	50.2	2.4	49.4	

F.1 Results with $\hat{m}_{ij} = \hat{\sigma}_i \hat{\omega}_j$

The results based on $\hat{m}_{ij} = \hat{\sigma}_i \hat{\omega}_j$ are reported in Table F3. The DGP is identical to the one used in Table 1. The results with this $\hat{m}_{ij}$ are shown in Table F3. The e-BH result omitted. Bootstrap dFDR is virtually identical to bootstrap dFDR based on $\hat{m}_{ij} = \hat{\sigma}_i \sqrt{\hat{\omega}_j' \hat{\Sigma}_x \hat{\omega}_j}$, whilst asymptotic dFDR often becomes more conservative than bootstrap dFDR. As a result, the power is often lower than the bootstrap one. These results suggest that Procedure 2 is expected to provide stable performance regardless of the choice of $\hat{m}_{ij}$.

Table F3: Directional FDR and power using asymptotic and bootstrap thresholds for $q = 0.1$, with cross-sectionally uncorrelated errors, $\hat{m}_{ij} = \hat{\sigma}_i \hat{\omega}_j$

$m = 2$ ($\max s_i = 5$)								
	$T = 200$				$T = 300$			
	asymptotic		bootstrap		asymptotic		bootstrap	
	dFDR	PWR	dFDR	PWR	dFDR	PWR	dFDR	PWR
Standard Normal Error								
$N = 50$	6.0	96.6	6.7	96.9	6.0	99.7	6.9	99.7
$N = 100$	6.3	92.8	6.9	93.2	6.1	99.1	7.3	99.2
$N = 200$	6.0	89.6	7.5	90.7	6.0	98.5	8.2	98.8
$N = 300$	5.7	86.7	6.7	87.4	5.5	98.0	8.5	98.4
Mixture Normal Error								
$N = 50$	6.8	92.7	5.8	92.0	6.6	98.7	6.2	98.7
$N = 100$	7.7	86.8	5.6	84.7	7.3	97.2	6.5	96.9
$N = 200$	8.5	84.2	6.7	82.5	7.7	96.3	7.7	96.3
$N = 300$	8.4	80.8	5.5	77.3	7.7	95.1	7.7	95.1
$m = 4$ ($\max s_i = 9$)								
	$T = 200$				$T = 300$			
	asymptotic		bootstrap		asymptotic		bootstrap	
	dFDR	PWR	dFDR	PWR	dFDR	PWR	dFDR	PWR
Standard Normal Error								
$N = 50$	5.8	87.1	6.7	87.9	5.8	96.0	6.8	96.4
$N = 100$	5.8	81.2	8.3	83.6	5.9	93.7	8.4	94.8
$N = 200$	5.9	75.3	9.3	78.3	5.6	90.5	9.4	92.2
$N = 300$	6.1	69.3	8.5	71.7	5.6	87.3	9.7	89.5
Mixture Normal Error								
$N = 50$	6.0	82.9	5.8	82.6	5.9	93.8	6.1	94.0
$N = 100$	6.4	77.3	7.1	78.1	6.3	91.1	7.6	91.8
$N = 200$	7.2	71.6	8.2	72.6	6.8	87.2	9.0	88.5
$N = 300$	7.4	65.8	6.8	65.1	6.8	83.7	8.9	85.0
$m = 7$ ($\max s_i = 15$)								
	$T = 200$				$T = 300$			
	asymptotic		bootstrap		asymptotic		bootstrap	
	dFDR	PWR	dFDR	PWR	dFDR	PWR	dFDR	PWR
Standard Normal Error								
$N = 50$	5.1	66.4	6.0	67.8	5.2	78.8	6.1	79.9
$N = 100$	5.6	58.1	8.0	61.0	5.6	72.3	8.0	74.7
$N = 200$	5.7	50.2	8.8	53.3	5.4	65.4	9.1	68.7
$N = 300$	5.8	47.1	8.8	49.8	5.2	62.6	9.8	66.2
Mixture Normal Error								
$N = 50$	5.0	63.1	4.9	62.9	4.9	76.2	5.1	76.5
$N = 100$	5.9	55.5	6.7	56.6	5.9	69.6	7.2	71.0
$N = 200$	7.5	48.2	7.5	48.0	6.6	62.8	8.1	64.3
$N = 300$	6.8	45.7	7.3	46.2	6.2	60.4	9.0	62.8

F.2 Results with cross-sectionally correlated errors

We have chosen the subset of the design in Table 1, $\{m = 2, T = 200, \text{mixture normal errors}\}$, which gives the worst FDR control by the asymptotic threshold therein. The DGP is identical to that used in Tables 1 and 2, except that the errors are cross-sectionally correlated.

We consider three designs. In the first design, for a given cross section unit i , the maximum number of cross-correlated errors is an integer part of $2N^{0.6} - 1$. Specifically, the cross-correlated errors are generated as follows: $\Sigma_u = (\sigma_{ij})$ with $\sigma_{ij} = k \cdot 1\{|i - j + 1| \leq \varphi\}$ with $k = i/j$ for $i \leq j$ and $k = j/i$ for $i \geq j$, and φ is an integer part of N^η . We set $\eta = 0.6$. The results are reported in Table F4. The performance of the asymptotic and bootstrap thresholds, as well as that of the e-BH thresholds, is very similar to the case with cross-sectionally uncorrelated errors reported in Table 1. For $q = 0.05$, the comparative analysis of the results from different thresholds is very similar to that for $q = 0.10$.

The second and the third designs are more challenging with respect to Condition 6, in which all cross section units are correlated. In the second design, $\sigma_{ij} = \rho_u^{|i-j|}$ with $\rho_u = 0.8$. In the third design, we consider the correlation implied by an error factor model, $u_{ti} = (f_t b_i + v_{ti}) / \sqrt{\sigma_f^2 + 1}$, where $f_t \sim IIDN(0, \sigma_f^2)$ and $v_{ti} \sim IIDN(0, 1)$ are drawn for each replication but $b_i \sim IIDU(-\sqrt{3}, \sqrt{3})$ is fixed over the replications. This structure gives pervasive cross-correlations of u_{ti} , as $\sigma_{ij} = E[u_{ti} u_{tj}] = \varphi_{\sigma_f^2} b_i b_j$ for $i \neq j$ and $\varphi_{\sigma_f^2} (b_i^2 + 1)$ for $i = j$, given $\mathbf{b} = (b_1, \dots, b_N)'$ and σ_f^2 , where $\varphi_{\sigma_f^2} = \sigma_f^2 / (\sigma_f^2 + 1)$. Rather than generating u_{ti} by the factor model, we generate $\mathbf{u}_t = \Sigma_u^{1/2} \boldsymbol{\varepsilon}_t$, where $\boldsymbol{\varepsilon}_t \sim IIDN(\mathbf{0}, \mathbf{I}_N)$, constructing Σ_u using $\mathbf{b}$ and $\sigma_f^2 = 1/4$. The corresponding results are summarized in Tables F5 and F6, respectively. As can be seen, for $T = 200$, both the asymptotic and the bootstrap dFDR exceed the level $q = 0.1$ and deviate further as N increases, while both versions of the e-BH thresholds control the FDR much better, keeping it well below 10%.

Table F4: FDR and power using asymptotic, bootstrap and e-BH thresholds for $q = 0.10, 0.05$, for a given i the maximum number of cross-correlated errors is $2N^{0.6} - 1$

$m = 2$ ($\max s_i = 5$), $T = 200$, Mixture normal, cross-correlated errors								
	asymptotic		bootstrap		$ x ^p$		$\exp(c x ^\alpha)$	
	dFDR	PWR	dFDR	PWR	FDR	PWR	FDR	PWR
$q = 0.10$								
$N = 50$	9.6	90.1	6.5	88.0	1.9	80.0	1.4	77.3
$N = 100$	10.2	83.9	6.5	80.6	1.9	70.6	1.3	67.2
$N = 200$	13.2	79.0	7.9	75.0	2.1	64.7	1.6	62.1
$N = 300$	14.5	77.0	7.4	71.6	2.1	62.1	1.6	60.2
$q = 0.05$								
$N = 50$	5.2	86.5	2.1	76.3	1.0	74.7	0.7	71.6
$N = 100$	5.5	79.4	2.9	74.5	0.8	63.5	0.6	60.5
$N = 200$	7.5	74.7	3.6	68.8	0.9	57.6	0.8	56.2
$N = 300$	8.3	72.8	3.2	64.0	0.8	54.9	0.8	54.5

Table F5: FDR and power using asymptotic, bootstrap and e-BH thresholds for $q = 0.10$, where $\sigma_{ij} = \rho_u^{|i-j|}$ with $\rho_u = 0.8$

$m = 2$ ($\max s_i = 5$), $T = 200$, Mixture normal, cross-correlated errors								
	asymptotic		bootstrap		$ x ^p$		$\exp(c x ^\alpha)$	
	dFDR	PWR	dFDR	PWR	FDR	PWR	FDR	PWR
$q = 0.10$								
$N = 50$	10.9	71.8	10.3	71.2	2.1	53.5	1.4	49.0
$N = 100$	13.0	66.3	11.6	65.2	2.4	48.5	1.7	44.7
$N = 200$	16.3	61.7	15.1	60.9	2.8	45.7	2.3	43.6
$N = 300$	17.6	63.1	15.4	61.8	3.0	47.7	2.5	46.4

Table F6: FDR and power using asymptotic, bootstrap and e-BH thresholds for $q = 0.10$, with a factor model, $u_{ti} = (f_t b_i + v_{ti}) / \sqrt{\sigma_f^2 + 1}$

$m = 2$ ($\max s_i = 5$), $T = 200$, Mixture normal, cross-correlated errors								
	asymptotic		bootstrap		$ x ^p$		$\exp(c x ^\alpha)$	
	dFDR	PWR	dFDR	PWR	FDR	PWR	FDR	PWR
$q = 0.10$								
$N = 50$	10.6	93.3	7.5	91.8	2.2	85.2	1.6	82.9
$N = 100$	14.5	87.7	10.0	85.2	3.1	76.6	2.1	73.7
$N = 200$	23.3	83.6	15.6	80.3	4.8	70.7	3.6	68.2
$N = 300$	30.3	81.1	17.1	75.4	5.8	66.4	4.6	64.3

G List of Variable Names in Empirical Applications

Table G1: Full list of FRED-MD macroeconomic variables.

FREDMD ID and Variable Descriptions			
Group 1: Output and Income		Group 5: Interest and Exchange Rate	
1	Real Personal Income	68	Federal Funds Rate
2	Real personal income ex transfer receipts	69	3-Month AA Financial Commercial Paper Rate
3	IP Index	70	3-Month Treasury Bill
4	IP: Final Products and Nonindustrial Supplies	71	6-Month Treasury Bill
5	IP: Final Products (Market Group)	72	1-Year Treasury Rate
6	IP: Consumer Goods	73	5-Year Treasury Rate
7	IP: Durable Consumer Goods	74	10-Year Treasury Rate
8	IP: Nondurable Consumer Goods	75	Moody's Seasoned Aaa Corporate Bond Yield Aaa bond
9	IP: Business Equipment	76	Moody's Seasoned Baa Corporate Bond Yield Baa bond
10	IP: Materials	77	3-Month Commercial Paper Minus FEDFUNDS CP-FF spread
11	IP: Durable Materials	78	3-Month Treasury C Minus FEDFUNDS 3 mo-FF spread
12	IP: Nondurable Materials	79	6-Month Treasury C Minus FEDFUNDS 6 mo-FF spread
13	IP: Manufacturing (SIC)	80	1-Year Treasury C Minus FEDFUNDS 1 yr-FF spread
14	IP: Residential Utilities	81	5-Year Treasury C Minus FEDFUNDS 5 yr-FF spread
15	IP: Fuels	82	10-Year Treasury C Minus FEDFUNDS 10 yr-FF spread
16	Capacity Utilization: Manufacturing	83	Moody's Aaa Corporate Bond Minus FEDFUNDS
Group 2: Labor Market		84	Moody's Baa Corporate Bond Minus FEDFUNDS
17	Help-Wanted Index for United States	85	Trade Weighted U.S. Dollar Index: Major Currencies Ex rate
18	Ratio of Help Wanted/No. Unemployed	86	Switzerland / U.S. Foreign Exchange Rate
19	Civilian Labor Force	87	Japan / U.S. Foreign Exchange Rate
20	Civilian Employment	88	U.S. / U.K. Foreign Exchange Rate
21	Civilian Unemployment Rate	89	Canada / U.S. Foreign Exchange Rate
22	Average Duration of Unemployment (Weeks)	Group 6: Prices	
23	Civilians Unemployed -Less Than 5 Weeks	90	PPI: Finished Goods
24	Civilians Unemployed for 5-14 Weeks	91	PPI: Finished Consumer Goods
25	Civilians Unemployed -15 Weeks & Over	92	PPI: Intermediate Materials
26	Civilians Unemployed for 15-26 Weeks	93	PPI: Crude Materials
27	Civilians Unemployed for 27 Weeks and Over	94	Crude Oil
28	Initial Claims	95	PPI: Metals and metal products:
29	All Employees: Total nonfarm	96	CPI : All Items
30	All Employees: Goods-Producing Industries	97	CPI : Apparel
31	All Employees: Mining and Logging: Mining	98	CPI : Transportation
32	All Employees: Construction	99	CPI : Medical Care
33	All Employees: Manufacturing	100	CPI : Commodities
34	All Employees: Durable goods	101	CPI : Durables
35	All Employees: Nondurable goods	102	CPI : Services
36	All Employees: Service-Providing Industries	103	CPI : All Items Less Food
37	All Employees: Trade	104	CPI : All items less shelter
38	All Employees: Wholesale Trade	105	CPI : All items less medical care
39	All Employees: Retail Trade	106	Personal Cons. Expend.: Chain Index
40	All Employees: Financial Activities	107	Personal Cons. Exp: Durable goods
41	All Employees: Government	108	Personal Cons. Exp: Nondurable goods
42	Avg Weekly Hours : Goods-Producing	109	Personal Cons. Exp: Services
43	Avg Weekly Overtime Hours : Manufacturing	Group 7: Money and Credit	
44	Avg Weekly Hours : Manufacturing	110	M1 Money Stock
45	Avg Hourly Earnings : Goods-Producing	111	M2 Money Stock
46	Avg Hourly Earnings : Construction	112	Real M2 Money Stock
47	Avg Hourly Earnings : Manufacturing	113	St. Louis Adjusted Monetary Base
Group 3: Consumption, Orders and Inventories		114	Total Reserves of Depository Institutions
48	Real personal consumption expenditures	115	Reserves Of Depository Institutions
49	Real Manu. and Trade Industries Sales	116	Commercial and Industrial Loans
50	Retail and Food Services Sales	117	Real Estate Loans at All Commercial Banks
51	New Orders for Consumer Goods	118	Total Nonrevolving Credit
52	New Orders for Durable Goods	119	Nonrevolving consumer credit to Personal Income
53	New Orders for Nondefense Capital Goods	120	MZM Money Stock
54	Un.lled Orders for Durable Goods	121	Consumer Motor Vehicle Loans Outstanding
55	Total Business Inventories	122	Total Consumer Loans and Leases Outstanding
56	Total Business: Inventories to Sales Ratio	123	Securities in Bank Credit at All Commercial Banks
57	Consumer Sentiment Index	Group 8: Stock Market	
Group 4: Housing		124	S&P's Common Stock Price Index: Composite
58	Housing Starts: Total New Privately Owned	125	S&P's Common Stock Price Index: Industrials
59	Housing Starts Northeast	126	S&P's Composite Common Stock: Dividend Yield
60	Housing Starts Midwest	127	S&P's Composite Common Stock: Price-Earnings Ratio
61	Housing Starts South	128	VXO
62	Housing Starts West		
63	New Private Housing Permits (SAAR)		
64	New Private Housing Permits Northeast		
65	New Private Housing Permits Midwest		
66	New Private Housing Permits South		
67	New Private Housing Permits West		

Table G2: Full list of the Local Area Districts (LADs) for the UK regional house price

Scotland		London Area		Wales	
ID	Region Names	ID	Region Names	ID	Region Names
1	Clackmannanshire	32	City.of.London	65	Isle.of.Anglesey
2	Dumfries.and.Galloway	33	Barking.and.Dagenham	66	Gwynedd
3	East.Ayrshire	34	Barnet	67	Conwy
4	East.Loathian	35	Bexley	68	Denbighshire
5	Na.h-Eileanan.Siar	36	Brent	69	Flintshire
6	Falkirk	37	Bromley	70	Wrexham
7	Highland	38	Camden	71	Ceredigion
8	Inverclyde	39	Croydon	72	Pembrokeshire
9	Midlothian	40	Ealing	73	Carmarthenshire
10	Moray	41	Enfield	74	Swansea
11	North.Ayrshire	42	Greenwich	75	Neath.Port.Talbot
12	Orkney.Islands	43	Hackney	76	Bridgend
13	Scottish.Borders	44	Hammersmith.and.Fulham	77	Vale.of.Glamorgan
14	Shetland.Islands	45	Haringey	78	Cardiff
15	South.Ayrshire	46	Harrow	79	Rhondda.Cynon.Taf
16	South.Lanarkshire	47	Havering	80	Caerphilly
17	Stirling	48	Hillingdon	81	Blaenau.Gwent
18	Aberdeen.City	49	Hounslow	82	Torfaen
19	Aberdeenshire	50	Islington	83	Monmouthshire
20	Argyll.and.Bute	51	Kensington.and.Chelsea	84	Newport
21	City.of.Edinburgh	52	Kingston.upon.Thames	85	Powys
22	Renfrewshire	53	Lambeth	86	Merthyr.Tydfil
23	West.Dunbartonshire	54	Lewisham		
24	West.Loathian	55	Merton		
25	Angus	56	Newham		
26	Dundee.City	57	Redbridge		
27	East.Dunbartonshire	58	Richmond.upon.Thames		
28	Fife	59	Southwark		
29	Perth.and.Kinross	60	Sutton		
30	Glasgow.City	61	Tower.Hamlets		
31	North.Lanarkshire	62	Waltham.Forest		
		63	Wandsworth		
		64	Westminster		

H Additional Results of Empirical Applications

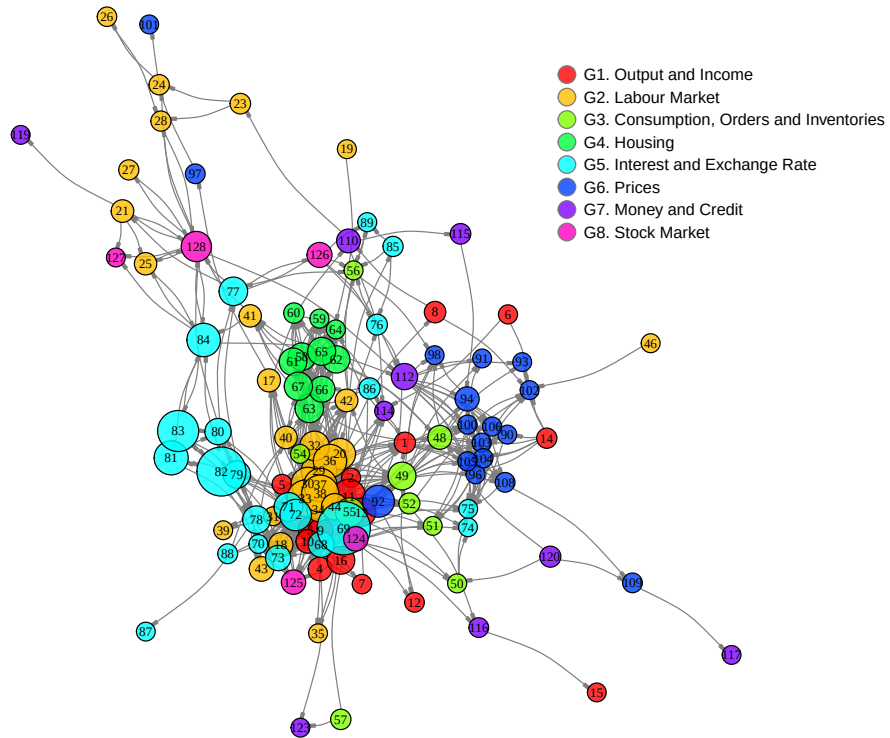


Figure H1: Dynamic network visualization: macroeconomic variables, asymptotic t_0 , $q = 0.05$

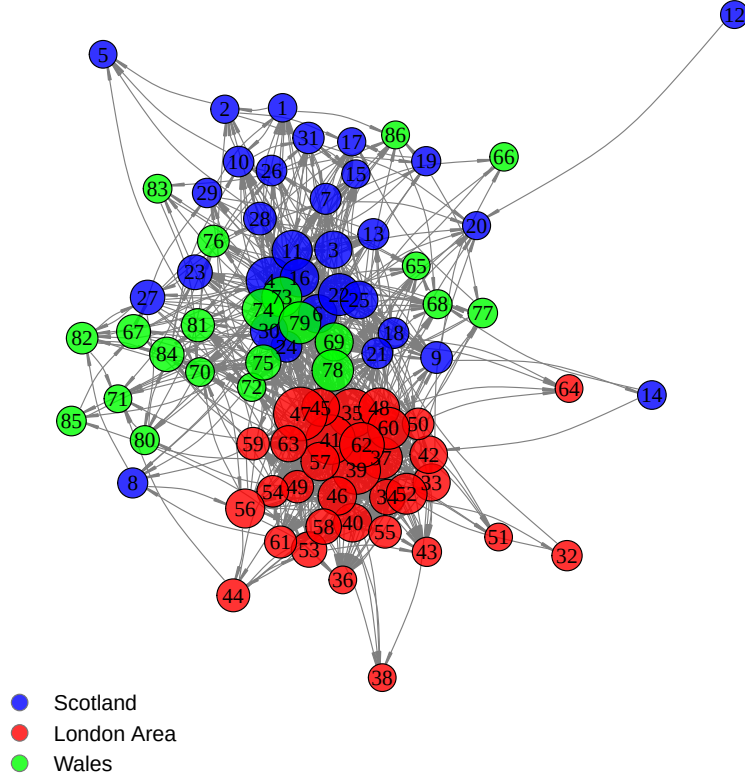


Figure H2: Dynamic network visualization: UK regional house prices, asymptotic t_0 , $q = 0.05$

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